CHAPTER

52

DOORS

CHAPTER 52
DOORS

Subject/Page	Date	COC
52-EFFECTIVE PAGES		
1 thru 2	Jun 15/2026	
52-HOW TO USE THE FIM		
1	Feb 15/2013	
2	Feb 15/2013	
3	Feb 15/2013	
4	Feb 15/2013	
5	Feb 15/2013	
6	Feb 15/2013	
52-FAULT CODE INDEX		
101	Oct 15/2024	
102	Oct 15/2024	
52-MAINT MSG INDEX		
101	Sep 15/2023	
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52-10 TASKS		
201	Feb 15/2026	
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206	Jun 15/2025	
207	Feb 15/2020	
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211	Feb 15/2020	
212	Oct 15/2023	
213	Oct 15/2023	
214	Oct 15/2023	
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217	Feb 15/2024	
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Subject/Page	Date	COC
52-20 TASKS		
201	Feb 15/2026	
202	Feb 15/2026	
203	Feb 15/2026	
204	Feb 15/2026	
205	Feb 15/2026	
206	Feb 15/2026	
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208	Feb 15/2026	
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231	Feb 15/2026	
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233	Feb 15/2026	
234	BLANK	

Subject/Page	Date	COC
52-30 TASKS		
201	Jun 15/2013	
202	Feb 15/2013	
203	Oct 15/2021	
204	Oct 15/2021	
205	Jun 15/2024	
206	Jun 15/2024	
207	Jun 15/2024	
208	BLANK	
52-32 TASKS		
201	Sep 15/2023	
202	Sep 15/2023	
203	Sep 15/2023	
204	Sep 15/2023	
205	Feb 15/2026	
206	Sep 15/2023	
207	Sep 15/2023	
208	Sep 15/2023	
209	Sep 15/2023	
210	Sep 15/2023	
211	Sep 15/2023	
212	Sep 15/2023	
213	Sep 15/2023	
214	Sep 15/2023	
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221	Feb 15/2026	
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227	Sep 15/2023	

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52-EFFECTIVE PAGES

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Subject/Page	Date	COC
52-32 TASKS (cont)		
228	Sep 15/2023	
229	Sep 15/2023	
230	Sep 15/2023	
231	Sep 15/2023	
232	Sep 15/2023	
52-40 TASKS		
201	Oct 15/2019	
202	Oct 15/2019	
203	Oct 15/2019	
204	Oct 15/2019	
52-50 TASKS		
201	Jun 15/2013	
202	Jun 15/2016	
203	Feb 15/2024	
204	Feb 15/2026	
205	Feb 15/2024	
206	Feb 15/2026	
207	Feb 15/2026	
208	Feb 15/2026	
209	Feb 15/2026	
210	Feb 15/2026	
211	Feb 15/2026	
212	Feb 15/2026	
213	Feb 15/2026	
214	Feb 15/2026	
215	Feb 15/2026	
216	BLANK	
52-99 TASKS		
201	Feb 15/2013	
202	BLANK	

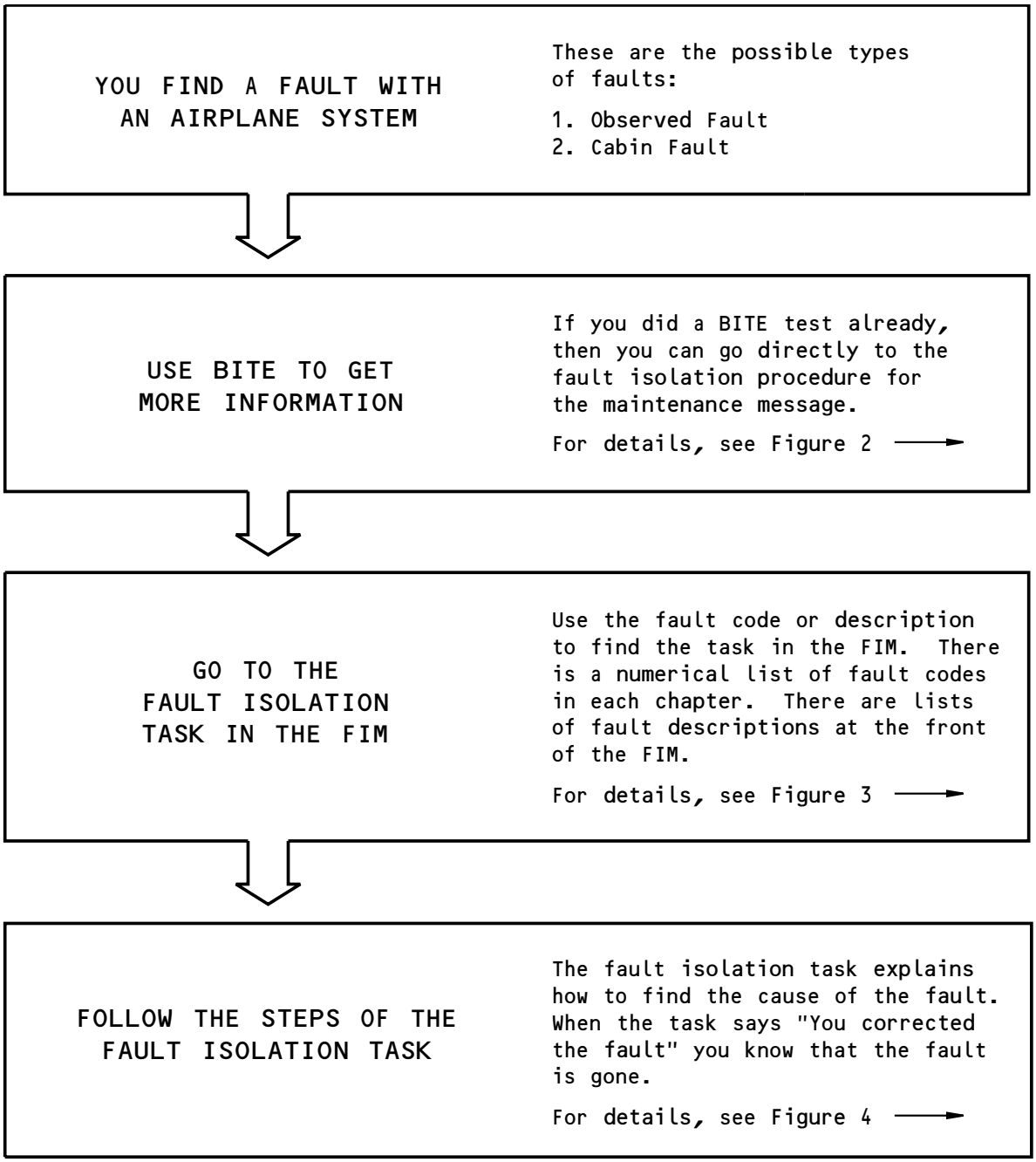
Subject/Page	Date	COC

Subject/Page	Date	COC

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52-EFFECTIVE PAGES

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G04902 S0000148576_V1

Basic Fault Isolation Process
Figure 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Some airplane systems have built-in test equipment (BITE). IF the system finds a fault when you do a BITE test, it will give you a maintenance message.

A maintenance message can be any of these:

- a code
- a text message
- a light
- an indication.

To find the fault isolation task for a maintenance message, go to the Maintenance Message Index in the chapter for the applicable system.

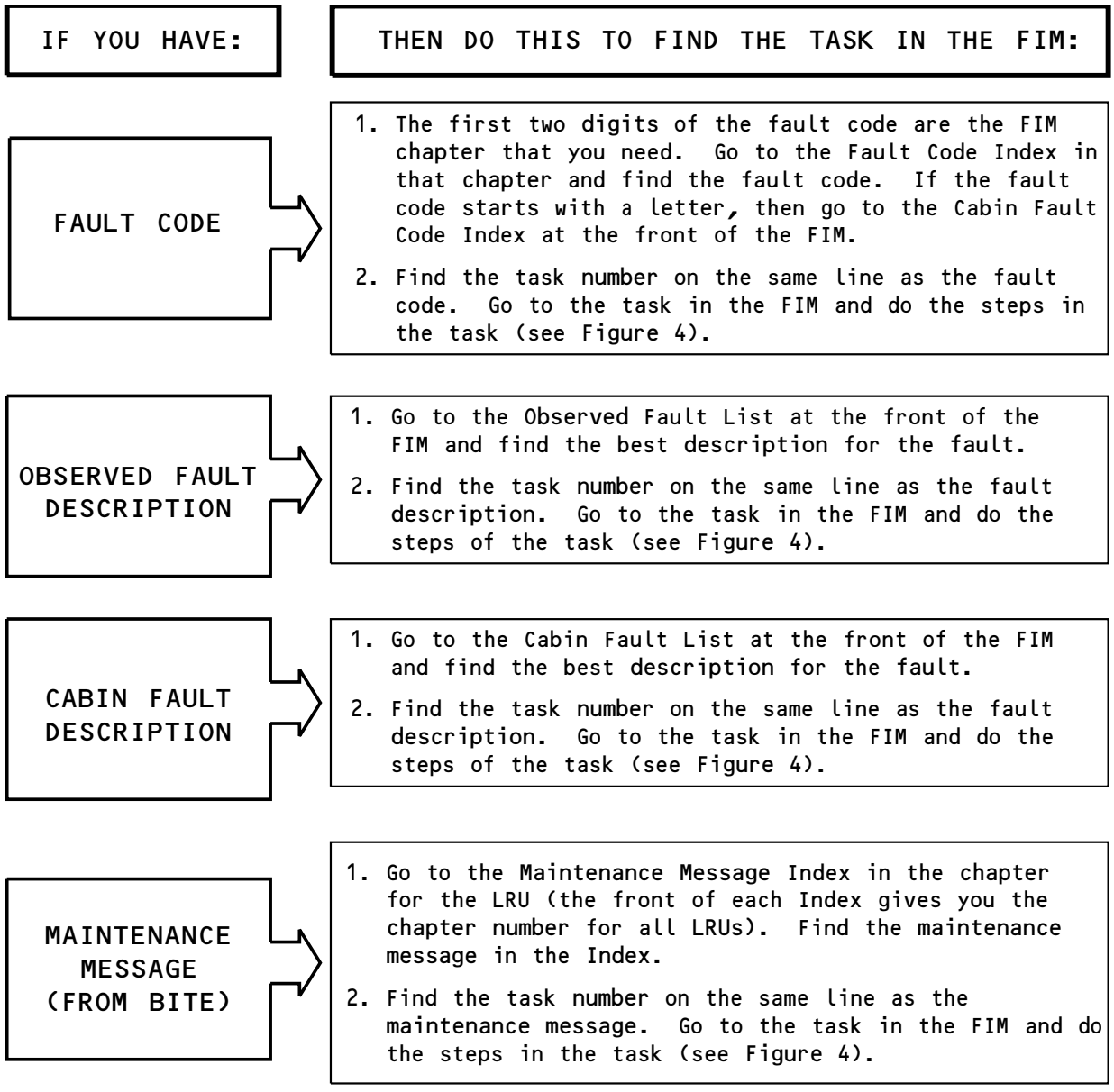
If you do not know which chapter is the correct one, look at the list at the front of any Maintenance Message Index. For each system or component (LRU) that has BITE, this list gives the chapter number where you can find the Index that you need.

Find the maintenance message for the applicable LRU or system in the Index. Then find the task number on the same line as the maintenance message. Go to the task in the FIM and do the steps of the task (see Figure 4).

G04950 S0000148578_V1

Getting Fault Information from BITE
Figure 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details



G04979 S0000148579_V2

Finding the Fault Isolation Task in the FIM
Figure 3

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ASSUMED CONDITIONS AT START OF TASK

- External electrical power is ON
- Hydraulic power and pneumatic power are OFF
- Engines are shut down
- No equipment in the system is deactivated

POSSIBLE CAUSES

- The list of possible causes has the most likely cause first and the least likely cause last.
- You can use the maintenance records of your airline to determine if the fault occurred before. Compare the list of possible causes to the past maintenance actions. This will help prevent repetition of the same maintenance actions.

INITIAL EVALUATION PARAGRAPH

- The primary purpose of the Initial Evaluation paragraph at the start of the task is to help you find out if you can detect the fault right now:
  - If you cannot detect the fault right now, then the task cannot isolate the fault and the Initial Evaluation paragraph will say that there was an intermittent fault.
  - If you have an intermittent fault, you must use your judgement (and follow your airline's policy) to decide which maintenance action to take. Then monitor the airplane to see if the fault happens again on subsequent flights.
- The Initial Evaluation paragraph can also help you find out which Fault Isolation Procedure to use to isolate and correct the fault.

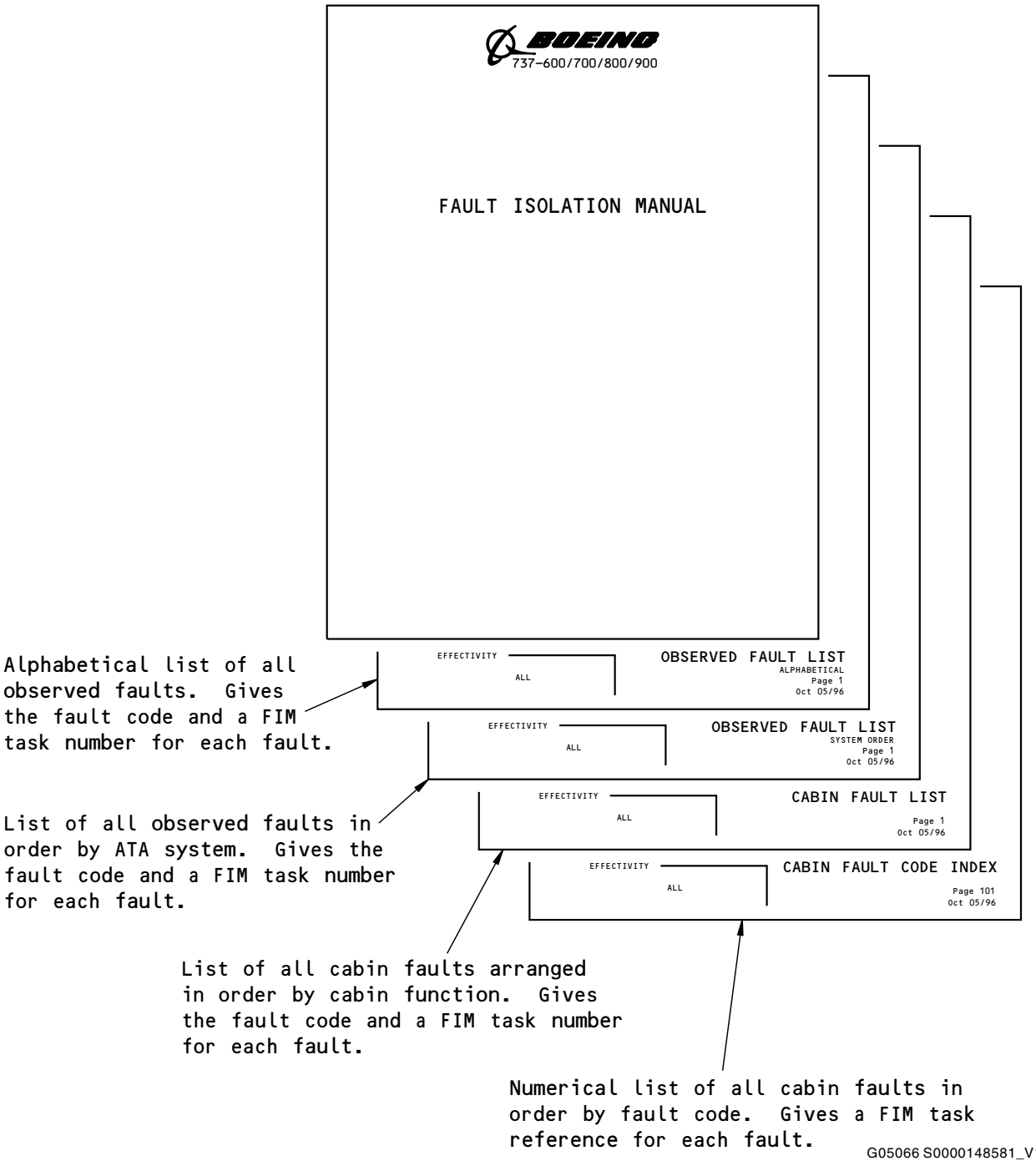
FAULT ISOLATION STEPS

- The FIM task steps are presented in a specified order. The "If... then" statements will guide you along a logical path. But if you do not plan to follow the FIM task exactly, make sure that you read it before you start to isolate the fault. Some FIM procedures start with important steps that have an effect on the other steps in the procedure.
- When you are at the endpoint of the path, the step says "...you corrected the fault." Complete the step and exit the procedure.

G05009 S0000148580_V3

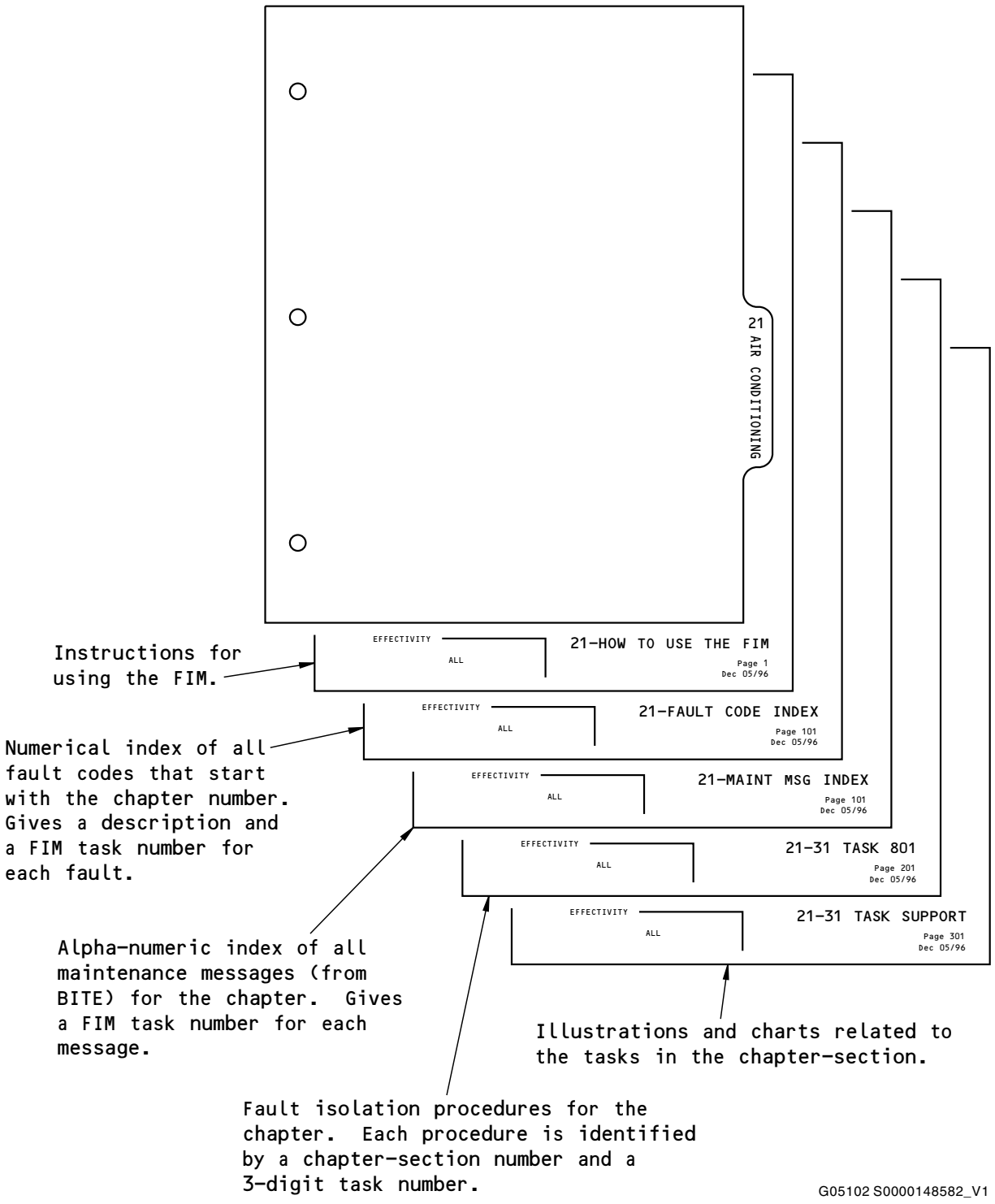
Doing the Fault Isolation Task
Figure 4

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Subjects at Front of FIM
Figure 5

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Subjects in Each FIM Chapter
Figure 6

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAULT CODE	FAULT DESCRIPTION	GO TO FIM TASK
521 010 44	Entry door exterior handle: Difficult to operate - forward.	52-10 TASK 811
521 010 45	Entry door exterior handle: Difficult to operate - aft.	52-10 TASK 811
521 011 44	Entry Door: Handle Movement - forward.	52-10 TASK 815
521 015 45	Aft entry door: Difficult to operate between cocked and fully open position - door jamming or binding.	52-10 TASK 813
521 611 44	Door warning annunciator is on - FWD ENTRY.	52-10 TASK 801
521 611 45	Door warning annunciator is on - AFT ENTRY.	52-10 TASK 801
523 010 00	Cargo Door: Difficult to open/close.	52-30 TASK 801
523 020 00	Cargo Door: does not unlatch/latch.	52-30 TASK 802
523 220 00	Cargo Door (Main Deck): does not raise.	52-32 TASK 805
523 230 00	Cargo Door (Main Deck): does not stop at canopy position when lowered.	52-32 TASK 806
523 240 00	Cargo Door (Main Deck): does not stop at canopy position when raised.	52-32 TASK 804
523 283 00	Cargo Door (Main Deck): does not latch.	52-32 TASK 817
523 284 00	Cargo Door (Main Deck): does not lock.	52-32 TASK 816
523 285 00	Cargo Door (Main Deck): does not unlatch.	52-32 TASK 815
523 286 00	Cargo Door (Main Deck): does not unlock.	52-32 TASK 814
523 611 44	Door warning annunciator is on - FWD CARGO.	52-30 TASK 803
523 611 45	Door warning annunciator is on - AFT CARGO.	52-30 TASK 803
524 010 00	Galley service door exterior handle: Difficult to operate.	52-10 TASK 811
524 015 45	Aft galley door: Difficult to operate between cocked and fully open position - door jamming or binding.	52-10 TASK 814
524 611 44	Door warning annunciator is on - FWD SERVICE.	52-10 TASK 801
524 611 45	Door warning annunciator is on - AFT SERVICE.	52-10 TASK 801
524 612 00	Door warning annunciator is on - EQUIP.	52-40 TASK 801 or 52-40 TASK 802
525 010 00	Control Cabin Door: CAB DOOR UNLOCKED light does not come on when the door is unlocked.	52-50 TASK 801
525 020 00	Control Cabin Door: Difficult to open/close.	52-50 TASK 802
525 030 00	Control Cabin Door: key is missing.	52-99 TASK 801
525 032 00	Control Cabin Door: key is damaged.	52-99 TASK 801
525 040 00	Control Cabin Door: does not release electrically.	52-50 TASK 803
525 121 00	Flight deck door (enhanced security door): chime does not sound with switch set to AUTO and correct code in keypad.	52-50 TASK 808
525 131 00	Flight deck door (enhanced security door): chime sounds with switch set to DENY.	52-50 TASK 807

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FAULT CODE	FAULT DESCRIPTION	GO TO FIM TASK
525 141 00	Flight deck door (enhanced security door): does not lock with switch set to AUTO.	52-50 TASK 804
525 151 00	Flight deck door (enhanced security door): does not unlock with switch set to AUTO and correct code in keypad.	52-50 TASK 806
525 161 00	Flight deck door (enhanced security door): does not unlock with switch set to UNLKD.	52-50 TASK 805
525 171 00	Flight deck door (enhanced security door): LOCK FAIL light is on.	52-50 TASK 809
525 181 00	Flight deck door (enhanced security door): unlocks with switch set to DENY.	52-50 TASK 807
526 560 00	Airstair, forward: uncommanded motion.	Reference Not Currently Available

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LRU/SYSTEM	SHORT NAME	CHAPTER
Air Data Inertial Reference System	ADIRS	34
Air Traffic Controller Transponder - 1 (Left)	ATC XPDR - 1 (L)	34
Air Traffic Controller Transponder - 2 (Right)	ATC XPDR - 2 (R)	34
Airborne Vibration Monitor System Signal Conditioner	AVM SIG COND	77
Antiskid Control Unit	ANTISKID	32
Attendant Control Panel	ACP	23
Automatic Direction Finder Receiver - 1	ADF RECVR - 1	34
Automatic Direction Finder Receiver - 2	ADF RECVR - 2	34
Autothrottle Computer	A/T COMPUTER	22
Auxiliary Power Unit	APU	49
Auxiliary Power Unit Generator Control Unit	APU GCU	24
Bus Power Control Unit	BPCU	24
Cabin Pressure Controller	CAB PRESS CON	21
Cabin Temperature Controller	CAB TEMP CONT	21
Cargo Electronic Unit - Lower Aft	CEU - LWR AFT	26
Cargo Electronic Unit - Lower Forward	CEU - LWR FWD	26
Cargo Electronic Unit - Main Aft	CEU - MAIN AFT	26
Cargo Electronic Unit - Main Forward	CEU - MAIN FWD	26
Cargo Fire Control Panel	CFCP	26
Common Display System	CDS	31
Compartment Overheat Detection Control Module	WING/BODY OHT	26
Digital Flight Control System	DFCS	22
Distance Measurement Equipment Interrogator	DME INTRROGTR	34
Electrical Meters, Battery, and Galley Power Module	P5-13	24
Electronic Engine Controller - 1	ENGINE - 1	73
Electronic Engine Controller - 2	ENGINE - 2	73
Emergency Locator Transmitter	ELT	23
Engine Accessory Unit	EAU	78
Engine Accessory Unit/TR DEPLOY ENG 1	EAU/TR DPLOY-ENG 1	78
Engine Accessory Unit/TR DEPLOY ENG 2	EAU/TR DPLOY-ENG 2	78
Engine Accessory Unit/TR STOW ENG 1	EAU/TR STOW-ENG 1	78
Engine Accessory Unit/TR STOW ENG 2	EAU/TR STOW-ENG 2	78
Engine and Auxiliary Power Unit Fire Detection Control Module	ENG/APU FIRE	26
Enhanced Digital Flight Control Computer-A	EDFCC-A	22
Enhanced Digital Flight Control System	EDFCS	22

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LRU/SYSTEM	SHORT NAME	CHAPTER
Flap/Slat Electronics Unit	FSEU	27
Flight Data Acquisition Unit	FDAU	31
Flight Management Computer System	FMCS	34
Fuel Quantity Indicating System	FQIS	28
Generator Control Unit - 1	GCU - 1	24
Generator Control Unit - 2	GCU - 2	24
Ground Proximity Computer	GROUND PROX	34
High Frequency Transceiver	HF XCVR	23
Low Limit (35 Degree F) Controller - Left	35 DEG CONT L	21
Low Limit (35 Degree F) Controller - Right	35 DEG CONT R	21
Multi-Mode Receiver	MMR	34
Nitrogen Generation System BITE Display Unit	NGS	47
Pack Flow Temperature Controller	PFTC	21
Pack/Zone Temperature Controller - Left	PACK/ZN CON - L	21
Pack/Zone Temperature Controller - Right	PACK/ZN CON - R	21
Proximity Switch Electronics Unit	PSEU	32
Radio Altimeter Receiver/Transmitter	RADIO ALTIMTR	34
Stall Management Yaw Damper Computer - 1	SMYD - 1	27
Stall Management Yaw Damper Computer - 2	SMYD - 2	27
Supplemental Proximity Switch Electronics Unit	SPSEU	32
Traffic Alert and Collision Avoidance System Computer	TCAS COMPUTER	34
VHF Omnidirectional Ranging Marker Beacon Receiver	VOR/MKR RCVR	34
Very High Frequency Transceiver	VHF XCVR	23
Waste Tank Logic Control Module	WASTE TANK	38
Weather Radar Receiver/Transmitter	WEATHER RADAR	34
Window Heat Control Unit - Left Forward	WHCU - L FWD	30
Window Heat Control Unit - Left Side	WHCU - L SIDE	30
Window Heat Control Unit - Right Forward	WHCU - R FWD	30
Window Heat Control Unit - Right Side	WHCU - R SIDE	30
Window Heat Control Unit 1 - Left Forward and Right Side	WHCU1 - L FWD/R SIDE	30
Window Heat Control Unit 2 - Right Forward and Left Side	WHCU2 - R FWD/L SIDE	30

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LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
PSEU	52-71001 FWD ENTR DR OPEN	52-10 TASK 801
PSEU	52-71002 FWD SERV DR OPEN	52-10 TASK 801
PSEU	52-71003 AFT ENTR DR OPEN	52-10 TASK 801
PSEU	52-71004 AFT SER DR OPEN	52-10 TASK 801
PSEU	52-72001 FWD CGO DR OPEN	52-30 TASK 803
PSEU	52-72002 AFT CGO DR OPEN	52-30 TASK 803
PSEU	52-72003 FWD ACC DR OPEN	52-40 TASK 801
PSEU	52-72004 EE ACC DR OPEN	52-40 TASK 802
PSEU	52-72005 L FL SW FAULT	52-20 TASK 801
PSEU	52-72006 L FWD FL SW FAULT	52-20 TASK 801
PSEU	52-72007 L FL SW FAULT	52-20 TASK 801
PSEU	52-72008 R FL SW FAULT	52-20 TASK 801
PSEU	52-72009 R FWD FL SW FAULT	52-20 TASK 801
PSEU	52-72010 L OW SW A FAULT	52-20 TASK 802
PSEU	52-72011 L OW SW B FAULT	52-20 TASK 802
PSEU	52-72012 L FWD OW SW A FLT	52-20 TASK 802
PSEU	52-72013 L FWD OW SW B FLT	52-20 TASK 802
PSEU	52-72014 R OW SW A FAULT	52-20 TASK 802
PSEU	52-72015 R OW SW B FAULT	52-20 TASK 802
PSEU	52-72016 R FWD OW SW A FLT	52-20 TASK 802
PSEU	52-72017 R FWD OW SW B FLT	52-20 TASK 802
PSEU	52-72018 ENG RUN R FAULT	52-20 TASK 803
PSEU	52-72019 OVWG OPT FAULT	52-20 TASK 804
PSEU	52-72020 L OW SW DISAGREE	52-20 TASK 805
PSEU	52-72021 L FWD OW SW DSGR	52-20 TASK 805
PSEU	52-72022 R OW SW DISAGREE	52-20 TASK 805
PSEU	52-72023 R FWD OW SW DSGR	52-20 TASK 805
PSEU	52-72024 FOUR OW OPT FLT	52-20 TASK 804
PSEU	52-72101 MDCD CL SW FAULT	52-32 TASK 811
PSEU	52-72102 MDCD LCH SW FAULT	52-32 TASK 811
PSEU	52-72103 MDCD LKD DISAGREE	52-32 TASK 811
PSEU	52-72104 MDCD TOW DISABLED	31-51 TASK 823
PSEU	52-72105 INVALID OPT PINS	52-20 TASK 809
PSEU	52-72106 L FWD FL SW FAULT	52-20 TASK 801
PSEU	52-72107 L FL SW FAULT	52-20 TASK 801

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LRU/SYSTEM	MAINTENANCE MESSAGE	GO TO FIM TASK
PSEU	52-72108 R FL SW FAULT	52-20 TASK 801
PSEU	52-72109 R FWD FL SW FAULT	52-20 TASK 801
PSEU	52-74001 L OVWG OPEN	52-20 TASK 806
PSEU	52-74002 L FWD OW OPEN	52-20 TASK 806
PSEU	52-74003 R OVWG OPEN	52-20 TASK 806
PSEU	52-74004 R FWD OW OPEN	52-20 TASK 806
PSEU	52-76001 EQPT WARN FLT	52-10 TASK 802
PSEU	52-76003 AFT CGO WARN FLT	52-10 TASK 802
PSEU	52-76004 AFT ENTR WARN FLT	52-10 TASK 802
PSEU	52-76005 AFT SERV WARN FLT	52-10 TASK 802
PSEU	52-76012 AIRSTAIR WARN FLT	52-10 TASK 802
PSEU	52-76014 FWD CGO WARN FLT	52-10 TASK 802
PSEU	52-76015 FWD ENTR WARN FLT	52-10 TASK 802
PSEU	52-76016 FWD SERV WARN FLT	52-10 TASK 802
PSEU	52-76017 FL RELAY 1 FAULT	52-20 TASK 807
PSEU	52-76018 FL RELAY 2 FAULT	52-20 TASK 807
PSEU	52-76019 LOW WARN FLT	52-20 TASK 808
PSEU	52-76020 L FWD OW WARN FLT	52-20 TASK 808
PSEU	52-76021 R OW WARN FLT	52-20 TASK 808
PSEU	52-76022 R FWD OW WARN FLT	52-20 TASK 808
PSEU	52-76101 MDCD WRN FAULT	52-32 TASK 808
PSEU	52-76102 MDCD UNLKD FAULT	52-32 TASK 808

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801. PSEU Entry and Galley Service Door Monitored Problem - Fault Isolation

A. Description

(1) This task is for these maintenance messages:

(a) 52-71001 FWD ENTR DR OPEN

(b) 52-71002 FWD SERV DR OPEN

(c) 52-71003 AFT ENTR DR OPEN

(d) 52-71004 AFT SERV DR OPEN

NOTE: The Proximity Switch Electronic Unit (PSEU) does not record these messages when the airplane is on the ground.

(2) This task is for P5 Door Warning Light indication during taxi or takeoff.

(3) For a detailed description of the system refer to SDS SUBJECT 52-71-00.

B. Possible Causes

(1) Applicable sensor for the maintenance message shown above:

(a) Forward Entry Door Indication Sensor, S199

(b) Aft Entry Door Indication Sensor, S200

(c) Forward Galley Service Entry Door Indication Sensor, S194

(d) Aft Galley Service Entry Door Indication Sensor, S195

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 802 POST SB 737-52-1159

(2) Forward Entry Door Switch, S1147

OMR ALL

(3) Entry or Galley Interior Door Handle out of position

(4) Service Door out of adjustment

(a) Forward Entry Door

(b) Aft Entry Door

(c) Galley Service Door

(5) Wiring

(6) Proximity Switch Electronic Unit (PSEU), M2061

C. Related Data

(1) WDM 52-71-11

(2) SSM 52-71-11

D. Initial Evaluation

OMR 701-799, 802 PRE SB 737-52-1159

(1) Look for obvious damage to the applicable sensor, target, and adjacent structure.

(a) If you find damage, then do the Fault Isolation Procedure below.

(b) If you do not find damage, then continue.

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OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 802 POST SB 737-52-1159

(2) For the Forward Entry Door, look for obvious damage to the Forward Entry Door Switch, S1147 and adjacent structure.

(a) If you find damage, then do the Fault Isolation Procedure below.

(b) If you do not find damage, then continue.

OMR ALL

(3) Do a check at the PSEU as follows:

(a) Make sure that the Main Entry Door is closed.

NOTE: Airplanes with Door Closed and Latched Switch, S1147:

If the Main Entry Door is not closed, the PSEU Replacement Test or Self-test will set a NO FLIGHT fault. You must erase the fault in the PSEU before a subsequent flight.

(b) Get access to the PSEU.

(c) Push the MENU switch until EXISTING FAULTS shows.

(d) Push the down switch until OTHER FUNCTNS? shows.

(e) Push the YES switch to select OTHER FUNCTNS?.

(f) Push the down switch until I/O MONITOR? shows.

(g) Push the YES switch to select I/O MONITOR.

(h) Push the down switch until SENSORS? shows.

(i) Push the YES switch to select SENSORS.

(j) Push the Down Switch until the applicable entry or Galley Service Door Indication Sensor shows:

1) For the Forward Entry Door Sensor, S199.

2) For the Aft Entry Door Sensor, S200.

3) For the Forward Galley Service Door Sensor, S194.

4) For the Aft Galley Service Door Sensor, S195.

(k) Push the YES Switch to show the Sensor Status.

1) If the sensor status is not FAILSAFE, then do this task: Entry and Galley Service Door Indication Sensor Adjustment (S194, S195, S199, or S200), AMM TASK 52-71-11-820-801, SUBTASK 52-71-11-820-005.

NOTE: Only completion of SUBTASK 52-71-11-820-005 is necessary.

2) If the Sensor Status is FAILSAFE, then do the Fault Isolation Procedure below.

E. Fault Isolation Procedure

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999; OMR 701-799, 802 POST SB 737-52-1159

(1) If you found obvious damage to the Forward Entry Door Switch, S1147, then do these steps:

(a) Repair the damage.

(b) Do a test of the Forward Entry Door Switch (Forward Entry Door Switch Test (S1147), AMM TASK 52-71-11-710-802).

1) Adjust the Forward Entry Door Switch, S1147 as necessary (Forward Entry Door Switch Adjustment (S1147), AMM TASK 52-71-11-820-802).

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(c) Open the Forward Entry Door.

(d) Close and latch the Forward Entry Door.

(e) If the FWD ENTRY light on the P5 panel is OFF, then you corrected the problem.

(f) If the FWD ENTRY light on the P5 panel stays ON, then do these steps:

1) Replace the Forward Entry Door Switch, S1147. These are the tasks:

- Forward Entry Door Switch Removal (S1147), AMM TASK 52-71-11-000-802
- Forward Entry Door Switch Installation (S1147), AMM TASK 52-71-11-400-802

2) If the FWD ENTRY light on the P5 panel stays ON, then continue.

OMR ALL

(2) If you found obvious damage to the applicable Sensor, Target, and adjacent structure, then do these steps:

(a) Repair the damage.

(b) Do the PSEU Check located in the Initial Evaluation.

1) If a maintenance message still shows, then continue.

(3) If you found that the Entry or Galley Interior Door Handle out of position:

(a) Do the Entry or Galley Interior Door Handle Does Not Stay In Position - Fault Isolation, 52-10 TASK 808.

(4) If you found that a Door is out of adjustment, do the applicable task below:

- For the Forward Entry Door, do the Forward Entry Door Adjustment, AMM TASK 52-11-00-820-801.
- For the Aft Entry Door, do the Aft Entry Door Adjustment, AMM TASK 52-13-00-820-801.
- For the Galley Service Door, do the Galley Service Door Adjustment, AMM TASK 52-41-00-820-801.

(a) Do the Repair Confirmation at the end of this task.

(5) If the Sensor Status is TGT NEAR, with a De-actuator installed, then do these steps:

(a) For the Forward Entry Door, do these steps:

1) Do this task: Forward Entry Door Indication Sensor Plunger Adjustment (S199), AMM TASK 52-71-11-820-803.

2) Do the Repair Confirmation at the end of this task.

(b) Do this task: Entry and Galley Service Door Indication Sensor Adjustment (S194, S195, S199, or S200), AMM TASK 52-71-11-820-801.

1) Do the Repair Confirmation at the end of this task.

(c) If the maintenance message still shows, then replace the Service Door Indication Sensor. These are the tasks:

- Entry and Galley Service Door Sensor Removal (S194, S195, S199, or S200), AMM TASK 52-71-11-000-801
- Entry and Galley Service Door Indication Sensor Installation (S194, S195, S199, or S200), AMM TASK 52-71-11-400-801

1) Do the Repair Confirmation at the end of this task.

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(d) Replace the PSEU, M2061. These are the tasks:

- Proximity Switch Electronics Unit (PSEU) Removal, AMM TASK 32-09-10-000-801
- Proximity Switch Electronics Unit (PSEU) Installation, AMM TASK 32-09-10-400-801

1) Do the Repair Confirmation at the end of this task.

(6) If the Sensor Status is TGT FAR without a De-actuator installed, then do these steps for the applicable Entry or Galley Service Door Indication Sensor:

(a) Do this task: Entry and Galley Service Door Indication Sensor Adjustment (S194, S195, S199, or S200), AMM TASK 52-71-11-820-801.

1) Do the Repair Confirmation at the end of this task.

(b) If the maintenance message still shows, then do these steps:

1) Replace the Service Door Indication Sensor. These are the tasks:

- Entry and Galley Service Door Sensor Removal (S194, S195, S199, or S200), AMM TASK 52-71-11-000-801
- Entry and Galley Service Door Indication Sensor Installation (S194, S195, S199, or S200), AMM TASK 52-71-11-400-801

2) Do the Repair Confirmation at the end of this task.

(7) If the Sensor Status is FAILSAFE, then do this PSEU Wiring Check between the PSEU and the applicable Entry or Galley Service Door Indication Sensor:

(a) Disconnect the applicable connector from the PSEU:

1) Connector D10988 for the Forward Entry Door or Forward Galley Service Door.
2) Connector D10986 for the Aft Entry Door or Aft Galley Service Door.

(b) Actuate the applicable Entry or Galley Service Door Indication Sensor.

(c) Find the wiring splice for the sensor.

(d) Do a wiring check between these pins of the applicable PSEU connector and sensor and make sure that the Resistance values are within these limits:

1) Between the Blue and Yellow Leads, do these steps to measure the Resistance between the pins of the PSEU connector (D10986 or D10988):

a) Connect the LCR meter (Inductance, Capacitance, Resistance), COM-1741 to the Blue (White/Blue) and Yellow (White/Yellow) Sensor Leads.

b) Push the L/C/R Mode Switch to set the LCR meter (Inductance, Capacitance, Resistance), COM-1741 in R (Resistance Mode).

c) Push the 1 kHz/120 Hz button to select the 120 Hz Excitation Frequency.

d) Push the L/C/R meter DATA HOLD button once.

e) Push and hold the L/C/R button on the LCR meter (Inductance, Capacitance, Resistance), COM-1741 until the meter changes to the Series Mode.

NOTE: "Ser" shows in the Upper Right corner of the display to indicate that the meter is in the Series Mode.

f) Read the Resistance value on the LCR meter (Inductance, Capacitance, Resistance), COM-1741.

<1> The Nominal Resistance reading at Normal Room Temperature (CMM DC Resistance, Bench Test condition) is referred to as R1. R1 should be 344.5 to 351.5 Ohms.

NOTE: The Line and Interface Resistance can add up to 2 Ohms.

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g) The permitted Resistance reading for in-service sensors exposed to varying conditions is 338 to 361 Ohms.

NOTE: For On-wing Resistance readings deviations from the shown limits, you should consider the increased/ lower Resistance associated with a Higher/Lower Temperature of the Proximity Sensor at the time of the measurement, before you make a decision about the condition of the Proximity Sensor.

Temperatures can deviate significantly from Normal Room Temperature, which is always the case with airplanes with systems ON, or airplanes that just have landed, or airplanes that have been sitting out in Hot or Cold Weather.

You need to know that the Resistance values can vary and that the Proximity Sensor could still be in good condition.

Copper has a Temperature Coefficient of Resistance which is approximately 0.40 Ohm/ °C.

The deviation of the Coil Circuit (generally the Yellow-Red Circuit) can be calculated at the rate of 0.40 Ohm/ °C from StandardTemperature (25 °C)

2) Between the Red and Yellow leads, do these steps to measure the Resistance between the pins of the PSEU connector (D10986 or D10988):

a) Connect the LCR meter (Inductance, Capacitance, Resistance), COM-1741 to the Red (White/Red) and Yellow (White/Yellow) Sensor leads.

b) Push the L/C/R Mode Switch to set the LCR meter (Inductance, Capacitance, Resistance), COM-1741 in R (Resistance Mode).

c) Push the 1 kHz/120 Hz button to select the 120 Hz Excitation Frequency.

d) Push the L/C/R meter DATA HOLD button once.

e) Push and hold the L/C/R button on the LCR meter (Inductance, Capacitance, Resistance), COM-1741 until the meter changes to the Series Mode.

NOTE: “Ser” appears in the Upper Right corner of the display to indicate that the meter is in the Series Mode.

f) Read the Resistance value on the LCR meter (Inductance, Capacitance, Resistance), COM-1741.

g) The Nominal Resistance reading at Normal Room Temperature (CMM DC Resistance, Bench Test condition) referred to as L1. L1 should be 27 to 33 Ohms.

NOTE: The Line and Interface Resistance can add up to 2 Ohms.

h) The permitted Resistance reading for in-service sensors exposed to varying conditions is 24 to 38 Ohms.

NOTE: For On-wing Resistance readings deviations from the shown limits, you should consider the increased/ lower Resistance associated with a Higher/Lower Temperature of the Proximity Sensor at the time of the measurement, before you make a decision about the condition of the Proximity Sensor.

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Temperatures can deviate significantly from Normal Room Temperature, which is always the case with airplanes with systems ON, or airplanes that just have landed, or airplanes that have been sitting out in Hot or Cold Weather.

You need to know that the Resistance values can vary and that the Proximity Sensor could still be in good condition.

Copper has a Temperature Coefficient of Resistance which is approximately 0.40 Ohm/ °C.

The deviation of the Coil Circuit (generally the Yellow-Red Circuit) can be calculated at the rate of 0.40 Ohm/ °C from StandardTemperature (25 °C)

D10988	**S199**
pin 19	Red
pin 36	Blue
Ground	Yellow

D10986	**S200**
pin 19	Red
pin 36	Blue
Ground	Yellow

D10988	**S194**
pin 58	Red
pin 57	Blue
Ground	Yellow

D10986	**S195**
pin 58	Red
pin 57	Blue
Ground	Yellow

(f) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.
2) Re-connect the applicable connector to the PSEU.
3) Re-actuate the sensor.
4) Do the Repair Confirmation at the end of this task.

F. Repair Confirmation

(1) Open the applicable FWD Entry (AFT Entry, Galley Service) Door.

(2) Close and latch the applicable Door.

(3) If the maintenance message does not show, then you corrected the problem.

(4) If the maintenance message still shows, then continue the Fault Isolation Procedure at the subsequent step.

END OF TASK

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802. PSEU Door Warning Light Monitored Problem - Fault Isolation

A. Description

(1) This task is for these maintenance messages:

(a) 52-76001 EQPT WARN FLT

(b) 52-76003 AFT CARGO WARN FLT

(c) 52-76004 AFT ENTR WARN FLT

(d) 52-76005 AFT SERV WARN FLT

OMR 802

(e) 52-76012 AIRSTAIR WARN FLT

OMR ALL

(f) 52-76014 FWD CGO WARN FLT

(g) 52-76015 FWD ENTR WARN FLT

(h) 52-76016 FWD SERV WARN FLT

(2) These maintenance messages show only after the PSEU replacement test or self test is done.

(3) (SDS SUBJECT 52-71-00)

B. Possible Causes

(1) The applicable light for the maintenance message listed above:

(a) Forward cargo door light, L1

(b) Aft cargo door light, L2

(c) Forward service galley door light, L3

(d) Aft service galley door light, L4

(e) Equipment access door light, L5

(f) Forward entry door light, L7

(g) Aft entry door light, L8

OMR 802

(h) Airstairs light, L9

OMR ALL

(2) Entry or galley interior door handle out of position

(3) Service door out of adjustment

(a) Forward entry door

(b) Aft entry door

(c) Galley service door

(4) Wiring

(5) Proximity switch electronics unit (PSEU), M2061

C. Related Data

(1) (SSM 52-71-11)

(2) (SSM 52-71-12)

(3) (WDM 52-71-11)

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(4) (WDM 52-71-12)

D. Initial Evaluation

(1) Do the PSEU replacement test or self test.

(a) Make sure that the main entry door is closed.

NOTE: On airplanes with door closed and latched switch S1147;

If the main entry door is not closed the PSEU replacement test or self test will set a no flight fault. The fault in the PSEU must be erased before subsequent flight.

(b) Do this task: Proximity Switch Electronics Unit (PSEU) - Operational Test, AMM TASK 32-09-10-710-801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

E. Fault Isolation Procedure

(1) Do this check of the door warning light on the forward overhead panel, P5:

(a) Push the applicable door warning light on the forward overhead panel, P5:

1) 52-76001 EQPT WARN FLT/EQUIP, L5.

2) 52-76003 AFT CGO WARN FLT/AFT CARGO, L2.

3) 52-76004 AFT ENTR WARN FLT/AFT ENTRY, L8.

4) 52-76005 AFT SERV WARN FLT/AFT SERVICE, L4.

OMR 802

5) 52-76012 AIRSTAIR WARN FLT/AIRSTAIR, L9.

OMR ALL

6) 52-76014 FWD CGO WARN FLT/FWD CARGO, L1.

7) 52-76015 FWD ENTR WARN FLT/FWD ENTRY, L7.

8) 52-76016 FWD SERV WARN FLT/FWD SERVICE, L3.

(b) Make sure the lamps in the indicator come on.

(c) If both lamps do not come on, then do these steps:

1) Do this task: Indicator Light - Lamp Replacement, AMM TASK 33-18-00-960-801.

2) Do the PSEU replacement test or self test. To do this, do this task: Proximity Switch Electronics Unit (PSEU) - Operational Test, AMM TASK 32-09-10-710-801.

3) If the maintenance message does not show, then you corrected the fault.

4) If the maintenance message shows, then do these steps:

a) Replace the applicable door warning light. To replace it, do this task: Indicator Light - Light Assembly Replacement, AMM TASK 33-18-00-960-802.

b) Do the PSEU replacement test or self test. To do this, do this task: Proximity Switch Electronics Unit (PSEU) - Operational Test, AMM TASK 32-09-10-710-801.

c) If the maintenance message does not show, then you corrected the fault.

(d) If either lamp or both lamps do come on, then continue.

(2) Do this check of the wiring:

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(a) Disconnect the applicable connector from the PSEU:

NOTE: The PSEU is in the forward electrical equipment bay.

(b) Disconnect the applicable connector from the door warning light.

(c) Do a wiring check between these pins of the applicable connector at the PSEU and the applicable connector at the door warning light:

Table 201

DOOR WARNING LIGHT	PSEU CONNECTOR		LIGHT CONNECTOR
EQUIPMENT, L5	D10986		D1406
	pin 37	-------	pin 6
AFT CARGO, L2	D10986		D482
	pin 33	-------	pin 12
AFT ENTRY, L8	D10986		D1406
	pin 60	-------	pin 5
AFT SERVICE, L4	D10986		D1406
	pin 59	-------	pin 3
OMR 802			
AIRSTAIRS, L9	D10988		D1406
	pin 37	-------	pin 4
OMR ALL			
FWD CARGO, L1	D10988		D1406
	pin 33	-------	pin 1
FWD ENTRY, L7	D10988		D1406
	pin 20	-------	pin 8
FWD SERVICE, L3	D10988		D1406
	pin 59	-------	pin 2

(d) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.
2) Re-connect the connector to the PSEU.
3) Re-connect the connector to the door warning light.
4) Do the PSEU replacement test or self test. To do this, do this task: Proximity Switch Electronics Unit (PSEU) - Operational Test, AMM TASK 32-09-10-710-801.
5) If the maintenance message does not show, then you corrected the fault.
6) If the maintenance message does show, then continue.

(e) If you did not find a problem with the wiring, then continue.

(3) If you found that the entry or galley interior door handle out of position:

(a) Do the Entry or Galley Interior Door Handle Does Not Stay In Position - Fault Isolation, 52-10 TASK 808.

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(4) If you found that a service door is out of adjustment:

(a) For the forward entry door, do the Forward Entry Door Adjustment, AMM TASK 52-11-00-820-801.

(b) For the aft entry door, do the Aft Entry Door Adjustment, AMM TASK 52-13-00-820-801.

(c) For the galley service door, do the Galley Service Door Adjustment, AMM TASK 52-41-00-820-801.

(5) Replace the PSEU, M2061.

These are the tasks:

Proximity Switch Electronics Unit (PSEU) Removal, AMM TASK 32-09-10-000-801,

Proximity Switch Electronics Unit (PSEU) Installation, AMM TASK 32-09-10-400-801.

(a) If the maintenance message does not show after the PSEU replacement test, then you corrected the fault.

——— END OF TASK ———

807. Entry or Galley Door Cannot Be Opened or Closed Easily - Fault Isolation

A. Description

(1) (SDS SUBJECT 52-10-00)

B. Possible Causes

(1) Lubrication

(2) Door Adjustment

(3) Snubber

(4) Hinge arms

(5) Fuselage hinge torque tube

C. Fault Isolation Procedure

(1) Do the visual inspection of the applicable door:

(a) Do this task: Forward Entry Door Check, AMM TASK 52-11-00-200-801.

(b) Do this task: Aft Entry Door Check, AMM TASK 52-13-00-200-801.

(c) Do this task: Galley Service Door Check, AMM TASK 52-41-00-200-801.

(d) If a problem is found, repair it.

1) Do the Repair Confirmation at the end of this task.

2) If the Repair Confirmation is not satisfactory, then continue.

(2) Lubricate the applicable door:

(a) Do this task: Forward Entry Door Servicing - Mechanism, AMM TASK 12-25-11-640-802.

(b) Do this task: Aft Entry Door Servicing - Mechanism, AMM TASK 12-25-12-640-802.

(c) Do this task: Forward Galley Service Door Lubrication - Mechanism, AMM TASK 12-25-13-640-802.

1) Do the Repair Confirmation at the end of this task.

2) If the Repair Confirmation is not satisfactory, then continue.

(3) Do the applicable door adjustment:

(a) Do this task: Forward Entry Door Adjustment, AMM TASK 52-11-00-820-801.

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(b) Do this task: Aft Entry Door Adjustment, AMM TASK 52-13-00-820-801.

(c) Do this task: Galley Service Door Adjustment, AMM TASK 52-41-00-820-801.

(d) If a problem is found, repair it.

1) Do the Repair Confirmation at the end of this task.

2) If the Repair Confirmation is not satisfactory, then continue.

(4) Do this check of the snubber:

(a) Examine the snubber.

1) Look for damage or leaks.

2) If you find a problem, repair it.

a) Do the Repair Confirmation at the end of this task.

b) If the Repair Confirmation is not satisfactory, then continue.

(5) Do this check of the hinge arms:

(a) Examine the guide arm and roller.

1) Look for corrosion or too much wear.

2) If you find a problem, repair it.

a) Do the Repair Confirmation at the end of this task.

b) If the Repair Confirmation is not satisfactory, then continue.

(b) Examine the hinge arms.

1) Look for corrosion or too much wear.

2) If you find a problem, repair it.

a) Do the Repair Confirmation at the end of this task.

b) If the Repair Confirmation is not satisfactory, then continue.

(c) Examine the door hinge torque tube.

1) Look for corrosion or too much wear.

2) If you find a problem, repair it.

a) Do the Repair Confirmation at the end of this task.

b) If the Repair Confirmation is not satisfactory, then continue.

(6) Do this check of the fuselage hinge torque tube:

(a) Examine the fuselage hinge torque tube.

1) Look for corrosion or too much wear.

2) If you find a problem, repair it.

a) Do the Repair Confirmation steps below.

D. Repair Confirmation

(1) Open and close the door.

(2) If it is easy to open and close the door, you have corrected the fault.

END OF TASK

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808. Entry or Galley Interior Door Handle Does Not Stay In Position - Fault Isolation

A. Description

(1) Interior entry door does not stay in correct position when latched.

(2) Interior galley door does not stay in correct position when latched.

B. Possible Causes

(1) Handle Box Assembly

C. Fault Isolation Procedure

(1) Remove the applicable Door Lining. These are the tasks:

- Forward Entry Door Lining - Removal, AMM TASK 52-11-31-000-802
- Aft Entry Door Lining Removal, AMM TASK 52-13-31-000-802
- Galley Service Door Lining Removal, AMM TASK 52-41-31-000-802

(2) Do this visual inspection of the Handle Box Assembly as follows:

(a) Make sure that the two Cam Follower Crank Assemblies are correctly installed.

(b) Make sure that all Handle Fasteners to the Hub are installed and tight.

(c) Make sure that Hub Fasteners to the Cam Plate are installed and tight.

(d) Make sure that the Cam Plate is correctly connected to the Cam Follower Crank Assemblies.

(e) If a problem is found, then repair it.

(f) Do the Repair Confirmation steps below.

D. Repair Confirmation

(1) Install the applicable Door Lining. These are the tasks:

- Forward Entry Door Lining - Installation, AMM TASK 52-11-31-400-802
- Aft Entry Door Lining Installation, AMM TASK 52-13-31-400-802
- Galley Service Door Lining Installation, AMM TASK 52-41-31-400-802

(2) LATCH and UNLATCH the Door with the Interior Handle.

(3) If the Interior Handle stays in position for both LATCH and UNLATCHED positions, you have corrected the problem.

——— END OF TASK ———

809. Door Hold Open Latch Hard To Operate - Fault Isolation

A. Description

(1) Refer to SDS SUBJECT 52-10-00 for a detailed description of the Passenger/Crew Doors.

B. Possible Causes

(1) Guide Arm

(2) Snubber

C. Fault Isolation Procedure

(1) To fault isolate the Hold Open Latch for the Forward Entry Door, do these steps:

(a) Open the door fully to engage the Door Hold Open Latch.

(b) Push the Yellow Release Lever on the Guide Arm to release the Hold Open Latch.

1) If the Hold Open Latch operates easily, then the fault was intermittent.

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2) If the Hold Open Latch is hard to operate, then continue.

(c) Remove or disassemble the Hold Open Mechanism, This is the task: Forward Entry Door Guide Arm and Roller Removal, AMM TASK 52-11-21-000-802.

1) Look for wear or damage to the Hold Open Mechanism or the Roller.

2) Make sure that the Hold Open Mechanism Spring is correctly installed.

a) If you find a problem, repair as necessary and then install the Fwd Entry Door Guide Arm and Roller and adjust the Guide Arm. These are the applicable tasks:

NOTE: Only do the Guide Arm adjustment.

- Forward Entry Door Guide Arm and Roller Installation, AMM TASK 52-11-21-400-802
- Forward Entry Door Adjustment, AMM TASK 52-11-00-820-801

<1> Do the Repair Confirmation at the end of this task.

(2) To fault isolate the Hold Open Latch for the Aft Entry Door or the Galley Door, do these steps:

(a) Open the door fully to engage the Door Hold Open Latch.

(b) Push the Yellow Release button on the Guide Arm to release the Hold Open Latch.

1) If the Hold Open Latch operates easily, then the fault was intermittent.

2) If the Hold Open Latch is hard to operate, then continue.

(c) Remove the Roller for the applicable Aft Entry (Galley) Door. These are the applicable tasks:

- Aft Entry Door Guide Arm and Roller Removal, AMM TASK 52-13-21-000-802
- Galley Service Door Guide Arm and Roller Removal, AMM TASK 52-41-21-000-801

(d) Remove the Snubber for the applicable Aft Entry (Galley) Door. These are the applicable tasks:

- Aft Entry Door Snubber Removal, AMM TASK 52-13-51-000-801
- Galley Service Door Snubber Removal, AMM TASK 52-41-51-000-801

(e) Look for wear or damage to the Roller Assembly.

(f) Make sure that the Snubber is not bottomed out.

(g) If you find a problem, repair as necessary and then install the Roller for the applicable Aft Entry (Galley) Door. These are the tasks:

- Aft Entry Door Guide Arm and Roller Installation, AMM TASK 52-13-21-400-802
- Galley Service Door Guide Arm and Roller Installation, AMM TASK 52-41-21-400-801

(h) Install the Snubber for the applicable Aft Entry (Galley) Door. These are the applicable tasks:

- Aft Entry Door Snubber Installation, AMM TASK 52-13-51-400-801
- Galley Service Door Snubber Installation, AMM TASK 52-41-51-400-801

(i) Adjust the Aft Entry Door. This is the task:

- Aft Entry Door Adjustment, AMM TASK 52-13-00-820-801

(j) Adjust the Galley Door Guide Arm and Roller. This is the task:

NOTE: Only do the Guide Arm and Snubber adjustment.

- Galley Service Door Adjustment, AMM TASK 52-41-00-820-801

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D. Repair Confirmation

(1) Open the door fully, until the Hold Open Latch engages.

(2) On the Forward Entry Door, push the Yellow Lever on the Guide Arm.

(3) On Aft Entry Door or Galley Door, push the Yellow Release button on the Guide Arm.

(4) If the Hold Open Latch operates correctly, you corrected the problem.

(5) If the Hold Open Latch does not operate correctly, continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

810. Door Latch Mechanism Hard To Operate - Fault Isolation

A. Description

(1) (SDS SUBJECT 52-10-00)

B. Possible Causes

(1) Adjustment

(2) Handle box

C. Fault Isolation Procedure

(1) Do the adjustment for the applicable door:

(a) Do this task: Forward Entry Door Adjustment, AMM TASK 52-11-00-820-801.

(b) Do this task: Aft Entry Door Adjustment, AMM TASK 52-13-00-820-801.

(c) Do this task: Galley Service Door Adjustment, AMM TASK 52-41-00-820-801.

NOTE: Only do latch and horizontal control rod adjustment.

(d) Do the Repair Confirmation at the end of this task.

(e) If the Repair Confirmation is not satisfactory, then continue.

(2) Do a check on the handle box:

(a) Remove the applicable door lining:

1) Do this task: Forward Entry Door Lining - Removal, AMM TASK 52-11-31-000-802.

2) Do this task: Aft Entry Door Lining Removal, AMM TASK 52-13-31-000-802.

3) Do this task: Galley Service Door Lining Removal, AMM TASK 52-41-31-000-802.

(b) Look for wear or damage on the latch crank and the latch crank bearings.

(c) Look for wear or damage on the horizontal control rod.

(d) If a problem is found, repair it.

(e) Install the applicable door lining:

1) Do this task: Forward Entry Door Lining - Installation, AMM TASK 52-11-31-400-802.

2) Do this task: Aft Entry Door Lining Installation, AMM TASK 52-13-31-400-802.

3) Do this task: Galley Service Door Lining Installation, AMM TASK 52-41-31-400-802.

(f) Do the Repair Confirmation steps below.

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D. Repair Confirmation

(1) Try to latch and unlatch the door.

(2) If it is easy to latch and unlatch the door, then you have corrected the fault.

——— END OF TASK ———

811. Forward Entry, Aft Entry, Forward Galley, or Aft Galley Doors Exterior Handle Difficult to Operate - Fault Isolation

A. Description

(1) Exterior handle is hard to pull out.

(2) Exterior handle is hard to turn.

(3) Exterior handle does not stow.

B. Possible Causes

(1) Lubrication

(2) Dirt and unwanted material

(3) Worn/Defective O-rings

(4) Horizontal Control Rod

(5) Bearing

(6) Spring

C. Related Data

(1) Forward Entry Door (SDS SUBJECT 52-10-00)

(2) Aft Entry Door (SDS SUBJECT 52-10-00)

(3) Galley Doors (SDS SUBJECT 52-40-00)

D. Fault Isolation Procedure

(1) Do this check of the exterior door handle:

(a) Remove the applicable door lining. These are the tasks:

- Forward Entry Door Lining - Removal, AMM TASK 52-11-31-000-802.
- Aft Entry Door Lining Removal, AMM TASK 52-13-31-000-802.
- Galley Service Door Lining Removal, AMM TASK 52-41-31-000-802.

(b) Examine the handle box and exterior handle assembly for damage or loose hardware.

(c) Remove dirt and unwanted material from the handle box.

(d) If the exterior handle is hard to pull out, then do these steps:

1) Lubricate the applicable handle shaft. These are the tasks:

- Forward Entry Door Servicing - Mechanism, AMM TASK 12-25-11-640-802.
- Aft Entry Door Servicing - Mechanism, AMM TASK 12-25-12-640-802.
- Forward Galley Service Door Lubrication - Mechanism, AMM TASK 12-25-13-640-802.

a) Do the Repair Confirmation at the end of this task.

b) If the Repair Confirmation is not satisfactory, then continue.

2) Remove the applicable exterior door handle. These are the tasks:

- Forward Entry Door Hinge Arm Removal, AMM TASK 52-11-11-000-802.

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- Aft Entry Door Hinge Arm Removal, AMM TASK 52-13-11-000-802.
- Galley Service Door Hinge Arm Removal, AMM TASK 52-41-11-000-801.

3) Do a check of both handle housing O-rings for wear, damage and correct installation.
   a) If necessary, adjust or replace the handle housing O-rings.
   b) Do the Repair Confirmation at the end of this task.
   c) If the Repair Confirmation is not satisfactory, then continue.
4) Do a check of the handle shaft spring for damage and release force and replace the spring if it is necessary.
   a) Do the Repair Confirmation at the end of this task.

(e) If the exterior handle is hard to turn, then do these steps:
1) Examine the handle shaft bearing.
   a) Make sure that the shaft turns freely in the bearing and replace the bearing if necessary.
   b) Do the Repair Confirmation at the end of this task.
   c) If the Repair Confirmation is not satisfactory, then continue.
2) Do the centering guide adjustment for the applicable door. These are the tasks:
   - Forward Entry Door Adjustment, AMM TASK 52-11-00-820-801.
   - Aft Entry Door Adjustment, AMM TASK 52-13-00-820-801.
   - Galley Service Door Adjustment, AMM TASK 52-41-00-820-801.
   a) Do the Repair Confirmation at the end of this task.
   b) If the Repair Confirmation is not satisfactory, then continue.
3) Do an inspection of the mating surfaces between the centering cam on the handle assembly and the handle housing. Repair or replace as necessary.
   a) Do the Repair Confirmation at the end of this task.

(f) If the exterior handle does not stow, then do these steps:
1) Do a check of the exterior handle O-rings for wear, damage and correct installation.
   a) If necessary, adjust or replace the exterior handle O-rings.
   b) Do the Repair Confirmation at the end of this task.
   c) If the Repair Confirmation is not satisfactory, then continue.
2) Do a check of the handle shaft spring for compression and replace the spring if necessary.
   a) Do the Repair Confirmation at the end of this task.
   b) If the Repair Confirmation is not satisfactory, then continue.
3) Do the horizontal rod adjustment for the applicable door. These are the tasks:
   - Forward Entry Door Adjustment, AMM TASK 52-11-00-820-801.
   - Aft Entry Door Adjustment, AMM TASK 52-13-00-820-801
   - Galley Service Door Adjustment, AMM TASK 52-41-00-820-801
   a) Do the Repair Confirmation at the end of this task.
   b) If the Repair Confirmation is not satisfactory, then continue.

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4) Do an inspection of the mating surfaces between the centering cam on the handle shaft assembly and the handle housing. Repair or replace as necessary.

5) Install the applicable exterior door handle. These are the tasks:

- Forward Entry Door Hinge Arm Installation, AMM TASK 52-11-11-400-802.
- Aft Entry Door Hinge Arm Installation, AMM TASK 52-13-11-400-802.
- Galley Service Door Hinge Arm Installation, AMM TASK 52-41-11-400-801.

a) Do the Repair Confirmation steps below.

E. Repair Confirmation

(1) Pull out the exterior handle.

(2) Open the door with the exterior handle.

(3) Close the door with the exterior handle.

(4) Stow the exterior handle.

(5) If it is easy to operate the exterior handle, you have corrected the problem.

(a) Install the applicable door lining. These are the tasks:

- Forward Entry Door Lining - Installation, AMM TASK 52-11-31-400-802.
- Aft Entry Door Lining Installation, AMM TASK 52-13-31-400-802.
- Galley Service Door Lining Installation, AMM TASK 52-41-31-400-802.

——— END OF TASK ———

812. Door Seal Leaking, Noisy, Torn - Fault Isolation

A. Description

(1) (SDS SUBJECT 52-09-00)

B. Possible Causes

(1) Delamination of a seal or joint.

(2) Cut or split in the seal joint or in the blade or bulb section of the seal.

(3) Damage of seal flanges or seal splices.

(4) Separation of splice bonds.

C. Fault Isolation Procedure

(1) Do the visual inspection of the seal as follows:

(a) Unlock, open partially, close and lock the door, and do these checks:

1) Make sure the seals are clear of the edge of all surrounding structure when the door opens and closes.

2) Make sure that the blades of the seals touch the seal depressors around the edge of the door when the door closes.

3) Make sure that the seal does not flip outboard at the corners when the door closes.

4) Make sure there are no cuts, delamination, or damage to the seal.

(2) Repair the applicable seals if it is necessary. To do this, do this task: Seal Repair Procedures For Specified Types of Seal Damage, AMM TASK 52-09-10-350-801.

(3) Replace the blade seal if it is necessary.

These are the tasks:

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Blade Seals Removal, AMM TASK 52-09-11-000-801,

Blade Seals Installation, AMM TASK 52-09-11-400-801.

(4) Replace the blade and diaphragm seal if it is necessary.

These are the tasks:

Blade and Diaphragm Seals Removal, AMM TASK 52-09-12-000-801,

Blade and Diaphragm Seals Installation, AMM TASK 52-09-12-400-801.

(5) Replace the bulb and diaphragm seal if it is necessary.

These are the tasks:

Bulb and Diaphragm Seal Removal, AMM TASK 52-09-14-000-801,

Bulb and Diaphragm Seal Installation, AMM TASK 52-09-14-400-801.

(6) Replace the aerodynamic seal if it is necessary.

These are the tasks:

Aerodynamic Seals Removal, AMM TASK 52-09-15-000-801,

Aerodynamic Seals Installation, AMM TASK 52-09-15-400-801.

(7) Replace the acoustic seal if it is necessary.

These are the tasks:

Acoustic Seal Removal, AMM TASK 52-09-16-000-801,

Acoustic Seal Installation, AMM TASK 52-09-16-400-802.

(8) Replace the light seal if it is necessary.

These are the tasks:

Light Seals Removal, AMM TASK 52-09-17-000-801,

Light Seal Installation, AMM TASK 52-09-17-400-801.

——— **END OF TASK** ———

813. Aft Entry Door Difficult to Operate Between Cocked and Fully Open Position, Door Jamming or Binding - Fault Isolation

A. Description

(1) Aft entry door difficult to operate between cocked and fully open position

(2) Aft entry door jamming or binding

(a) This issue occurs when the bumpers migrate, causing the door to jam. The rubber bumper at the lower hinge can help prevent the binding condition.

B. Possible Causes

(1) Aft Entry Door Bumper

C. Related Data

(1) Aft Entry Door (SDS SUBJECT 52-10-00)

D. Fault Isolation Procedure

(1) Do a visual inspection of the bumpers. This is the task:

- Aft Entry Door Bumper - Inspection, AMM TASK 52-13-51-200-801

(2) If the bumpers are missing or the position is incorrect, remove them. These are the tasks:

- Aft Entry Door Bumper - Removal, AMM TASK 52-13-51-000-802

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- Aft Entry Door Bumper - Installation, AMM TASK 52-13-51-400-802

(a) Do the Repair Confirmation at the end of this task.

E. Repair Confirmation

(1) Pull out the exterior handle.

(2) Open the door with the exterior handle.

(3) Close the door with the exterior handle.

(4) Stow the exterior handle.

(5) If it is easy to operate the exterior handle, you have corrected the problem.

——— END OF TASK ———

814. Aft Galley Door Difficult to Operate Between Cocked and Fully Open Position, Door Jamming or Binding - Fault Isolation

A. Description

(1) Aft galley door difficult to operate between cocked and fully open position

(2) Aft galley door jamming or binding

B. Possible Causes

(1) Aft Galley Door Bumper

C. Related Data

(1) Aft Galley Door (SDS SUBJECT 52-40-00).

D. Fault Isolation Procedure

(1) Do a visual inspection of the bumpers. This is the task:

- Galley Service Door Bumper - Inspection, AMM TASK 52-41-51-200-801

(2) If the bumpers are missing or the position is incorrect, remove them. These are the tasks:

- Galley Service Door Bumper - Removal, AMM TASK 52-41-51-000-802
- Galley Service Door Bumper - Installation, AMM TASK 52-41-51-400-802

(a) Do the Repair Confirmation at the end of this task.

E. Repair Confirmation

(1) Pull out the exterior handle.

(2) Open the door with the exterior handle.

(3) Close the door with the exterior handle.

(4) Stow the exterior handle.

(5) If it is easy to operate the exterior handle, you have corrected the problem.

——— END OF TASK ———

815. Fwd Entry Door Handle Movement - Fault Isolation

A. Description

(1) This task is for this Observed Fault:

(a) Fwd Entry Door handle movement during taxi or takeoff (on ground).

1) This can be an intermittent fault due to an improperly closed Fwd Entry Door prior to departure.

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a) When the Forward Entry Door is closed from the outside, it needs to be verified from the inside by pushing the handle to its fully closed horizontal position.

b) The Forward Entry Door should be closed from the inside to make sure that the handle has turned to the full horizontal position before releasing the handle.

2) Soft-Unlatching.

(b) Fwd Entry door handle movement during climb out (in air).

1) If the door handle moves and there is no door indication on the flight deck, Boeing considers this is a safe condition for flight and the handle movement can be ignored.

2) This can be an intermittent fault due to an older design of the Sensor Assembly.

3) The maintenance message 52-71001 is related to this fault.

B. Possible Causes

(1) Door operation

(2) Door rigging

(3) Forward Entry Door Indication Sensor, S199

C. Initial Evaluation

(1) Do a check of flight crew reports and maintenance records for this Observed Fault symptom.

(a) If you find that the observed fault symptom occurred on ground, then do the Fault Isolation Procedure - Handle Movement Observed on Ground below.

(b) If you find that the observed fault symptom occurred in air, then do the Fault Isolation Procedure - Handle Movement Observed in Air below.

D. Fault Isolation Procedure - Handle Movement Observed on Ground

(1) If necessary, do this task: Forward Entry Door (Soft Unlatching), AMM TASK 52-11-00-820-802.

(a) If the observed fault symptom does not occur on the subsequent flight, then you corrected the problem.

E. Fault Isolation Procedure - Handle Movement Observed in Air

(1) Look for the maintenance messages 52-71001.

(a) If you find the related maintenance message, then do the applicable Fault Isolation Procedure.

(b) If the observed fault symptom does not occur on the subsequent flight, then you corrected the problem.

——— END OF TASK ———

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801. Overwing Exit Door Flight Lock Switch Problem - Fault Isolation

A. Description

(1) This task is for these maintenance messages:

(a) 52-72005 L FL SW FAULT

(b) 52-72007 L FL SW FAULT

(c) 52-72107 L FL SW FAULT

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(d) 52-72006 L FWD FL SW FAULT

(e) 52-72106 L FWD FL SW FAULT

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(f) 52-72008 R FL SW FAULT

(g) 52-72108 R FL SW FAULT

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(h) 52-72009 R FWD FL SW FAULT

(i) 52-72109 R FWD FL SW FAULT

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(2) These maintenance messages show when there are all of these conditions:

(a) A minimum of one engine is on.

(b) The Engine Thrust Lever is set at more than 53 degrees.

(c) The Overwing Exit Doors are closed.

(d) The applicable Flight Lock Switch indicates open.

(3) (SDS SUBJECT 52-22-00)

B. Possible Causes

(1) Left (Right) Flight Lock Solenoid, M2219 (M2221)

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(2) Left Forward (Right Forward) Flight Lock Solenoid, M2220 (M2222)

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(3) Wire problem

(4) Left (Right) Flight Lock Switch, S1112 (S1114)

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(5) Left Forward (Right Forward) Flight Lock Switch, S1113 (S1115)

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(6) Proximity Switch Electronic Unit (PSEU), M2061

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT

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(Continued)

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	1	C01515	OVERWING FLIGHT LOCK-RIGHT
D	2	C01514	OVERWING FLIGHT LOCK-LEFT
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

D. Related Data

(1) SSM 52-71-13

(2) SSM 52-71-14

(3) WDM 52-71-13

(4) WDM 52-71-14

E. Initial Evaluation

(1) Do these steps to prepare for the Initial Evaluation:

NOTE: Two persons are necessary to do parts of this task: one person in the control compartment and one person in the passenger compartment.

(a) Open these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	1	C01515	OVERWING FLIGHT LOCK-RIGHT
D	2	C01514	OVERWING FLIGHT LOCK-LEFT

(b) Stop for a minimum of ten seconds.

(c) Close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	1	C01515	OVERWING FLIGHT LOCK-RIGHT
D	2	C01514	OVERWING FLIGHT LOCK-LEFT

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(d) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

(e) Make sure that a minimum of 3 out of 4 entry/service doors are closed.

(f) Make sure that the airplane is in ground mode.

(g) Make sure that the Engine Start Switches on the P5-20 Panel are in the OFF position.

(h) Move the Engine Start Levers on the Control Stand to IDLE.

(i) Stop for 5 minutes.

(j) Move the Engine Thrust Levers to the fully forward position.

(2) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message shows, then do these steps:

NOTE: Use these steps to isolate the door problem to the door indication circuit or the Flight Lock Solenoid circuit.

WARNING MAKE SURE THAT THE DOOR OPENING PATH IS CLEAR BEFORE YOU RELEASE THE DOOR HANDLE. THE DOOR IS SPRING-LOADED TO OPEN AUTOMATICALLY AND INJURIES COULD OCCUR.

1) Make sure that the Flight Lock Solenoid is in the locked position:
   a) Pull the Overwing Exit Door handle to open the door.
   b) Make sure that the Overwing Wing Exit Door does not open.
2) Move the Engine Thrust Levers to idle.
3) The following steps must occur at the same time:
   a) Stand by the Overwing Doors and prepare to listen for a delay between the left door and right door flight lock actuation.
   b) Move the Thrust Levers forward.

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c) Listen for the left and right door flight lock actuation to occur at the same time.

<1> If a flight lock actuates 1 or more seconds after the others, replace the slow Actuating Solenoid Assembly.

4) Move the Engine Thrust Levers back to idle.

5) Move the Engine Start Levers to CUTOFF.

WARNING MAKE SURE THAT THE DOOR OPENING PATH IS CLEAR BEFORE YOU RELEASE THE DOOR HANDLE. THE DOOR IS SPRING-LOADED TO OPEN AUTOMATICALLY AND INJURIES COULD OCCUR.

6) Make sure that the Flight Lock Solenoid is not in the locked position:

a) Pull the Overwing Exit Door handle to open the door.

b) Make sure the Overwing Exit Door will open.

7) Do the Fault Isolation Procedure below.

a) Make sure that the problem is identified as a door indication problem or a Flight Lock Solenoid problem before you continue.

F. Fault Isolation Procedure

(1) If you identified the fault as a Flight Lock Solenoid mechanical problem, then do these steps:

(a) Remove the door lining to get access to the Flight Lock Solenoid. This is the task: Emergency Exit Door Lining Removal, AMM TASK 52-22-51-000-801.

(b) Do a continuity check between the terminals of the applicable Flight Lock Solenoid.

NOTE: Over time, moisture with possible freezing can affect the ground termination. When checking the wiring, it is recommended to inspect the applicable ground blocks for signs of corrosion.

Table 201/52-20-00-993-801

MAINTENANCE MESSAGE	SOLENOID	TERMINALS
52-72005 L FL SW FAULT 52-72007 L FL SW FAULT 52-72107 L FL SW FAULT	M2219	A, B
OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999		
52-72006 L FWD FL SW FAULT 52-72106 L FWD FL SW FAULT	M2220	A, B
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999		
52-72008 R FL SW FAULT 52-72108 R FL SW FAULT	M2221	A, B
OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999		
52-72009 R FWD FL SW FAULT 52-72109 R FWD FL SW FAULT	M2222	A, B
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999		

1) If there is no continuity, then replace the Flight Lock Solenoid. These are the tasks:

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- Emergency Exit Door Fight Lock Solenoid Removal, AMM TASK 52-22-41-000-802
- Emergency Exit Door Flight Lock Solenoid Installation, AMM TASK 52-22-41-400-802

a) Do the Repair Confirmation at the end of this task.

2) If there is continuity, then continue.

(c) Do this wire check between the Flight Lock Relay and the Flight Lock Solenoid:

1) Remove the applicable Flight Lock Relay.

2) Disconnect the connector for the applicable Flight Lock Solenoid.

(d) Do a wire check between these pins of the connector for the Flight Lock Relay and the connector for the Flight Lock Solenoid:

NOTE: Over time, moisture with possible freezing can affect the ground termination. When checking the wiring, it is recommended to inspect the applicable ground blocks for signs of corrosion.

Table 202/52-20-00-993-802

MAINTENANCE MESSAGE NUMBER	RELAY CONNECTOR		SOLENOID CONNECTOR
52-72005 52-72007 52-72107	D12756 pin A1	-------	D40196P pin 18
OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
52-72006 52-72106	D12756 pin B1	-------	D40196P pin 22
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
52-72008 52-72108	D12758 pin A1	-------	D40296P pin 24
OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
52-72009 52-72109	D12758 pin B1	-------	D40296P pin 27
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			

1) If you find a problem with the wire, then do these steps:

a) Repair or replace the wire.

b) Re-install the applicable Flight Lock Relay.

c) Re-connect the electrical connector for the applicable Flight Lock Solenoid.

d) Do the Repair Confirmation at the end of this task.

2) If you do not find a problem with the wires, then continue.

(e) Examine the Engine 1 (2) Running Relays, R737 (R738):

NOTE: The Engine 1 Running Relay, R737, is found in the J22 Junction Box.

The Engine 2 Running Relay, R738, is found in the J24 Junction Box.

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1) Do the Emergency Exit Door Flight Lock Mechanical Switch Operational Test. This is the task: Emergency Exit Door Flight Lock Mechanical Switch Operational Test, AMM TASK 52-22-00-710-802.

2) If the operational test is not in the limits, then examine the wires for the applicable Engine Running Relay:

   a) Disconnect the connector for the applicable Flight Lock Solenoid.

   b) Remove the applicable Engine Running Relay.

   c) Do a wire check between the applicable Flight Lock Solenoid and the applicable relay:

      NOTE: Over time, moisture with possible freezing can affect the ground termination. When checking the wiring, it is recommended to inspect the applicable ground blocks for signs of corrosion.

Table 203/52-20-00-993-803

MAINTENANCE MESSAGE NUMBER	SOLENOID CONNECTOR		R737 CONNECTOR
52-72005 52-72007 52-72107	D40196P pin 17	-------	D12538 pin A1
OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
52-72006 52-72106	D40196P pin 17	-------	D12538 pin A1
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
52-72008 52-72108	D40296P pin 23	-------	D12538 pin B1
OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
52-72009 52-72109	D40296P pin 23	-------	D12538 pin B1
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			

Table 204/52-20-00-993-804

MAINTENANCE MESSAGE NUMBER	SOLENOID CONNECTOR		R738 CONNECTOR
52-72005 52-72007 52-72107	D40196P pin 17	-------	D12540 pin A1
OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
52-72006 52-72106	D40196P pin 17	-------	D12540 pin A1
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
52-72008 52-72108	D40296P pin 23	-------	D12540 pin B1

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Table 204/52-20-00-993-804 (Continued)

MAINTENANCE MESSAGE NUMBER	SOLENOID CONNECTOR		R738 CONNECTOR
OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
52-72009 52-72109	D40296P pin 23	-------	D12540 pin B1
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			

<1> If you find a problem with the wires, then do these steps:

<a> Repair or replace the wires.

<b> Re-connect the connector for the applicable Flight Lock Solenoid.

<c> Re-install the applicable Engine Running Relay.

<d> Do the Repair Confirmation at the end of this task.

<2> If you do not find a problem with the wires, then replace the Engine 1 (2) Running Relay, R737 (R738).

<a> Do the Repair Confirmation at the end of this task.

(2) If you identified the fault as an indication problem, then do these steps:

(a) Do these steps to do a check of the applicable Flight Lock Switch:

1) If it is necessary, remove the lining from the door. This is the task: Emergency Exit Door Lining Removal, AMM TASK 52-22-51-000-801.

2) Disconnect the electrical connector from the applicable Flight Lock Switch.

3) Manually operate the Flight Lock Switch while you do the continuity check.

4) Do a continuity check of the connectors splices and the ground.

5) Do a continuity check between terminals of the Flight Lock Switch:

NOTE: Over time, moisture with possible freezing can affect the ground termination. When checking the wiring, it is recommended to inspect the applicable ground blocks for signs of corrosion.

Table 205/52-20-00-993-805

MAINTENANCE MESSAGE	SWITCH	TERMINALS
52-72005 L FL SW FAULT 52-72007 L FL SW FAULT 52-72107 L FL SW FAULT	S1112	A, B
OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999		
52-72006 L FWD FL SW FAULT 52-72106 L FWD FL SW FAULT	S1113	A, B
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999		
52-72008 R FL SW FAULT 52-72108 R FL SW FAULT	S1114	A, B

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Table 205/52-20-00-993-805 (Continued)

MAINTENANCE MESSAGE	SWITCH	TERMINALS
OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999		
52-72009 R FWD FL SW FAULT 52-72109 R FWD FL SW FAULT	S1115	A, B
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999		

a) If the continuity check is not correct, then replace the applicable Flight Lock Switch. These are the tasks:

- Flight Lock Switch Removal, AMM TASK 52-22-41-020-802
- Flight Lock Switch Installation, AMM TASK 52-22-41-420-802

<1> Make sure that the electrical connector is correctly installed on the Flight Lock Switch.

<2> Do the Repair Confirmation at the end of this task.

b) If the continuity check is correct, then continue.

(b) Do this wire check between the Flight Lock Switch and the PSEU:

1) Disconnect the applicable electrical connector from the PSEU.

2) Do a wire check between the ground of the Flight Lock Switch and the connector for the PSEU:

NOTE: Over time, moisture with possible freezing can affect the ground termination. When checking the wiring, it is recommended to inspect the applicable ground blocks for signs of corrosion.

Table 206/52-20-00-993-806

MAINTENANCE MESSAGE NUMBER	PSEU CONNECTOR		GROUND AT SWITCH
52-72005 52-72007 52-72107	D10986 pin 20	-------	S1112
OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
52-72006 52-72106	D10986 pin 53	-------	S1113
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
52-72008 52-72108	D10988 pin 2	-------	S1114
OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
52-72009 52-72109	D10988 pin 52	-------	S1115
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			

a) If you find a problem with the wire, then do these steps:

<1> Repair or replace the wire.

<2> Re-connect the electrical connector to the Flight Lock Switch.

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<3> Re-connect the electrical connector to the PSEU.

<4> Do the Repair Confirmation at the end of this task.

b) If you do not find a problem with the wiring, then replace the PSEU, M2061. These are the tasks:

- Proximity Switch Electronics Unit (PSEU) Removal, AMM TASK 32-09-10-000-801
- Proximity Switch Electronics Unit (PSEU) Installation, AMM TASK 32-09-10-400-801

<1> Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

NOTE: You must do the steps to prepare for the Initial Evaluation before you do these steps.

(1) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	1	C01515	OVERWING FLIGHT LOCK-RIGHT
D	2	C01514	OVERWING FLIGHT LOCK-LEFT

(2) Make sure that the connector for the applicable Flight Lock Switch is installed.

(3) Make sure that the applicable Flight Lock Relay is installed.

(4) Make sure that the connector for the applicable Flight Lock Solenoid is installed.

(5) Make sure that the applicable Engine Running Relay is installed.

(6) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(a) If the maintenance message does not show, then you corrected the problem.

1) Put the airplane back to its usual condition.

a) Move the Engine Start Levers to CUTOFF.

b) Move the Engine Thrust Levers to idle.

c) Make sure that the Engine Start Switches are in the OFF/AUTO position.

d) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

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F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

e) Make sure that the door linings are correctly installed. This is the task: Emergency Exit Door Lining Installation, AMM TASK 52-22-51-400-801.

(b) If the maintenance message shows, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

802. Overwing Exit Door Indication Switch Problem - Fault Isolation

A. Description

(1) This task is for these maintenance messages:

(a) 52-72010 L OW SW A FAULT

(b) 52-72011 L OW SW B FAULT

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(c) 52-72012 L FWD OW SW A FLT

(d) 52-72013 L FWD OW SW B FLT

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(e) 52-72014 R OW SW A FAULT

(f) 52-72015 R OW SW B FAULT

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(g) 52-72016 R FWD OW SW A FLT

(h) 52-72017 R FWD OW SW B FLT

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(2) This maintenance message shows when there are all of these conditions:

(a) The airplane is in flight.

(b) Any one of the door switches indicates open.

B. Possible Causes

(1) Left Overwing Exit Door Indication Switch, S1104 or S1105

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(2) Left Forward Overwing Exit Door Indication Switch, S1108 or S1109

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(3) Right Overwing Exit Door Indication Switch, S1106 or S1107

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(4) Right Forward Overwing Exit Door Indication Switch, S1110 or S1111

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(5) Wiring problem

(6) PSEU, M2061

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C. Related Data

(1) SSM 52-71-13

(2) SSM 52-71-14

(3) WDM 52-71-13

(4) WDM 52-71-14

D. Initial Evaluation

(1) Do these steps to prepare for the Initial Evaluation:

(a) Make sure that a minimum of 3 out of 4 entry/service doors are closed.

(b) Make sure all the Overwing Exit Doors are closed, latched and locked.

(c) Do this task: Put the Airplane in the Air Mode, AMM TASK 32-09-00-860-801.

(2) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message shows, then do these steps:

1) Do this task: Return the Airplane to the Ground Mode, AMM TASK 32-09-00-860-802.

2) Do the Fault Isolation Procedure below.

E. Fault Isolation Procedure

(1) Adjust the applicable switch. This is the task: Emergency Exit Door Indication Switch Adjustment, AMM TASK 52-71-22-820-805.

Table 207

MAINTENANCE MESSAGE	SWITCH
52-72010 L OW SW A FAULT	S1104
52-72011 L OW SW B FAULT	S1105
OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999	
52-72012 L FWD OW SW A FLT	S1108
52-72013 L FWD OW SW B FT	S1109
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999	
52-72014 R OW SW A FAULT	S1106
52-72015 R OW SW B FAULT	S1107
OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999	
52-72016 R FWD OW SW A FLT	S1110
52-72017 R FWD OW SW B FLT	S1111
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999	

(a) Do the Repair Confirmation at the end of this task.

(2) Replace the applicable switch.

(a) Do the Repair Confirmation at the end of this task.

(3) Do this wiring check between the Indication Switch and the PSEU:

(a) Disconnect the applicable connector from the PSEU.

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(b) Do a wiring check between the Indication Switch ground and the connector for the PSEU:

1) With all of the Overwing Exit Doors closed, make sure that there is continuity between the Indication Switch ground and the PSEU connector:

NOTE: Over time, moisture with possible freezing can affect the ground termination. When checking the wiring, it is recommended to inspect the applicable ground blocks for signs of corrosion.

Table 208

MAINTENANCE MESSAGE NUMBER	GROUND AT SWITCH		PSEU CONNECTOR
52-72010			D10986
	S1104	-------	pin 8
52-72011			D10988
	S1105	-------	pin 21
OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
52-72012			D10986
	S1108	-------	pin 52
52-72013			D10988
	S1109	-------	pin 35
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
52-72014			D10986
	S1106	-------	pin 61
52-72015			D10988
	S1107	-------	pin 1
OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
52-72016			D10986
	S1110	-------	pin 15
52-72017			D10988
	S1111	-------	pin 30
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			

a) If you find a problem with the wiring, then do these steps:

<1> Repair the wiring.

<2> Re-connect the connector to the applicable switch.

<3> Re-connect the applicable connector to the PSEU.

<4> Do the Repair Confirmation at the end of this task.

b) If you do not find a problem with the wiring, then continue.

(4) Replace the PSEU, M2061. These are the tasks:

- Proximity Switch Electronics Unit (PSEU) Removal, AMM TASK 32-09-10-000-801
- Proximity Switch Electronics Unit (PSEU) Installation, AMM TASK 32-09-10-400-801

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(a) Do the Repair Confirmation at the end of this task.

F. Repair Confirmation

NOTE: You must do the steps to prepare for the Initial Evaluation before you do these steps.

(1) Make sure the connector for the applicable Indication Switch is installed.

(2) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(a) If the maintenance message does not show, then you corrected the problem.

1) Put the airplane back to its usual condition. This is the task: Return the Airplane to the Ground Mode, AMM TASK 32-09-00-860-802.

(b) If the maintenance message shows, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

803. Right Engine Running Signal to PSEU Problem - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) 52-72018 ENG RUN R FAULT

(2) This maintenance message shows when these conditions occur:

(a) The right engine is on, but the right engine-running relay is not providing a ground signal to the Proximity Sensor Electronics Unit (PSEU) input.

(b) The right engine thrust lever is advanced more than 64 degrees.

B. Possible Causes

(1) Right engine-running relay, R563

(2) Proximity switch electronics unit (PSEU), M2061

(3) Hydraulic Shutoff Valve, V33 (Eng 1) V34 (Eng2)

(4) Bleed Air Shutoff Valve, V17 (Eng 1) V16 (Eng 2).

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR

D. Related Data

(1) SSM 29-11-11

(2) WDM 29-11-11

(3) SSM 31-62-24

(4) WDM 31-62-24

(5) SSM 36-11-11

(6) WDM 36-11-11

(7) SSM 52-71-14

(8) WDM 52-71-14

(9) SSM 73-22-31

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(10) WDM 73-22-31

E. Initial Evaluation

(1) Do these steps to prepare for the initial evaluation:

(a) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

(b) Make sure that a minimum of 3 out of 4 entry/service doors are closed.

(c) Make sure that the airplane is in ground mode.

(d) Make sure that the engine start switches on the P5-20 panel are in the OFF position.

(e) Move the engine start levers on the control stand to IDLE.

(f) Wait for 5 minutes, then move the engine thrust levers on the control stand to the fully advanced position.

(2) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message shows, do these steps then do the Fault Isolation Procedure below:

1) Move the engine thrust levers to idle.

2) Move the engine start levers to CUTOFF.

F. Fault Isolation Procedure

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR

(2) Replace the right engine-running relay R563.

NOTE: The right engine-running relay is installed in junction box J24.

(a) Do the Repair Confirmation at the end of this task.

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1) If the Repair Confirmation is not satisfactory, then continue.

(3) Do this wiring check:

(a) Disconnect connector D10986 from the PSEU.

(b) Remove the right engine-running relay R563 from the J24 junction box.

(c) Do a wiring check between these pins of connector D10986 and connector D10916 at the J24 junction box:

(d) Do a wiring check between the relay and the PSEU.

D10986	D10916
pin 47	pin A1

(e) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Reinstall the right engine-running relay R563.

3) Reconnect connector D10986 to the PSEU.

4) Do the Repair Confirmation at the end of this task.

(f) If you do not find a problem with the wiring, then continue.

(4) Replace the PSEU, M2061. These are the tasks:

- Proximity Switch Electronics Unit (PSEU) Removal, AMM TASK 32-09-10-000-801
- Proximity Switch Electronics Unit (PSEU) Installation, AMM TASK 32-09-10-400-801

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR

(2) Do these steps to prepare for the Repair Confirmation:

(a) Make sure that these circuit breakers are open and have safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

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F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

(b) Make sure that the right-engine running relay, R563 is installed.

(c) Make sure that a minimum of 3 out of 4 entry/service doors are closed.

(d) Make sure that the airplane is in ground mode.

(e) Make sure that the engine start switches on the P5-20 panel are in the OFF position.

(f) Move the engine start levers on the control stand to IDLE.

(g) Wait for 5 minutes, then move the engine thrust levers on the control stand to the fully advanced position.

(3) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(a) If the maintenance message shows, then continue the Fault Isolation Procedure at the subsequent step.

(b) If the maintenance message does not show, then you corrected the fault, continue to the procedure below.

H. Put the Airplane in its Usual Condition

(1) Do these steps to put the airplane back to its usual condition.

(a) Move the engine start levers to CUTOFF.

(b) Move the engine thrust levers to idle.

(c) Make sure the engine start switches are OFF.

(d) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

END OF TASK

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804. Overwing Exit Door Option Problem - Fault Isolation

A. Description

(1) This task is for these maintenance messages:

(a) 52-72019 OVWG OPT FAULT

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(b) 52-72024 FOUR OW OPT FLT

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(2) These maintenance messages show when there are all of these conditions:

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(a) The OVERWING OPT pin on the PSEU is not grounded.

(b) The FOUR OVERWING OPT pin on the PSEU is not grounded.

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(c) The left and right flight lock switches indicate closed.

B. Possible Causes

(1) Wiring problem

(2) Proximity switch electronics unit (PSEU), M2061

C. Related Data

(1) (SSM 52-71-13)

(2) (SSM 52-71-14)

(3) (WDM 52-71-13)

(4) (WDM 52-71-14)

D. Initial Evaluation

(1) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(2) If the maintenance message does not show, then there was an intermittent fault.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

E. Fault Isolation Procedure

(1) Do this wiring check:

(a) Disconnect connector D10986 from the PSEU.

(b) Make sure the applicable pin of connector D10986 goes to ground (WDM 52-71-13).

Table 209

MAINTENANCE MESSAGE	CONNECTOR	PIN NUMBER
52-72019 OVWG OPT FAULT	D10986	10
OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999		
52-72024 FOUR OVWG OPT FLT	D10986	11
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999		

(c) If the pin does not go to ground, then do these steps:

1) Repair the wiring.

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2) Re-connect connector D10986 to the PSEU.

3) Do the Repair Confirmation at the end of this task.

(d) If the pin goes to ground, then continue.

(2) Replace the PSEU, M2061.

These are the tasks:

Proximity Switch Electronics Unit (PSEU) Removal, AMM TASK 32-09-10-000-801,

Proximity Switch Electronics Unit (PSEU) Installation, AMM TASK 32-09-10-400-801.

(a) Do the Repair Confirmation at the end of this task.

F. Repair Confirmation

(1) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(2) If the maintenance message does not show, then you corrected the fault.

——— END OF TASK ———

805. Overwing Exit Door Switch Disagreement Problem - Fault Isolation

A. Description

(1) This task is for these maintenance messages:

(a) 52-72020 L OW SW DISAGREE

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(b) 52-72021 L FWD OW SW DSGR

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(c) 52-72022 R OW SW DISAGREE

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(d) 52-72023 R FWD OW SW DSGR

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(2) These maintenance messages show when one of the two indication switches for an overwing exit door indicates open and the other indicates closed.

B. Possible Causes

(1) Left Overwing Exit Door Indication Switch, S1104 or S1105

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(2) Left Forward Overwing Exit Door Indication Switch, S1108 or S1109

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(3) Right Overwing Exit Door Indication Switch, S1106 or S1107

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(4) Right Forward Overwing Exit Door Indication Switch, S1110 or S1111

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(5) Wiring problem

(6) PSEU, M2061

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C. Related Data

(1) SSM 52-71-13

(2) SSM 52-71-14

(3) WDM 52-71-13

(4) WDM 52-71-14

D. Initial Evaluation

(1) Make sure that all the overwing exit doors are closed, latched and locked.

(2) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

(4) If the maintenance message does not show, open all the overwing exit doors.

(5) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(6) If the maintenance message shows, then do the Fault Isolation Procedure below.

(7) If the maintenance message does not show, then there was an intermittent fault.

NOTE: If this message occurs frequently, then the fault can be a loose connection at the indication switch.

E. Fault Isolation Procedure

(1) Adjust the applicable indication switch, do this task: Emergency Exit Door Indication Switch Adjustment, AMM TASK 52-71-22-820-805.

Table 210

MAINTENANCE MESSAGE	SWITCHES
52-72020 L OW SW DISAGREE	S1104, S1105
OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999	
52-72021 L FWD OW SW DSGR	S1108, S1109
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999	
52-72022 R OW SW DISAREE	S1106, S1107
OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999	
52-72023 R FWD OW SW DSGR	S1110, S1111
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999	

(a) Do the Repair Confirmation at the end of this task.

(2) Do these steps from the PSEU BITE Display to check the exit door indication switches inputs to the PSEU:

(a) Push the ON button.

(b) Push the NO button until OTHER FUNCTNS? shows.

(c) Push the YES button.

(d) Push the NO button until I/O MONITOR? shows.

(e) Push the YES button until SENSORS? shows.

(f) Push the NO button until INPUTS? shows.

NOTE: Use the arrow down button if INPUT INACTIVE shows.

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(g) Push the YES button.

(h) Select a connector.

NOTE: Use the arrow buttons to move to the desired connector, then push the YES button to select a connector.

(i) Select a pin.

NOTE: Use the arrow buttons to move to the desired pin, then push the YES button to select the pin.

(j) Do a check of the applicable exit door indication switches inputs to the PSEU:

Table 211

MAINTENANCE MESSAGE NUMBER	SWITCH	CONNECTOR	PIN	TEST CONDITION	PSEU BITE DISPLAY
52-72020	S1104	D10986	8	DOOR LOCKED	GND
		D10986	8	DOOR NOT LOCKED	NO GND
	S1105	D10988	21	DOOR LOCKED	GND
		D10988	21	DOOR NOT LOCKED	NO GND
OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999					
52-72021	S1108	D10986	52	DOOR LOCKED	GND
		D10986	52	DOOR NOT LOCKED	NO GND
	S1109	D10988	35	DOOR LOCKED	GND
		D10988	35	DOOR NOT LOCKED	NO GND
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999					
52-72022	S1106	D10986	61	DOOR LOCKED	GND
		D10986	61	DOOR NOT LOCKED	NO GND
	S1107	D10988	1	DOOR LOCKED	GND
		D10988	1	DOOR NOT LOCKED	NO GND
OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999					
52-72023	S1110	D10986	15	DOOR LOCKED	GND
		D10986	15	DOOR NOT LOCKED	NO GND
	S1111	D10988	30	DOOR LOCKED	GND
		D10988	30	DOOR NOT LOCKED	NO GND
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999					

1) If you find a problem with the inputs for the test conditions in the table, then continue with the exit door indication switch replacement or wiring check.

2) If you do not find a problem with the inputs for the test conditions in the table, then continue with the PSEU replacement.

(3) Replace the applicable indication switch. These are the tasks:

- Emergency Exit Door Indication Switch Removal, AMM TASK 52-71-22-000-803
- Emergency Exit Door Indication Switch Installation, AMM TASK 52-71-22-420-801

(a) Do the Repair Confirmation at the end of this task.

(4) Do this wiring check:

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(a) Make sure that the applicable overwing exit door is closed, latched and locked.

(b) Disconnect the connector for the applicable indication switch from the PSEU.

(c) Do a wiring check between the ground of the connector for the indication switch and the connector for the PSEU:

Table 212

MAINTENANCE MESSAGE NUMBER	GROUND AT SWITCH		PSEU CONNECTOR
52-72020			D10986
	S1104	-------	pin 8
52-72020			D10988
	S1105	-------	pin 21
OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
52-72021			D10986
	S1108	-------	pin 52
52-72021			D10988
	S1109	-------	pin 35
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
52-72022			D10986
	S1106	-------	pin 61
52-72022			D10988
	S1107	-------	pin 1
OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
52-72023			D10986
	S1110	-------	pin 15
52-72023			D10988
	S1111	-------	pin 30
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			

1) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-install the applicable switch.
   c) Re-connect the applicable connector to the PSEU.
   d) Do the Repair Confirmation at the end of this task.
2) If you do not find a problem with the wiring, then continue.

(5) Replace the PSEU, M2061. These are the tasks:
- Proximity Switch Electronics Unit (PSEU) Removal, AMM TASK 32-09-10-000-801
- Proximity Switch Electronics Unit (PSEU) Installation, AMM TASK 32-09-10-400-801

(a) Do the Repair Confirmation at the end of this task.

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F. Repair Confirmation

(1) Make sure that the connector for the applicable switch is installed.

(2) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(a) If the maintenance message does not show, then you corrected the problem.

(b) If the maintenance message shows, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

806. Overwing Exit Door Open Problem - Fault Isolation

A. Description

(1) This task is for these maintenance messages:

(a) 52-74001 L OVWG OPEN

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(b) 52-74002 L FWD OW OPEN

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(c) 52-74003 R OVWG OPEN

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(d) 52-74004 R FWD OW OPEN

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(2) This maintenance message shows when both overwing exit door indication switches indicate open.

B. Possible Causes

(1) Left overwing exit door indication switch, S1104 or S1105

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(2) Left forward overwing exit door indication switch, S1108 or S1109

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(3) Right overwing exit door indication switch, S1106 or S1107

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(4) Right forward overwing exit door indication switch, S1110 or S1111

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(5) Wiring problem

(6) Proximity switch electronics unit (PSEU), M2061

C. Related Data

(1) (SSM 52-71-13)

(2) (SSM 52-71-14)

(3) (WDM 52-71-13)

(4) (WDM 52-71-14)

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D. Initial Evaluation

(1) Make sure all the overwing exit doors are closed, latched and locked.

(2) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(3) If the maintenance message does not show, then there was an intermittent fault.

NOTE: If this message occurs frequently, then the fault can be a loose connection at the switch.

(4) If the maintenance message shows, then do the Fault Isolation Procedure below.

E. Fault Isolation Procedure

(1) Adjust the applicable switch. To adjust it, do this task: Emergency Exit Door Indication Switch Adjustment, AMM TASK 52-71-22-820-805.

Table 213

MAINTENANCE MESSAGE	SWITCHES
52-74001 L OVWG OPEN	S1104, S1105
OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999	
52-74002 L FWD OW OPEN	S1108, S1109
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999	
52-74003 R OVWG OPEN	S1106, S1107
OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999	
52-74004 R FWD OW OPEN	S1110, S1111
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999	

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is not satisfactory, then continue.

(2) Replace the applicable switch.

These are the tasks:

Emergency Exit Door Indication Switch Removal, AMM TASK 52-71-22-000-803,

Emergency Exit Door Indication Switch Installation, AMM TASK 52-71-22-420-801.

(a) Do the Repair Confirmation at the end of this task.

1) If the Repair Confirmation is not satisfactory, then continue.

(3) Do a check of the wiring:

(a) Disconnect the connector for the applicable indication switch from the PSEU.

(b) Do this wiring check between the indication switch ground and the PSEU:

Table 214

MAINTENANCE MESSAGE NUMBER	GROUND AT SWITCH		PSEU CONNECTOR
52-74001			D10986
	S1104	------	pin 8
52-74001			D10988

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Table 214 (Continued)

MAINTENANCE MESSAGE NUMBER	GROUND AT SWITCH		PSEU CONNECTOR
	S1105	------	pin 21
OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
52-74002			D10986
	S1108	------	pin 52
52-74002			D10988
	S1109	------	pin 35
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
52-74003			D10986
	S1106	------	pin 61
52-74003			D10988
	S1107	------	pin 1
OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
52-74004			D10986
	S1110	------	pin 15
52-74004			D10988
	S1111	------	pin 30

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(c) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.
2) Re-install the applicable switch.
3) Reconnect the applicable connector to the PSEU.
4) Do the Repair Confirmation at the end of this task.

(d) If you do not find a problem with the wiring, then continue.

(4) Replace the PSEU, M2061.

These are the tasks:

Proximity Switch Electronics Unit (PSEU) Removal, AMM TASK 32-09-10-000-801,

Proximity Switch Electronics Unit (PSEU) Installation, AMM TASK 32-09-10-400-801.

(a) Do the Repair Confirmation at the end of this task.

F. Repair Confirmation

(1) Make sure the connector for the applicable switch is installed.

(2) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(3) If the maintenance message does not show, then you corrected the fault.

END OF TASK

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807. Overwing Exit Door Flight Lock Relay Problem - Fault Isolation

A. Description

(1) This task is for these maintenance messages:

(a) 52-76017 FL RELAY 1 FAULT.

(b) 52-76018 FL RELAY 2 FAULT.

(2) These maintenance messages show when the PSEU does not sense the expected load from the coil of relay R742 (left) or R743 (right) during the PSEU power-on-self-test.

(3) (SDS SUBJECT 52-22-00)

B. Possible Causes

(1) Flight lock relay, R742 (left)

(2) Flight lock relay, R743 (right)

(3) Wiring problem

(4) Proximity switch electronics unit (PSEU), M2061

(5) Proximity switch electronics unit (PSEU) nuisance message.

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	1	C01515	OVERWING FLIGHT LOCK-RIGHT
D	2	C01514	OVERWING FLIGHT LOCK-LEFT

D. Related Data

(1) (SSM 52-71-13)

(2) (SSM 52-71-14)

(3) (WDM 52-71-13)

(4) (WDM 52-71-14)

E. Initial Evaluation

(1) Do these steps to determine if the flight lock relay message is a PSEU nuisance indication:

NOTE: These nuisance indications are caused during airplane power transfers or system power-up.

(a) Make sure all Repair Confirmations for this task are complete.

(b) Open these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN

(c) Do this step after 10 seconds:

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1) Close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN

(d) After approximately 3 seconds, make sure the three landing gear green lights come on.

NOTE: The PSEU self test is complete when the landing gear lights illumiate.

(e) If the PSEU fault light does not come on, the previous flight lock relay message is a nuisance message.

(2) Do the EXISTING FAULTS test at the PSEU BITE. To do the test, do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(3) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Do these steps to prepare for the procedure:

(a) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

(b) Make sure that a minimum of 3 out of 4 entry/service doors are closed.

(c) Make sure all overwing exit doors are fully closed, locked and latched.

(d) Make sure that the airplane is in ground mode.

(e) Make sure that the engine start switches on the P5-20 panel are in the OFF position.

(f) Move the engine start levers on the control stand to IDLE.

(g) Wait for 5 minutes, then move the engine thrust levers on the control stand to the fully advanced position.

(2) Replace the applicable overwing exit door flight lock relay: R742 or R743.

(a) Do the Repair Confirmation at the end of this task.

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1) If the Repair Confirmation is not satisfactory, then continue.

(3) Do this check for 28 VDC at the flight lock relay:

(a) Remove the applicable flight lock relay R742 (left) or R743 (right)

NOTE: R742 is in junction box J22. R743 is in junction box J24.

(b) Do a check for 28 VDC between pin X1 of connector D12756 (left) or D12758 (right) and structure ground.

(c) If there is not 28 VDC at pin X1, then do these steps:

1) Do a wiring check between these pins of the connector for the flight lock relay and the circuit breaker:

Table 215

MAINTENANCE MESSAGE NUMBER	RELAY CONNECTOR		CIRCUIT BREAKER
52-76017	D12756		C1514
	pin X1	-------	term L
52-76018	D12758		C1515
	pin X1	-------	term L

2) If you find a problem with the wiring, then do these steps:

a) Repair the wiring.

b) Re-install the flight lock relay.

c) Do the Repair Confirmation at the end of this task.

(d) If there is 28 VDC at pin X1, then continue.

(4) Do this check of the ground circuit between the flight lock relay and the PSEU:

(a) Disconnect the applicable connector, D10986 (left) or D10988 (right) from the PSEU.

(b) Make sure pin X2 of connector D12756 (left) or D12758 (right) goes to ground.

(c) If pin X2 does not go to ground, then do these steps:

1) Do a wiring check between these pins on the connector for the applicable flight lock relay and the connector for the PSEU:

Table 216

MAINTENANCE MESSAGE NUMBER	RELAY CONNECTOR		PSEU CONNECTOR
52-76017	D12756		D10986
	pin X2	-------	pin 40
52-76018	D12758		D10988
	pin X2	-------	pin 53

(d) If you find a problem with the wiring, then do these steps:

1) Repair the wiring.

2) Re-install the flight lock relay.

3) Re-connect the connector at the PSEU.

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4) Do the Repair Confirmation at the end of this task.

(e) If you do not find a problem with the wiring, then re-install the flight lock relay and continue.

(5) Replace the PSEU, M2061.

These are the tasks:

Proximity Switch Electronics Unit (PSEU) Removal, AMM TASK 32-09-10-000-801,

Proximity Switch Electronics Unit (PSEU) Installation, AMM TASK 32-09-10-400-801.

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	1	C01515	OVERWING FLIGHT LOCK-RIGHT
D	2	C01514	OVERWING FLIGHT LOCK-LEFT

(2) Make sure the connector for the applicable circuit breaker is installed.

(3) Make sure the applicable flight lock relay is installed.

(4) Make sure the applicable connector is connected to the PSEU.

(5) Move the power switch for the PSEU to the OFF position and then move it to the ON position.

NOTE: This will start the power-on-self-test.

(6) Do the EXISTING FAULTS test at the PSEU BITE. To do the test, do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(7) If the maintenance message does not show, then you corrected the fault.

(8) Put the airplane back to its usual condition.

(a) Move the engine start Levers to CUTOFF.

(b) Move the engine thrust levers to idle.

(c) Make sure the engine start switches are OFF.

(d) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

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F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

——— END OF TASK ———

808. Overwing Exit Door Warning Light Problem - Fault Isolation

A. Description

(1) This task is for these maintenance messages:

(a) 52-76019 L OV WARN FLT

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(b) 52-76020 L FWD OW WARN FLT

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(c) 52-76021 R OW WARN FLT

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(d) 52-76022 R FWD OW WARN FLT

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(2) These maintenance messages show if the PSEU does not sense the expected load from the overwing exit door warning lights during the PSEU power-on-self-test.

(3) (SDS SUBJECT 52-22-00)

B. Possible Causes

(1) Overwing exit door warning light, L6 (left) or L10 (right)

OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(2) Forward overwing exit door warning light, L11 (left) or L12 (right)

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

(3) Wiring problem

(4) Proximity switch electronics unit (PSEU), M2061

C. Circuit Breakers

(1) This is the primary circuit breaker related to the fault:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
F	14	C01180	INDICATOR MASTER DIM SECT 8

D. Related Data

(1) (SSM 33-18-35)

(2) (SSM 52-71-13)

(3) (SSM 52-71-14)

(4) (WDM 33-18-35)

(5) (WDM 52-71-13)

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(6) (WDM 52-71-14)

E. Initial Evaluation

(1) Do the EXISTING FAULTS test at the PSEU BITE. To do the test, do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(2) If the maintenance message shows, then do the Fault Isolation Procedure below.

F. Fault Isolation Procedure

(1) Do these steps to prepare for the procedure:

(a) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

(b) Make sure that a minimum of 3 out of 4 entry/service doors are closed.

(c) Make sure all overwing exit doors are fully closed, locked and latched.

(d) Make sure that the airplane is in ground mode.

(e) Make sure that the engine start switches on the P5-20 panel are in the OFF position.

(f) Move the engine start levers on the control stand to IDLE.

(g) Wait for 5 minutes, then move the engine thrust levers on the control stand to the fully advanced position.

(2) Push the applicable door warning light on the P5 forward overhead panel.

(a) If the light does not come on, then do these steps:

1) Do this task: Flight Compartment Lighting Problem - Fault Isolation, 33-10 TASK 801.

2) Do the Repair Confirmation at the end of this task.

(b) If the light comes on, then continue.

(3) Do this wiring check between the overwing exit door warning light and the PSEU:

(a) Remove the door warning annunciator panel, P5-20.

(b) Disconnect the applicable connector from the PSEU.

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(c) Do a wiring check between these pins of the connector for the at the door warning light and the connector at the PSEU:

Table 217

MAINTENANCE MESSAGE NUMBER	WARNING LIGHT CONNECTOR		PSEU CONNECTOR
52-76019	D1406		D10982
	pin 10	-------	pin 5
OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
52-76020	D482		D11138
	pin 4	-------	pin 9
OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
52-76021	D1406		D10988
	pin 7	-------	pin 31
OMR 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999			
52-76022	D482		D10984
	pin 11	-------	pin 54

OMR 701-799, 802, 805, 809, 810, 812, 813, 815, 816, 818-826, 901-999

1) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Re-install the door warning annunciator panel, P5-20.
   c) Re-connect the applicable connector at the PSEU.
   d) Do the Repair Confirmation at the end of this task.
2) If you do not find a problem with the wiring, then continue.

(4) Replace the PSEU, M2061.

These are the tasks:

Proximity Switch Electronics Unit (PSEU) Removal, AMM TASK 32-09-10-000-801,

Proximity Switch Electronics Unit (PSEU) Installation, AMM TASK 32-09-10-400-801.

(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Make sure that this circuit breaker is closed:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
F	14	C01180	INDICATOR MASTER DIM SECT 8

(2) Make sure the door warning annunciator panel, P5-20 is installed.

(3) Make sure the applicable connector at the PSEU is installed.

(4) Move the power switch for the PSEU to the OFF position and then move it to the ON position.

NOTE: This will start the power-on-self-test.

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(5) Do the EXISTING FAULTS test at the PSEU BITE. To do the test, do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(6) If the maintenance message does not show, then you corrected the fault.

(7) Put the airplane back to its usual condition.

(a) Move the engine start levers to CUTOFF.

(b) Move the engine thrust levers to idle.

(c) Make sure the engine start switches are OFF.

(d) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	1	C00458	ENGINE 1 IGNITION RIGHT
A	3	C00153	ENGINE 1 IGNITION LEFT

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	4	C00459	ENGINE 2 IGNITION RIGHT
D	6	C00151	ENGINE 2 IGNITION LEFT

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

——— END OF TASK ———

809. PSEU Emergency Door Option Pins Incorrect - Fault Isolation

A. Description

(1) This task is for this maintenance message:

(a) 52-72105 INVALID OPT PINS

B. Possible Causes

(1) Program Pins for PSEU, M2061

(2) PSEU, M2061

C. Related Data

(1) WDM 32-64-21

(2) WDM 52-71-13

(3) SSM 32-61-21

(4) SSM 52-71-13

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D. Initial Evaluation

(1) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(a) If the maintenance message does not show, then there was an intermittent fault.

(b) If the maintenance message shows, then do the Fault Isolation Procedure below.

E. Fault Isolation Procedure

(1) Do this check of the program pins of the PSEU, M2061:

(a) Disconnect connectors D10984 and D10986 from the PSEU, M2061.

(b) Make sure that the program pin 18 of connector D10984 and pin 11 of connector D10986 are connected as specified in WDM 32-64-21 and WDM 52-71-13.

1) If the program pins are not connected correctly, then do these steps:

a) Connect the program pins as specified in the WDM.

b) Re-connect connectors D10984 and D10986 to the PSEU, M2061.

c) Do the Repair Confirmation at the end of this task.

2) If the program pins are connected correctly, then continue.

(2) Replace the PSEU, M2061. These are the tasks:

- Proximity Switch Electronics Unit (PSEU) Removal, AMM TASK 32-09-10-000-801
- Proximity Switch Electronics Unit (PSEU) Installation, AMM TASK 32-09-10-400-801

(a) Do the Repair Confirmation at the end of this task.

F. Repair Confirmation

(1) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(a) If the maintenance message does not show, then you corrected the problem.

(b) If the maintenance message shows, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

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801. Cargo Door Difficult to Open or Close - Fault Isolation

A. Description

(1) (SDS SUBJECT 52-30-00)

B. Possible Causes

(1) Counterbalance mechanism

(2) Hinge arms

(3) Snubber

C. Fault Isolation Procedure

(1) Do these checks of the counterbalance mechanism:

(a) Remove the door lining.

(b) Look for a broken counterbalance cable.

1) If the cable is broken, then do these steps:

a) Repair the cable. To repair it, do this task: Broken Cargo Door Counterbalance Cable Replacement, AMM TASK 52-31-12-000-802.

b) Do the Repair Confirmation at the end of this task.

2) If the cable is not broken, then continue.

(c) Examine the counterbalance cable for wear or damage.

1) If the cable is worn or damaged, then do these steps:

a) Replace the cable.

These are the tasks:

Cargo Door Counterbalance Removal, AMM TASK 52-31-12-000-801,

Cargo Door Counterbalance Installation, AMM TASK 52-31-12-400-801.

b) Do the Repair Confirmation at the end of this task.

c) If the Repair Confirmation is not satisfactory, then continue.

2) If the cable is not worn or damaged, then continue.

(d) Make sure the cable is installed correctly on the cargo compartment ceiling, over the pulley, and to the counterbalance.

1) If the cable is not installed correctly, then do these steps:

a) Re-install the cable.

These are the tasks:

Cargo Door Counterbalance Removal, AMM TASK 52-31-12-000-801,

Cargo Door Counterbalance Installation, AMM TASK 52-31-12-400-801.

b) Do the Repair Confirmation at the end of this task.

2) If the cable is installed correctly, then continue

(e) Look for debris in the counterbalance mechanism.

1) If there is debris, then do these steps:

a) Remove the debris.

b) Do the Repair Confirmation at the end of this task.

c) If the Repair Confirmation is not satisfactory, then continue.

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2) If there is no debris, then continue.

(f) Examine the counterbalance mechanism for wear or damage.

1) If the mechanism is worn or damaged, then do these steps:

a) Replace the counterbalance mechanism.

These are the tasks:

Cargo Door Counterbalance Removal, AMM TASK 52-31-12-000-801,

Cargo Door Counterbalance Installation, AMM TASK 52-31-12-400-801.

b) Do the Repair Confirmation at the end of this task.

c) If the Repair Confirmation is not satisfactory, then continue.

2) If the mechanism is not worn or damaged, then continue.

(g) Make sure the counterbalance idler crank is lubricated. To lubricate it, do this task: Cargo Door Servicing, AMM TASK 12-25-31-640-801.

1) Do the Repair Confirmation at the end of this task.

2) If the Repair Confirmation is not satisfactory, then continue.

(h) Adjust the counterbalance mechanism. To adjust it, do this task: Cargo Door Adjustment, AMM TASK 52-31-00-820-801.

1) Do the Repair Confirmation at the end of this task.

2) If the Repair Confirmation is not satisfactory, then continue.

(2) Do this check of the hinge arms:

(a) Examine the hinge arms for wear, damage, and loose fasteners.

1) If a problem is found, then do these steps:

a) Repair the hinge arms.

b) Do the Repair Confirmation at the end of this task.

c) If the Repair Confirmation is not satisfactory, then continue.

2) If you do find a problem, then continue.

(3) Do this check of the snubber:

(a) Examine the snubber for damage and leaks.

1) If a problem is found, then do these steps:

a) Replace the snubber.

These are the tasks:

Cargo Door Snubber Removal, AMM TASK 52-31-13-000-801,

Cargo Door Snubber Installation, AMM TASK 52-31-13-400-801.

b) Do the Repair Confirmation at the end of this task.

D. Repair Confirmation

(1) Do these steps to confirm the repair:

(a) Try to open and close the door.

(b) If it is easy to open and close the door, then you corrected the fault. Re-install the door lining.

END OF TASK

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802. Cargo Door Difficult to Unlatch or Latch - Fault Isolation

A. Description

(1) This task is for this Observed Fault:

(a) Cargo door difficult to unlatch or latch.

B. Possible Causes

(1) Blockage of the handle of the latch mechanism

(2) Latch mechanism and handle

(3) Handle spring

(4) Hinge arms

(5) Interference with fuselage frame

C. Related Data

(1) SDS SUBJECT 52-30-00

D. Fault Isolation Procedure

(1) Do this check for a blockage of the handle of the latch mechanism inside the airplane:

(a) If unable to unlatch the forward cargo door, get access to the forward cargo bay as follows:

1) Open the access panel that is on the right side of the airplane, at approximately the tenth window cutout from the front.

NOTE: The panel is approximately 20 inches long and 20 inches wide, (508 mm by 508 mm) with a serge around the edges of the carpet.

(b) If unable to unlatch the aft cargo door, get access to the aft cargo bay as follows:

1) Open the access panel that is on the right side of the airplane with its aft edge at STA 767.

NOTE: The panel is approximately 20 inches long and 20 inches wide, (508 mm by 508 mm) with a serge around the edges of the carpet. The access panel is aft of another panel of approximately the same size.

(c) If a blockage is found inside the cargo area at the handle of the latch assembly, move the blockage.

(d) Make sure that the cargo nets are installed correctly to prevent blockage due to cargo.

(e) Remove any cargo that is inside the area enclosed by the cargo nets.

(f) Do the Repair Confirmation at the end of this task.

(2) Do this check of the latch mechanism and handle:

(a) Remove the door lining.

(b) Remove the panel over the latch mechanism.

(c) Examine the latch mechanism and handle.

1) Look for debris in the latch mechanism or wear or damage. This is the task: Cargo Door Check, AMM TASK 52-31-00-200-801.

a) If you find a problem, then repair it.

<1> Do the Repair Confirmation at the end of this task.

2) Examine the handle spring for damage. This is the task: Cargo Door Check, AMM TASK 52-31-00-200-801.

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a) If you find a problem, replace handle spring. These are the tasks:
- Cargo Door Handle Mechanism Removal, AMM TASK 52-31-14-000-801
- Cargo Door Handle Mechanism Installation, AMM TASK 52-31-14-400-801

<1> Do the Repair Confirmation at the end of this task.

3) Make sure that the latch torque tube is lubricated. This is the task: Cargo Door Servicing, AMM TASK 12-25-31-640-801.
a) Do the Repair Confirmation at the end of this task.

(d) Adjust the latch mechanism. This is the task: Cargo Door Adjustment, AMM TASK 52-31-00-820-801.
1) Do the Repair Confirmation at the end of this task.

(3) Do this check of the hinge arms:
(a) Examine the hinge arms.
1) Look for wear or damage.
2) Look for loose fasteners.
3) If you find a problem, then repair it.
a) Do the Repair Confirmation at the end of this task.

(4) Do this check for interference between the door and the fuselage frame:
(a) Try to unlatch, open, close, and latch the door.
1) Look and listen for points of interference between the door and fuselage frame.
a) Make sure that the door seal does not cause interference.
<1> If the door seal causes interference, replace the blade seal. These are the tasks:
- Blade Seals Removal, AMM TASK 52-09-11-000-801
- Blade Seals Installation, AMM TASK 52-09-11-400-801

b) Repair the problems that you find.
<1> Do this task: Cargo Door Adjustment, AMM TASK 52-31-00-820-801.
<2> Do the Repair Confirmation at the end of this task.

E. Repair Confirmation

(1) Try to unlatch and latch the door.
(a) If it is easy to unlatch and latch the door, then you corrected the problem.
1) Re-install the door lining, if it is necessary.
(b) If it is difficult to unlatch and latch the door, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

803. PSEU Cargo Door Monitored Problem - Fault Isolation

A. Description

(1) This task is for these maintenance messages:
(a) 52-72001 FWD CGO DR OPEN

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(b) 52-72002 AFT CGO DR OPEN

NOTE: The Proximity Switch Electronic Unit (PSEU) does not record these messages when the airplane is on the ground.

(2) This task is also for FWD CARGO and AFT CARGO Door Warning Lights (P5) Indication during taxi or takeoff.

(3) For a detail description of the Door Warning System, refer to SDS SUBJECT 52-71-00.

B. Possible Causes

(1) Cargo Door out of adjustment

(2) Forward (Aft) Cargo Door Warning Switch, S960 (S961)

(3) Wiring

(4) PSEU, M2061

(5) Cargo Door Handle Mechanism spring.

C. Related Data

(1) WDM 52-71-12

(2) SSM 52-71-12

D. Initial Evaluation

(1) Open the applicable FWD (AFT) Cargo Door.

(2) Close, latch and lock the applicable Cargo Door.

(3) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801

(a) If the maintenance message does not show in EXISTING FAULTS, then there was an intermittent fault.

(b) If the maintenance message shows in EXISTING FAULTS, then do the Fault Isolation Procedure below.

E. Fault Isolation Procedure

(1) Do a check of the applicable Cargo Door adjustment. This is the task: Cargo Door Adjustment, AMM TASK 52-31-00-820-801.

(a) Do the Repair Confirmation at the end of this task.

(2) Do this check of the applicable FWD (AFT) Cargo Door Warning Switch, S960 (S961) adjustment. This is the task: Cargo Door Indication Switch - Adjustment, AMM TASK 52-71-31-820-801.

(a) Do the Repair Confirmation at the end of this task.

(3) Do this check of the PSEU:

(a) Disconnect the applicable connector for the FWD (AFT) Cargo Door Warning Switch from the PSEU (WDM 52-71-12).

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CARGO DOOR	PSEU CONNECTOR
Forward Cargo Door	D10988
Aft Cargo Door	D10986

(b) Close, latch and lock the applicable FWD (AFT) Cargo Door.

(c) Do a continuity check between pin 56 of the applicable connector and Structure Ground.

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1) If you find continuity between pin 56 and Structure Ground, then replace the PSEU, M2061. These are the tasks:
   - Proximity Switch Electronics Unit (PSEU) Removal, AMM TASK 32-09-10-000-801
   - Proximity Switch Electronics Unit (PSEU) Installation, AMM TASK 32-09-10-400-801

   a) Do the Repair Confirmation at the end of this task.

2) If you do not find continuity between pin 56 and Structure Ground, then continue.

(4) Do this check of the applicable FWD (AFT) Cargo Door Warning Switch, S960 (S961) Wiring (WDM 52-71-12):

FWD CRG DR WARN SWITCH

PSEU D10988	**S960 Terminal**
pin 56	term NO

S960 Terminal	
term C	GD84-DC

AFT CRG DR WARN SWITCH

PSEU D10986	**S961 Terminal**
pin 56	term NO

S961 Terminal	
term C	GD11794-DC

(a) If you find a problem in the wiring, then do these steps:
   1) Repair the wiring as necessary.
   2) Connect the applicable connector D10988 (D10986) to the PSEU.
   3) Do the Repair Confirmation at the end of this task.

(b) If you do not find a problem in the wiring, then replace the applicable FWD (AFT) Cargo Door Warning Switch, S960 (S961) Switch. These are the tasks:
   - Forward Cargo Door Indication Switch - Removal, AMM TASK 52-71-31-000-801
   - Forward Cargo Door Indication Switch - Installation, AMM TASK 52-71-31-400-801

   1) Connect the applicable connector D10988 (D10986) to the PSEU.
   2) Do the Repair Confirmation at the end of this task.

(5) Do a visual check of the cargo door handle mechanism spring:

(a) To get access to the cargo door handle mechanism, do this task: Cargo Door Handle Mechanism Removal, AMM TASK 52-31-14-000-801.
   1) If you find a problem with the spring, then replace the spring, these are the tasks:
      - Cargo Door Handle Mechanism Removal, AMM TASK 52-31-14-000-801

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- Cargo Door Handle Mechanism Installation, AMM TASK 52-31-14-400-801

a) Do the Repair Confirmation at the end of this task.

F. Repair Confirmation

(1) Close, latch and lock the applicable FWD (AFT) Cargo Door.

(2) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(a) If the maintenance message does not show in EXISTING FAULTS then you corrected the problem.

(b) If the maintenance message still shows in EXISTING FAULTS, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

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801. Main Deck Cargo Door Does Not Raise - Fault Isolation

A. Description

(1) This task is for these Observed Faults:

(a) The Main Deck Cargo Door (MDCD) does not raise when you hold the ARM Switch to the Up ARM position and put the UP TO CANOPY/CLOSE Control Switch, S1235 to the “UP TO CANOPY” position.

1) The CLOSED & LATCH Light is OFF and the IN TRANSIT Light is ON.

(b) The Main Deck Cargo Door (MDCD) does not open to FULL OPEN position when you hold the ARM Switch to the Up ARM position and put the FULL OPEN/DOWN TO CANOPY Control Switch, S1236 to the “FULL OPEN” position.

1) The OPENED Light is ON.

(2) For a detailed description of the MDCD, refer to SDS SUBJECT 52-32-00.

B. Possible Causes

(1) Check Valve
(2) Relief Valve
(3) Flow Limiter
(4) Latch Actuator
(5) Latch Mechanism
(6) Control Valve, V59
(7) Shut-off Valve, V60
(8) Canopy Position Switch B, S1163
(9) Door Unlocked Switch, S1162
(10) Open HYD Delay Relay, R1324
(11) AIR/GND SYS 1 Relay, R587
(12) Control Switch, S1235 (UP TO CANOPY/CLOSE)
(13) Control Switch, S1236 (FULL OPEN/DOWN TO CANOPY)
(14) ARM Switch, S1234
(15) Lock Actuator, M3280
(16) Wiring Problem

C. Related Data

(1) WDM 52-32-11
(2) SSM 52-32-11

D. Initial Evaluation

(1) On the Main Deck Cargo Door Control Panel, P32, put the UNLOCK/LOCK Switch to the “UNLOCK” position to unlock the door, hold the ARM Switch to the Up “ARM” position and put the UP TO CANOPY/CLOSE Switch to the “UP TO CANOPY” position to unlatch and raise the door electrically.

(a) Make sure that the Green CLOSED & LATCHED Light on the Main Deck Cargo Door Control Panel, P32, is ON.

NOTE: The Main Deck Cargo Door Control Panel, P32, is on the Rigid Cargo Barrier, adjacent to the Forward Entry Door.

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1) If the door does not unlock and the READY TO OPEN Light is OFF, do the Main Deck Cargo Door Does Not Unlock - Fault Isolation, 52-32 TASK 814.
2) If the CLOSED & LATCHED Light stays ON and the door does not raise, do the Main Deck Cargo Door (MDCD) Does Not Unlatch - Fault Isolation, 52-32 TASK 815.
3) If the CLOSED & LATCHED Light goes OFF and the door raises with the IN TRANSIT Light ON, then there was an intermittent fault.
4) If the CLOSED & LATCHED Light goes OFF and the door does not raise with the IN TRANSIT Light ON, do the “Fault Isolation - MDCD does not raise to CANOPY Position” below.
5) If the CLOSED & LATCHED Light goes OFF and the door does not raise with the OPENED Light ON, do the “Fault Isolation - MDCD does not raise to FULL OPEN Position” below.

E. Fault Isolation - MDCD does not raise to CANOPY Position (BCF)

(1) If the Main Deck Cargo Door (MDCD) does not Open to Canopy Position, do these steps (WDM 52-31-11):

Open this access panel:

Number	Name/Location
194DR	Aft Wing To Body Fairing Panel

(a) Remove the connector D1222 from the Shutoff Valve, V60, and do a check for ground at pin 3 of connector D1222.
1) If you find no ground, replace the AIR/GND SYS 1 Relay, R587.
a) Do the Repair Confirmation at the end of this task.
2) If you find ground, continue.

(b) Hold the ARM Switch to the up ARM position, put the UP TO CANOPY/CLOSE Switch, S1235 to the UP TO CANOPY position, and do a check for 28V DC at pin 1 of connector D1222.
1) If you find 28V DC at pin 1, replace the Shutoff Valve, V60. These are the tasks:
- Shutoff Valve Removal, AMM TASK 52-32-71-000-801
- Shutoff Valve Installation, AMM TASK 52-32-71-400-801

a) Do the Repair Confirmation at the end of this task.
2) If you do not find 28V DC at pin 1, continue.

(c) Remove the connector D43156P from the Main Deck Cargo Door Control Panel, P32, hold the ARM Switch to the Down ARM position, put the UP TO CANOPY/CLOSE Switch, S1235 to the UP TO CANOPY position, and do a check for continuity between pins 20 and 27 of connector D43156J.
1) If you find no continuity, repair the wiring between pins 20 and 27.
a) Do the Repair Confirmation at the end of this task.
2) If you find continuity, continue.

(d) Remove the connector D17553 from the Open HYD Delay Relay, R1324, hold the ARM Switch to the Up “ARM” position, put the UP TO CANOPY/CLOSE Switch, S1235 to the “UP TO CANOPY” position, and do a check for continuity between pin 20 of connector D43156J and pin C1 of connector D17553.

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1) If you find continuity, do a check for continuity between pin 15 of connector D43156J and pins A2 and X1 of connector D17553.
   a) If you find continuity, replace the Open HYD Relay, R1324.
      <1> Do the Repair Confirmation at the end of this task.
   b) If you find no continuity, repair the wiring between pin 15 to pins A2 and X1.
      <1> Do the Repair Confirmation at the end of this task.
2) If you find no continuity, continue.

(e) Hold the ARM Switch to the Up “ARM” position and do a check for continuity between terminals 7 and 8 of the ARM Switch, S1234.
1) If you find no continuity, replace the ARM Switch, S1234 on the Main Deck Cargo Door Control Panel, P32.
   a) Do the Repair Confirmation at the end of this task.
2) If you find continuity, replace the UP TO CANOPY/CLOSE Switch, S1235.
   a) Do the Repair Confirmation at the end of this task.

F. Fault Isolation - MDCD does not raise to FULL OPEN Position (BCF)

(1) If the Main Deck Cargo Door (MDCD) does not go to the FULL OPEN position from the CANOPY position, do these steps (WDM 52-31-11):
(a) Remove the connector D1222 from the Shutoff Valve, V60, and do a check for ground at pin 3 of connector D1222.
1) If you find no ground, replace the AIR/GRND SYS 1 Relay, R587.
   a) Do the Repair Confirmation at the end of this task.
2) If you find ground, continue.

(b) Hold the ARM Switch to the Up “ARM” position, put the UP TO CANOPY/CLOSE Switch, S1235 to the “UP TO CANOPY” position, and do a check for 28V DC at pin 1 of connector D1222.
1) If you find 28V DC at pin 1, replace the Shutoff Valve, V60. These are the tasks:
   - Shutoff Valve Removal, AMM TASK 52-32-71-000-801
   - Shutoff Valve Installation, AMM TASK 52-32-71-400-801
   a) Do the Repair Confirmation at the end of this task.
2) If you do not find 28V DC at pin 1, continue.

(c) Remove the connector D43156P from the Main Deck Cargo Door Control Panel, P32, and do a check for continuity between pins 31 and 20 of connector D43156P.
1) If you find no continuity, repair the wiring between pins 31 and 20.
   a) Do the Repair Confirmation at the end of this task.
2) If you find continuity, continue.

(d) Hold the ARM Switch to the Up ARM position, put the UP TO CANOPY/CLOSE Switch, S1235 to the UP TO CANOPY position, and do a check for continuity between pins 15 and 31 of connector D43156J.
1) If you find no continuity, replace the FULL OPEN/DOWN TO CANOPY Switch, S1236.
   a) Do the Repair Confirmation at the end of this task.
2) If you find continuity, continue.

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(e) Remove the connector D1220 and do a check for continuity between pin 29 of connector D43156P and pin A of connector D1220.

1) If you find continuity, replace the CANOPY POSITION “B” Switch, S1163. These are the tasks:

- Canopy Position Switches Removal, AMM TASK 52-32-34-000-801
- Canopy Position Switches Installation, AMM TASK 52-32-34-000-802

a) Do the Repair Confirmation at the end of this task.

2) If you find no continuity, continue.

(f) Do a check for continuity between pin 31 of connector D43156P and pin A of connector D1220.

1) If you find no continuity, replace the CANOPY POSITION B Switch, S1163. These are the tasks:

- Canopy Position Switches Removal, AMM TASK 52-32-34-000-801
- Canopy Position Switches Installation, AMM TASK 52-32-34-000-802

a) Do the Repair Confirmation at the end of this task.

2) If you find continuity, replace the Control Valve, V59. These are the tasks:

- Control Valve Removal, AMM TASK 52-32-61-000-801
- Control Valve Installation, AMM TASK 52-32-61-400-801

a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) On the Main Deck Cargo Door Control Panel, P32, hold the ARM Switch to the “ARM” position and put the UP TO CANOPY/CLOSE or FULL OPEN/DOWN TO CANOPY Control Switch, S1235 (S1236) to the “UP TO CANOPY” or “FULL OPEN” position to raise the door electrically.

(a) If the door raises correctly, then you corrected the problem.

1) Close this access panel:

Number	Name/Location
194DR	Aft Wing To Body Fairing Panel

(b) If the door does not raise correctly, continue the applicable Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

804. Main Deck Cargo Door Does Not Stop at Canopy Position When Raised - Fault Isolation

A. Description

(1) This task is for this Observed Fault:

(a) The Main Deck Cargo Door (MDCD) does not stop at the Canopy position when you hold the ARM Switch to the Up ARM position and put the UP TO CANOPY/CLOSE Control Switch, S1235 to the “UP TO CANOPY” position.

(2) For a detailed description of the MDCD, refer to SDS SUBJECT 52-32-00.

B. Possible Causes

(1) Motor Control Valve, V59

(2) Solenoid Operated Shut-off Valve, V60

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(3) Canopy Position Switch B, S1163

(4) Wiring

(5) FULL OPEN/DOWN TO CANOPY Switch, S1236

C. Related Data

(1) WDM 52-32-11

(2) SSM 52-32-11

D. Initial Evaluation

(1) On the Main Deck Cargo Door Control Panel, P32, hold the ARM Switch to the Up “ARM” position and put the UP TO CANOPY/CLOSE Switch to the “UP TO CANOPY” position to raise the door electrically.

NOTE: The Main Deck Cargo Door Control Panel, P32, is on the Rigid Cargo Barrier, adjacent to the Forward Entry Door.

(a) If the door raises and stops at the Canopy position correctly, then there was an intermittent fault.

(b) If the door does not stop at the Canopy position correctly, do the Fault Isolation Procedure below.

E. Fault Isolation Procedure

(1) Remove the connector D43156P from the Main Deck Cargo Door Control Panel, P32.

(a) Hold the ARM Switch to the Up ARM position, put the UP TO CANOPY/CLOSE Switch, S1235 to the UP TO CANOPY position, and do a check for continuity between pins 15 and 31 of connector D43156J.

1) If you find continuity, replace the FULL OPEN/DOWN TO CANOPY Switch, S1236.

a) Do the Repair Confirmation at the end of this task.

2) If you do not find continuity, continue.

(2) Remove the connector D1220 from the Motor Control Valve, V59.

(a) Do a check for continuity between pin 31 of connector D43156P and pin A of connector D1220.

1) If you find continuity, replace the Motor Control Valve, V59. These are the tasks:

- Control Valve Removal, AMM TASK 52-32-61-000-801
- Control Valve Installation, AMM TASK 52-32-61-400-801

a) Do the Repair Confirmation at the end of this task.

2) If you do not find continuity, repair the wiring as necessary.

a) Do the Repair Confirmation at the end of this task.

(3) If the problem continues, replace the Shutoff Valve, V60. These are the tasks:

- Shutoff Valve Removal, AMM TASK 52-32-71-000-801
- Shutoff Valve Installation, AMM TASK 52-32-71-400-801

(a) Do the Repair Confirmation at the end of this task.

F. Repair Confirmation

(1) On the Main Deck Cargo Door Control Panel, P32, hold the ARM Switch to the “ARM” position and put the UP TO CANOPY/CLOSE Control Switch, S1235 to the “UP TO CANOPY” position to raise the door electrically.

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(a) If the door raises and stops at the Canopy position correctly, then you corrected the problem.

(b) If the door does not stop at canopy position correctly, continue the applicable Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

805. Main Deck Cargo Door Does Not Lower - Fault Isolation

A. Description

(1) This task is for these Observed Faults:

(a) The Main Deck Cargo Door (MDCD) does not move to the DOWN TO CANOPY position when you hold the ARM Switch, S1234 to the Down ARM position and put the FULL OPEN/DOWN TO CANOPY Control Switch, S1236 to the “DOWN TO CANOPY” position.

(b) The Main Deck Cargo Door (MDCD) does not move to the CLOSE position when you hold the ARM Switch, S1234 to the Down ARM position and put the UP TO CANOPY/CLOSE Control Switch, S1235 to the “CLOSE” position.

(2) For a detailed description of the MDCD, refer to SDS SUBJECT 52-32-00.

B. Possible Causes

(1) Control Valve, V59

(2) Shut-off Valve, V60

(3) Canopy Position Switch A, S1164

(4) Latch Switch, S1226

(5) Open HYD Relay, R1324

(6) AIR/GROUND Relay, R587

(7) Control Switch, S1235 (UP TO CANOPY/CLOSE)

(8) Control Switch, S1236 (FULL OPEN/DOWN TO CANOPY)

(9) ARM Switch, S1234

(10) Lock Actuator, M3280

(11) Wiring Problem

C. Related Data

(1) WDM 52-32-11

(2) SSM 52-32-11

D. Initial Evaluation

(1) On the Main Deck Cargo Door Control Panel, P32, hold the ARM Switch, S1234 to the Down ARM position and put the FULL OPEN/DOWN TO CANOPY or UP TO CANOPY/CLOSE switch, S1236 (S1235) to the “DOWN TO CANOPY” or “CLOSE” position to lower the door electrically.

NOTE: The Main Deck Cargo Door Control Panel, P32, is on the Rigid Cargo Barrier, adjacent to the Forward Entry Door.

(a) If the door lowers correctly, then there was an intermittent fault.

(b) If the door does not lower to the Canopy position, do the “Fault Isolation - MDCD Does Not Lower to CANOPY Position” below.

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(c) If the door does not lower to the Close position, do the “Fault Isolation - MDCD Does Not Lower to CLOSE Position” below.

E. Fault Isolation - MDCD Does Not Lower to CANOPY Position (BCF)

(1) If the Main Deck Cargo Door (MDCD) does not go to the CANOPY position from the FULL OPEN position, do these steps (WDM 52-31-11):

(a) Open this access panel:

Number	Name/Location
194DR	Aft Wing To Body Fairing Panel

(b) Make sure that the Control Valve, V59 is in the Position 2.

1) If the valve is not in the Position 2, remove the connector, D1220 from the Control Valve, V59.

a) Hold the ARM Switch, S1234 to the Down “ARM” position, put the FULL OPEN/DOWN TO CANOPY Switch, S1236 to the “DOWN TO CANOPY” position, and do a check for 28V DC at pin B of connector D1220.

<1> If you find 28V DC, do a check for ground at pin E.

<a> If you find ground, replace the Control Valve, V59. These are the tasks:

- Control Valve Removal, AMM TASK 52-32-61-000-801
- Control Valve Installation, AMM TASK 52-32-61-400-801

I. Do the Repair Confirmation at the end of this task.

<b> If you do not find ground, repair the wiring as necessary.

I. Do the Repair Confirmation at the end of this task.

<2> If you do not find 28V DC, repair the wiring between pin D of connector D1220 and pin 27 of connector D43156P.

<a> Do the Repair Confirmation at the end of this task.

2) If the valve is in the Position 2, continue.

(c) Remove the connector D1222 from the Shutoff Valve, V60, and do a check for ground at pin 3 of connector D1222.

1) If you find no ground, replace the AIR/GRND SYS 1 Relay, R587.

a) Do the Repair Confirmation at the end of this task.

2) If you find ground, continue.

(d) Hold the ARM Switch to the Down “ARM” position, put the FULL OPEN/DOWN TO CANOPY Switch, S1236 to the “DOWN TO CANOPY” position, and do a check for 28V DC at pin 1 of connector D1222.

1) If you find 28V DC at pin 1, replace the Shutoff Valve, V60. These are the tasks:

- Shutoff Valve Removal, AMM TASK 52-32-71-000-801
- Shutoff Valve Installation, AMM TASK 52-32-71-400-801

a) Do the Repair Confirmation at the end of this task.

2) If you do not find 28V DC at pin 1, continue.

(e) Remove the connector D43156P from the Main Deck Cargo Door Control Panel, P32, and do a check for continuity between pin 25 of the connector D43156P to pin B of the connector D1220.

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1) If you do not find continuity, do a check of continuity between pin 24 of connector D43156P and pin B of connector D1220.
   a) If you find continuity, replace the Canopy Position “A” Switch, S1164. These are the tasks:
      • Canopy Position Switches Removal, AMM TASK 52-32-34-000-801
      • Canopy Position Switches Installation, AMM TASK 52-32-34-000-802
      <1> Do the Repair Confirmation at the end of this task.
   b) If you do not find continuity, repair the wiring between pin 25 of the connector D43156P and pin B of the connector D1220.
      <1> Do the Repair Confirmation at the end of this task.
2) If you find continuity, continue.

(f) Do a check for continuity between pin 14 of connector D43156P and pin D of connector D1220.
1) If you do not find continuity, do a check of continuity between pin 13 of connector D43156P and pin D of connector D1220.
   a) If you find continuity, replace the Latch Switch, S1226.
      <1> Do the Repair Confirmation at the end of this task.
   b) If you do not find continuity, repair the wiring between pin 14 of connector D43156P and pin D of connector D1220.
      <1> Do the Repair Confirmation at the end of this task.
2) If you find continuity, continue.

(g) Hold the ARM Switch to the Down “ARM” position, put the FULL OPEN/DOWN TO CANOPY Switch, S1236 to the “DOWN TO CANOPY” position, and do a check for continuity between pins 15 and 25 of connector D43156J.
1) If you do not find continuity, replace the FULL OPEN/DOWN TO CANOPY Switch, S1236.
   a) Do the Repair Confirmation at the end of this task.
2) If you find continuity, do a check for continuity between pin 14 to pin 27 of connector D43156J.
   a) If you do not find continuity, replace the ARM Switch, S1234.
      <1> Do the Repair Confirmation at the end of this task.
   b) If you find continuity, repair the wiring between pin 27 of connector D43156P and pin 1 of connector D1222.
      <1> Do the Repair Confirmation at the end of this task.

F. Fault Isolation - MDCD Does Not Lower to CLOSE Position (BCF)

(1) If the Main Deck Cargo Door (MDCD) does not Open to Canopy Position, do these steps (WDM 52-31-11):

Open this access panel:

Number	Name/Location
194DR	Aft Wing To Body Fairing Panel

(a) Make sure that the Control Valve, V59 is in the Position 2.

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1) If the valve is not in the Position 2, remove the connector, D1220 from the Control Valve, V59.
   a) Hold the ARM Switch, S1234 to the Down “ARM” position, put the UP TO CANOPY/CLOSE Switch, S1235 to the “CLOSE” position, and do a check for 28V DC at pin B of connector D1220.
      <1> If you find 28V DC, do a check for ground at pin E.
         <a> If you find ground, replace the Control Valve, V59. These are the tasks:
            • Control Valve Removal, AMM TASK 52-32-61-000-801
            • Control Valve Installation, AMM TASK 52-32-61-400-801
               I. Do the Repair Confirmation at the end of this task.
         <b> If you do not find ground, repair the wiring as necessary.
               I. Do the Repair Confirmation at the end of this task.
      <2> If you do not find 28V DC, repair the wiring between pin D of connector D1220 and pin 27 of connector D43156P.
         <a> Do the Repair Confirmation at the end of this task.
2) If the valve is in the Position 2, continue.

(b) Do a check for continuity between pin 14 of connector D43156P and pin D of connector D1220.
1) If you do not find continuity, do a check of continuity between pin 13 of the connector D43156P to pin D of the connector D1220.
   a) If you find continuity, replace the Latch Switch, S1226.
      <1> Do the Repair Confirmation at the end of this task.
   b) If you do not find continuity, repair the wiring between pin 14 of the connector D43156P to pin D of the connector D1220.
      <1> Do the Repair Confirmation at the end of this task.
2) If you find continuity, continue.

(c) Remove the connector D1222 from the Shutoff Valve, V60, and do a check for ground at pin 3 of connector D1222.
1) If you find no ground, replace the AIR/GND SYS 1 Relay, R587.
   a) Do the Repair Confirmation at the end of this task.
2) If you find ground, continue.

(d) Hold the ARM Switch to the Down “ARM” position, put the UP TO CANOPY/CLOSE Switch, S1235 to the “CLOSE” position, and do a check for 28V DC at pin 1 of connector D1222.
1) If you find 28V DC at pin 1, replace the Shutoff Valve, V60. These are the tasks:
   • Shutoff Valve Removal, AMM TASK 52-32-71-000-801
   • Shutoff Valve Installation, AMM TASK 52-32-71-400-801
   a) Do the Repair Confirmation at the end of this task.
2) If you do not find 28V DC at pin 1, continue.

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(e) Remove the connector D43156P from the Main Deck Cargo Door Control Panel, P32, and do a check for continuity between pin 24 of connector D43156P and pin B of connector D1220.

1) If you do not find continuity, do a check of continuity between pin 25 of connector D43156P and pin B of the connector D1220.

a) If you find continuity, replace the Canopy Position “A” Switch, S1164. These are the tasks:

- Canopy Position Switches Removal, AMM TASK 52-32-34-000-801
- Canopy Position Switches Installation, AMM TASK 52-32-34-000-802

<1> Do the Repair Confirmation at the end of this task.

b) If you do not find continuity, repair the wiring between pin 24 of connector D43156P and pin B of connector D1220.

<1> Do the Repair Confirmation at the end of this task.

2) If you find continuity, do a check of continuity between pin 14 of connector D43156P and pin D of connector D1220.

a) If you do not find continuity, do a check of continuity between pin 13 of connector D43156P and pin D of the connector D1220.

<1> If you find continuity, replace the Latch Switch, S1226.

<a> Do the Repair Confirmation at the end of this task.

<2> If you do not find continuity, repair the wiring between pin 14 of connector D43156P and pin D of connector D1220.

<a> Do the Repair Confirmation at the end of this task.

b) If you find continuity, continue.

(f) Hold the ARM Switch to the Down “ARM” position, put the UP TO CANOPY/CLOSE Switch, S1235 to the “CLOSE” position, and do a check for continuity between pins 15 and 24 of connector D43156J.

1) If you do not find continuity, replace the UP TO CANOPY/CLOSE Switch, S1235.

a) Do the Repair Confirmation at the end of this task.

2) If you find continuity, do a check for continuity between pins 14 and 27 of connector D43156J.

a) If you do not find continuity, replace the ARM Switch, S1234.

<1> Do the Repair Confirmation at the end of this task.

b) If you find continuity, repair the wiring between pin 27 of connector D43156P and pin 1 of connector D1222.

<1> Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) On the Main Deck Cargo Door Control Panel, P32, hold the ARM Switch, S1234 to the down ARM position and put the FULL OPEN/DOWN TO CANOPY or UP TO CANOPY/CLOSE Switch, S1236 (S1235) to the DOWN TO CANOPY or CLOSE position to lower the door electrically.

(a) If the door lowers correctly, then you corrected the problem.

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1) Close this access panel:

Number	Name/Location
194DR	Aft Wing To Body Fairing Panel

(b) If the door does not lower correctly, continue the applicable Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

806. Main Deck Cargo Door Does Not Stop at Canopy Position When Lowered - Fault Isolation

A. Description

(1) This task is for these Observed Faults:

(a) The Main Deck Cargo Door (MDCD) does not stop at the Canopy position when you hold the ARM Switch to the Up ARM position and put the FULL OPEN/DOWN TO CANOPY Control Switch, S1236 to the “DOWN TO CANOPY” position.

(2) For a detailed description of the MDCD, refer to SDS SUBJECT 52-32-00.

B. Possible Causes

(1) Canopy Position Switch A, S1164

(2) Wiring

(3) Door Closed A Switch, S1156

(4) Motor Control Valve, V59

(5) Main Deck Cargo Door Control Panel, P32

C. Related Data

(1) WDM 52-32-11

(2) SSM 52-32-11

D. Initial Evaluation

(1) On the Main Deck Cargo Door Control Panel, P32, hold the ARM Switch to the Down “ARM” position and put the FULL OPEN/DOWN TO CANOPY Control Switch to the “DOWN TO CANOPY” position to lower the door electrically.

NOTE: The Main Deck Cargo Door Control Panel, P32, is on the Rigid Cargo Barrier, adjacent to the Forward Entry Door.

(a) If the door lowers and stops at the Canopy position correctly, then there was an intermittent fault.

(b) If the door does not stop at the Canopy position correctly, do the Fault Isolation Procedure below.

E. Possible Causes

(1) Canopy Position Switch A, S1164

(2) Closed A Switch, S1156

(3) Motor Control Valve, V59

(4) Main Deck Cargo Door Control Panel, P32

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F. Fault Isolation Procedure

(1) Remove connector D1220 from the Motor Control Valve, V59 and D43156P from the Main Deck Cargo Door Control Panel, P32 (WDM 52-32-11).

(2) Do a continuity check from the MDCD Control Panel, P32 through the Canopy Position A Switch, S1164 to the Motor Control Valve, V59 as follows:

MDCD CTRL PNL D43156P	MOTOR CTRL VLV D1220
pin 24	pin B

(a) If there is no continuity, do a continuity check through the Canopy Position A Switch, S1164 as follows:

D43156P	D1220
pin 25	pin B

1) If there is still no continuity, repair the wiring between connector D43156P, pin 24 and connector D1220, pin B (WDM 52-32-11).

a) Reconnect connector D43156P to the MCDC Control Panel, P32 and connector D1220 to the Motor Control Valve, V59.

b) Do the Repair Confirmation at the end of this task.

2) If there is continuity, adjust or replace the Canopy Position A Switch, S1164 as necessary. These are the applicable tasks:

- Canopy Position Switches Adjustment, AMM TASK 52-32-34-000-803
- Canopy Position Switches Removal, AMM TASK 52-32-34-000-801
- Canopy Position Switches Installation, AMM TASK 52-32-34-000-802

a) Reconnect connector D43156P to the MCDC Control Panel, P32 and connector D1220 to the Motor Control Valve, V59.

b) Do the Repair Confirmation at the end of this task.

(b) If there is continuity between D43156P, pin 25 and D1220, pin B, then continue.

(3) Reconnect connector D43156P to the MCDC Control Panel, P32 and connector D1220 to the Motor Control Valve, V59.

(4) Remove the connector D1222 from the Solenoid Operated Shutoff Valve, V60.

(5) Do a check for 28V DC at pin 1 of connector D1222 (WDM 52-32-11).

(a) If you find 28V DC at pin 1, disconnect D43156P from the MDCD Control Panel, P32.

1) Do a continuity check across the MDCD Control Panel, P32, as follows:

MDCD PNL D43156J	MDCD PNL D43156J
pin 15	pin 27

a) If you find continuity, replace the ARM Switch, S1234.

<1> Do the Repair Confirmation at the end of this task

b) If you do not find continuity, continue.

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(b) If you do not find 28V DC at pin 1, reconnect D43156P to the MDCD Control Panel and continue.

(6) Do a check for 28V DC at pin A of the Motor Control Valve connector, D1220 (WDM 52-32-11).

(a) If you do not find 28V DC, do a continuity check between the MDCD Control Panel, P32 and the Motor Control Valve, V59 as follows:

MDCD CTRL PANEL D43156P	MOTOR CTRL VALVE D1220
pin 28	pin A

1) If there is no continuity replace the CLOSED A Switch, S1156.

2) If there is continuity, do a continuity check between the MDCD Control Panel, P32 and the Motor Control Valve, V59 as follows:

MDCD CTRL PANEL D43156P	MOTOR CTRL VALVE D1220
pin 20	pin C

a) If there is continuity, replace the Motor Control Valve, V59

<1> Do the Repair Confirmation at the end of this task

b) If there is no continuity, repair the wiring between pins 15 and 20 of the Main Deck Cargo Door Control Panel, P32.

<1> Do the Repair Confirmation at the end of this task

(b) If there is 28V DC at pin A, replace the Motor Control Valve, V59.

1) Do the Repair Confirmation at the end of this task

G. Repair Confirmation

(1) On the Main Deck Cargo Door Control Panel, P32, hold the ARM Switch to the Down “ARM” position and put the FULL OPEN/DOWN TO CANOPY Control Switch to the “DOWN TO CANOPY” position to lower the door electrically.

(a) If the door lowers and stops at the Canopy position correctly, then you corrected the problem.

(b) If the door does not stop at the Canopy position correctly, continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

808. Main Deck Cargo Door Warning Indication Lights - Fault Isolation

A. Description

(1) This task is for these maintenance messages:

(a) 52-76101 MDCD WRN FAULT

1) The Proximity Switch Electronic Unit (PSEU) has detected a problem with the MAIN CARGO Light, L13 circuit.

2) When this maintenance message shows, the MAIN CARGO Warning Light in the flight compartment also comes ON.

(b) 52-76102 MDCD UNLKD FAULT

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1) The PSEU has detected a problem for the UNLOCKED Light, L1591 circuit.
2) When this maintenance message shows, the MAIN CARGO Warning Light in the flight compartment also comes ON.
3) The UNLOCKED Light normally comes ON when the Main Deck Cargo Door is unlocked.

(2) For a detailed description of the MDCD, refer to SDS SUBJECT 52-32-00.

B. Possible Causes

(1) MAIN CARGO Light, L13
(2) Wiring
(3) UNLOCKED Light, L1591
(4) PSEU, M2061

C. Related Data

(1) WDM 52-32-11
(2) WDM 52-32-12
(3) WDM 52-71-13
(4) WDM 52-71-15
(5) WDM 32-64-21
(6) WDM 33-18-35
(7) SSM 52-32-11
(8) SSM 52-32-12

D. Initial Evaluation

(1) Do the Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.
(a) If a maintenance message does not show, then there was an intermittent fault.
(b) If the 52-76101 maintenance message shows, do the “Fault Isolation Procedure - MDCD WRN FAULT” below.
(c) If the 52-76102 maintenance message shows, do the “Fault Isolation Procedure - MDCD UNLKD FAULT” below.

NOTE: Make sure that the Main Deck Cargo Door is still unlocked (Open the Main Deck Cargo Door, AMM TASK 52-32-00-860-801).

E. Fault Isolation Procedure - MDCD WRN FAULT

(1) On the P5-20 DOOR WARNING ANNUNCIATOR Panel, push the TEST Switch.
(a) If the MAIN CARGO Light does not come ON, replace the Light Bulb. This is the task: Master Dim and Test - Operational Test, AMM TASK 33-18-00-710-802.
1) Do the Repair Confirmation at the end of this task.
a) If the problem continues, make sure that there is 28V DC at terminal 1 of the MAIN CARGO Light, L13 and continuity to Ground at terminal 4 (WDM 52-71-15, WDM 33-18-35):
b) If you find a problem, repair the wiring as necessary.
(b) If the MAIN CARGO Light comes ON, continue.
(2) Examine the wiring between the PSEU and the MAIN CARGO Light as follows (WDM 52-71-15):

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PSEU	MAIN CARGO LIGHT
D10984	D1406
pin 31	pin 9

(a) Repair the problems that you find.

(b) Do the Repair Confirmation at the end of this task.

F. Fault Isolation Procedure - MDCD UNLKD FAULT

(1) Examine the wiring between the PSEU, M2061, and the J48 Junction Box as follows (WDM 52-71-15):

PSEU	JUNCTN BOX
D10984	D48116P
pin 30	pin A8

(a) If you find a short or open circuit, repair the wiring.

(b) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) Do the Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(a) If the maintenance message does not show, then you corrected the problem.

(b) If the maintenance message still shows, continue the applicable Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

811. Main Deck Cargo Door Switch A and B DISAGREE Faults - Fault Isolation

A. Description

(1) This task is for these maintenance messages:

(a) 52-72101 MDCD CL SW FAULT

1) The Proximity Sensor Electronics Unit (PSEU) detects that the Door Closed A and B Switches disagree.

2) When this maintenance message shows, the MAIN CARGO Warning Light in the flight compartment comes ON.

3) The Main Deck Cargo Door (MDCD) may or may not be closed.

(b) 52-72102 MDCD LCH SW FAULT

1) The PSEU detects that the Door Latched A and B Switches disagree.

2) When this maintenance message shows, the MAIN CARGO Warning Light in the flight compartment comes ON.

3) The MDCD may or may not be latched.

4) The CLOSED & LATCHED Light on the MDCD Door Control Panel, P32 normally comes ON when the MDCD latches.

(c) 52-72103 MDCD LKD DISAGREE

1) The PSEU detects that the Door Locked A and B Switches disagree.

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2) When this maintenance message shows, the MAIN CARGO Warning Light in the flight compartment comes ON.

3) The MDCD may or may not be locked.

4) The LOCKED Light on the MDCD Door Control Panel, P32 normally comes ON when the MDCD is locked.

B. Possible Causes

(1) Wiring Problem

(2) Door Closed A Switch, S1156

(3) Door Closed B Switch, S1159

(4) Door Latched A Switch, S1230

(5) Door Latched B Switch, S1231

(6) Door Locked A Switch, S1158

(7) Door Locked B Switch, S1161

(8) PSEU, M02061

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	1	C01399	PSEU PRI
D	18	C00451	LANDING GEAR AURAL WARN

F/O Electrical System Panel, P6-11

Row	Col	Number	Name
C	7	C01577	MAIN DECK CARGO DOOR CTRL & IND

D. Related Data

(1) SSM 52-71-15

(2) WDM 52-71-15

E. Initial Evaluation

(1) Open and then close the MDCD. These are the tasks:

- Open the Main Deck Cargo Door, AMM TASK 52-32-00-860-801
- Close the Main Deck Cargo Door, AMM TASK 52-32-00-860-802

(2) Do the Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(a) If none of the maintenance messages indicated in the description shows, then there was an intermittent fault.

(b) If the 52-72101 MDCD CL SW FAULT maintenance message shows, do the “Fault Isolation Procedure - MDCD CL SW FAULT” below.

(c) If the 52-72102 MDCD LCH SW FAULT maintenance message shows, do the “Fault Isolation Procedure - MDCD LCH SW FAULT” below.

(d) If the 52-72103 MDCD LKD DISAGREE maintenance message shows, do the “Fault Isolation Procedure - MDCD LKD DISAGREE FAULT” below.

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F. Fault Isolation Procedure - MDCD CL SW FAULT

(1) Do a check of the wiring between the PSEU and MDCD Closed A and B Switches S1156 and S1159 as follows:

(a) Disconnect connectors D10986 and D10988 from the PSEU, M2061 (WDM 52-71-15).

(b) With the MDCD closed, do a check for Ground from the PSEU connectors, D10986 and D10988 as follows:

DOOR CLOSED A SWITCH

PSEU
D10986
pin 52 . . . . . GND

DOOR CLOSED B SWITCH

PSEU
D10988
pin 35 . . . . . GND

1) If you do not find continuity to Ground, do these steps:

a) Replace the applicable MDCD Closed A (B) Switch, S1156 (S1159).

b) Connect connectors D10986 and D10988 to the PSEU.

c) Do the Repair Confirmation at the end of this task.

<1> If the problem continues, repair the wiring as necessary.

2) If you find continuity to Ground, then continue.

(2) Make sure that the Door Closed A Switch, S1156 and Door Closed B Switch, S1159 are adjusted correctly. These are the tasks:

- Door Closed Switch Removal, AMM TASK 52-71-32-000-801
- Door Closed Switch Installation, AMM TASK 52-71-32-400-804

NOTE: For the correct adjustment, it is necessary to first remove the door closed switches.

(a) Do the Repair Confirmation at the end of this task.

(3) If a Door Closed Switch is stuck OPEN or CLOSED, then replace the applicable Door Closed A (B) Switch S1156 (S1159). These are the tasks:

- Door Closed Switch Removal, AMM TASK 52-71-32-000-801
- Door Closed Switch Installation, AMM TASK 52-71-32-400-804

(a) Do the Repair Confirmation at the end of this task.

(4) Replace the PSEU, M2061. These are the tasks.

- Proximity Switch Electronics Unit (PSEU) Removal, AMM TASK 32-09-10-000-801
- Proximity Switch Electronics Unit (PSEU) Installation, AMM TASK 32-09-10-400-801

(a) Do the Repair Confirmation at the end of this task.

G. Fault Isolation Procedure - MDCD LCH SW FAULT

(1) Do a check of the wiring between the PSEU and the MDCD Latched A and B Switches as follows:

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(a) Open this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	1	C01399	PSEU PRI

(b) Disconnect connectors D10986 and D10988 from the PSEU, M2061 (WDM 52-71-15).

(c) Disconnect connectors D16884 and D16886 from the MDCD Latched A and B Switches, S1230 and S1231.

(d) With the MDCD closed, locked and latched, do a check for Ground from the PSEU connectors, D10986 and D10988 as follows:

DOOR LATCHED A SWITCH

PSEU
D10986
pin 53 . . . . . GND

DOOR LATCHED B SWITCH

PSEU
D10988
pin 52 . . . . . GND

1) If you do not find continuity to Ground, examine the wiring as follows:

DOOR LATCHED A SWITCH

PSEU **S1230**
D10986 **D16884**
pin 53 . . . . . pin C

S1230
D16884
pin B . . . . . GND

DOOR LATCHED B SWITCH

PSEU **S1231**
D10988 **D16886**
pin 52 . . . . . pin C

S1231
D16886
pin B . . . . . GND

a) If you find a problem with the wiring, do these steps:

<1> Repair the wiring as necessary.

<2> Connect connectors D10986 and D10988 to the PSEU.

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<3> Close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	1	C01399	PSEU PRI

<4> Do the Repair Confirmation at the end of this task.

b) If you do not find a problem in the wiring, then do these steps:

<1> Replace the applicable MDCD Latched A (B) Switch. These are the task:

- Door Latched Switch Removal, AMM TASK 52-71-32-000-802
- Door Latched Switch Installation, AMM TASK 52-71-32-400-805

<2> Connect connectors D10986 and D10988 to the PSEU.

<3> Close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	1	C01399	PSEU PRI

<4> Do the Repair Confirmation at the end of this task.

2) If you find continuity to Ground, then continue.

(2) Do a check for 28V DC Power at the MDCD Latched A and B Switches, S1230 and S1231.

(a) Close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	1	C01399	PSEU PRI

(b) Do a voltage check as follows (WDM 52-71-15):

DOOR LATCHED A SWITCH

S1230		
D16884		**VOLTAGE**
pin A	GND	28V DC

DOOR LATCHED B SWITCH

S1231		
D16886		**VOLTAGE**
pin A	GND	28V DC

1) If you do not find 28V DC, at pin A of one MDCD Latched Switch connector, but found the correct voltage at the other, do these steps for the Latched A (B) Switch that 28V DC Power was not available:

a) Examine the wiring between the Junction Box, J48 and the applicable Latched A (B) Switch connector as follows:

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DOOR LATCHED A SWITCH

J48		**S1230**
D40956		**D16884**
pin A7	. . . . .	pin A

DOOR LATCHED B SWITCH

J48		**S1231**
D40956		**D16886**
pin B7	. . . . .	pin A

<1> Repair the problems that you find.

<2> Connect connectors D10986 and D10988 to the PSEU.

<3> Connect connectors D16884 and D16886 to the MDCD Latched A and B Switches.

<4> Do the Repair Confirmation at the end of this task.

(3) Make sure that the Door Latched A Switch, S1230 and Door Latched B Switch, S1231 are adjusted correctly. Do this task: Door Latched Switch Adjustment, AMM TASK 52-71-32-820-809.

(a) Do the Repair Confirmation at the end of this task.

(4) If a Door Latched Switch is stuck OPEN or CLOSED, then replace the applicable Door Latched A (B) Switch, S1230 (S1231). These are the tasks:

- Door Latched Switch Removal, AMM TASK 52-71-32-000-802
- Door Latched Switch Installation, AMM TASK 52-71-32-400-805

(a) Do the Repair Confirmation at the end of this task.

(5) Replace the PSEU, M2061. These are the tasks.

- Proximity Switch Electronics Unit (PSEU) Removal, AMM TASK 32-09-10-000-801
- Proximity Switch Electronics Unit (PSEU) Installation, AMM TASK 32-09-10-400-801

(a) Do the Repair Confirmation at the end of this task.

H. Fault Isolation Procedure - MDCD LKD DISAGREE

(1) Do a check of the wiring between the PSEU and the MDCD Locked A and B Switches as follows:

(a) Disconnect connectors D10986 and D10988 from the PSEU (WDM 52-71-15).

(b) With the MDCD closed and locked, do a check for Ground from the PSEU connectors, D10986 and D10988 as follows.

DOOR LOCKED A SWITCH

PSEU		
D10986		
pin 15	. . . . .	GND

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DOOR LOCKED B SWITCH

PSEU
D10988
pin 30 . . . . . GND

1) If you do not find continuity to Ground, do these steps:
   a) Replace the applicable MDCD Locked A (B) Switch, S1158 (S1161).
   b) Connect connectors D10986 and D10988 to the PSEU.
   c) Do the Repair Confirmation at the end of this task.
      <1> If the problem continues, repair the wiring as necessary.
2) If you find continuity to Ground, then continue.

(2) Make sure that the Door Locked A Switch, S1158 and Door Locked B Switch, S1161 are adjusted correctly. Do this task: Door Locked Switch Adjustment, AMM TASK 52-71-32-820-810.
   (a) Do the Repair Confirmation at the end of this task.

(3) If a Door Locked Switch is stuck OPEN or CLOSED, then replace the applicable Door Locked A (B) Switch, S1158 (S1161). These are the tasks:
   • Door Locked Switch Removal, AMM TASK 52-71-32-000-803
   • Door Locked Switch Installation, AMM TASK 52-71-32-400-806
   (a) Do the Repair Confirmation at the end of this task.

(4) Replace the PSEU, M2061. These are the tasks.
   • Proximity Switch Electronics Unit (PSEU) Removal, AMM TASK 32-09-10-000-801
   • Proximity Switch Electronics Unit (PSEU) Installation, AMM TASK 32-09-10-400-801
   (a) Do the Repair Confirmation at the end of this task.

I. Repair Confirmation

(1) Open and then close the MDCD. These are the tasks:
   • Open the Main Deck Cargo Door, AMM TASK 52-32-00-860-801
   • Close the Main Deck Cargo Door, AMM TASK 52-32-00-860-802

(2) Do the Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.
   (a) If the maintenance message does not show, then you corrected the problem.
   (b) If the maintenance message still shows, continue the applicable Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

814. Main Deck Cargo Door Does Not Unlock - Fault Isolation

A. Description

(1) This task is for these Observed Faults:
   (a) The Main Deck Cargo Door (MDCD) does not unlock when you hold the LOCK/UNLOCK Switch to the “UNLOCK” position.
   (b) The UNLOCKED Light is ON and the READY TO OPEN Light is OFF.

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(c) The UNLOCKED Light is OFF, the LOCKED Light is OFF and the READY TO OPEN Light is OFF.

B. Possible Causes

(1) MDCD Lock Switch, S1237

(2) Lock Actuator, M3280

(3) Lock Enable Switch, S1233

(4) Unlocked Light Relay, R1301

(5) Wiring

C. Related Data

(1) WDM 52-32-12

(2) SSM 52-32-12

D. Initial Evaluation

(1) On the Main Deck Cargo Door Control Panel, P32, put the UNLOCK/LOCK Switch to the “UNLOCK” position and hold it to unlock the door.

(a) Make sure that the READY TO OPEN Light on the Main Deck Cargo Door Control Panel, P32, is ON.

NOTE: The Main Deck Cargo Door Control Panel, P32, is on the Rigid Cargo Barrier, adjacent to the Forward Entry Door.

1) If the door unlocks and the READY TO OPEN Light is ON, then there was an intermittent fault.

2) If the UNLOCKED Light is ON and the READY TO OPEN Light is OFF, do the Fault Isolation - “UNLOCKED Light is ON and READY TO OPEN Light is OFF” below.

3) If the UNLOCKED Light is OFF and the READY TO OPEN Light is OFF, do the Fault Isolation - “UNLOCKED Light is OFF and READY TO OPEN Light is OFF” below.

E. Fault Isolation - UNLOCKED Light is ON and READY TO OPEN Light is OFF

(1) Remove connector D16838 from the Lock Actuator, M3280 (WDM 52-32-12).

(2) Hold the UNLOCK/LOCK Switch to the “UNLOCK” position, and do a check for 115V AC at pin 6 of the connector D16838.

(a) If you find 115V AC, do a check for Ground at pin 8 of connector D16838.

1) If you find ground, do a check for continuity between pins 11 and 12 of the connector D16838.

a) If you find continuity, replace the Lock Actuator, M3280. These are the tasks:

- Lock Mechanism - Removal, AMM TASK 52-32-24-000-801
- Lock Mechanism - Installation, AMM TASK 52-32-24-400-801

<1> Do the Repair Confirmation at the end of this task.

b) If you do not find continuity, repair the wiring between pins 11 and 12 (WDM 52-32-12).

<1> Do the Repair Confirmation at the end of this task.

(b) If you do not find 115V AC at pin 6 of the connector D16838, continue.

(3) Remove the connector D43158P from the MDCD Control Panel, P32, and do a check for continuity between pin 3 of connector D43158P and pin 6 of connector D16838.

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(a) If you do not find continuity, do a check for continuity between pins 1 and 3 of the Lock Enable Switch, S1233.

1) If you do not find continuity, replace the Lock Enable Switch, S1233 (WDM 52-32-12).

a) Do the Repair Confirmation at the end of this task.

2) If you find continuity between pins 1 and 3 of the Lock Enable Switch, S1233, repair the wiring between pin 3 of connector D43158P and pin 6 of connector D16838 (WDM 52-32-12).

a) Do the Repair Confirmation at the end of this task.

(b) If you find continuity between pin 3 of connector D43158P and pin 6 of connector D16838, replace the MDCD Lock Switch, S1237 (WDM 52-32-12).

F. Fault Isolation - UNLOCKED Light is OFF and READY TO OPEN Light is OFF

(1) Remove the connector D17376 of the Unlocked Light Relay, R1301 (WDM 52-32-12).

(2) Do a check for 28V DC at pin A2 of connector D17376.

(a) If you find 28V DC at pin A2, do a check for continuity between pins A2 and A3 of the Unlocked Light Relay, R1301.

1) If you find no continuity, replace the Unlocked Light Relay, R1301 (WDM 52-32-12).

a) Do the Repair Confirmation at the end of this task.

2) If you find continuity, replace the Unlocked Light, L1591 (WDM 52-32-12).

a) Do the Repair Confirmation at the end of this task.

(b) If you do not find 28V DC at pin A2, repair the wiring between pin A2 of connector D17376, and pin 1 of connector D43158J (WDM 52-32-12).

1) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) On the Main Deck Cargo Door Control Panel, P32, put the UNLOCK/LOCK Switch to the “UNLOCK” position and hold it to unlock the door.

(a) If the door unlocks correctly, then you corrected the problem.

(b) If the door does not unlock, continue the applicable Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

815. Main Deck Cargo Door (MDCD) Does Not Unlatch - Fault Isolation

A. Description

(1) This task is for these Observed Faults:

(a) The Main Deck Cargo Door (MDCD) does not unlatch when you hold the ARM Switch to the Up ARM position and put the UP TO CANOPY/CLOSE Control Switch, S1235 to the “UP TO CANOPY” position.

(b) The CLOSED & LATCH Light is ON.

B. Possible Causes

(1) Latch Actuator

(2) Latch Mechanism

(3) Unlocked Limit (Latch Enable) Switch, S1162

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(4) Lock Actuator, M3280

(5) Canopy Position Switch B, S1163

(6) MDCD Latch Relay, R1231

(7) MDCD AIR/GND Relay, R1297

(8) Control Switch, S1235 (UP TO CANOPY/CLOSE)

(9) ARM Switch, S1234

(10) Control Valve, V59

(11) Unlatched Switch, S1227

(12) Wiring

C. Related Data

(1) WDM 52-32-11

(2) SSM 52-32-11

D. Initial Evaluation

(1) On the Main Deck Cargo Door Control Panel, P32, put the UNLOCK/LOCK Switch to the UNLOCK position to unlock the door.

(2) Hold the ARM Switch to the Up ARM position and put the UP TO CANOPY/CLOSE Switch to the “UP TO CANOPY” position to unlatch and raise the door electrically.

(a) Make sure that the Green CLOSED & LATCHED Light on the Main Deck Cargo Door Control Panel, P32, is ON.

NOTE: The Main Deck Cargo Door Control Panel, P32, is on the Rigid Cargo Barrier, adjacent to the Forward Entry Door.

1) If the door does not unlock and the READY TO OPEN Light is OFF, do the Main Deck Cargo Door Does Not Unlock - Fault Isolation, 52-32 TASK 814.

2) If the CLOSED & LATCHED Light goes OFF and the door raises with the IN TRANSIT Light ON, then there was an intermittent fault.

3) If the CLOSED & LATCHED Light stays ON and the door does not raise, do the Fault Isolation Procedure below.

E. Fault Isolation Procedure

(1) Remove the connector D40956P from the J48 Junction Box.

(2) Do a check for continuity between pins B9 and B10 of the connector D40956P at Junction Box J48.

(a) If you find no continuity, do a visual check of the Door Locks and see if the Door Locks clearly disengaged from the Latch Cam.

1) If the Door Locks clearly disengaged from the Latch Cam, do a visual check of the Unlocked Limit Switch, S1162 and see if the Switch Roller is in contact with Lock Shaft.

a) If the Switch Roller is in contact with the Lock Shaft, put a 0.05 in. (0.13 cm)-0.08 in. (0.20 cm) thick shim between the Switch Roller and Lock Shaft.

<1> If the door unlatches, adjust the Unlocked Limit Switch, S1162. This is the task: Door Latch Enable Switch Adjustment, AMM TASK 52-71-32-820-814

<a> Do the Repair Confirmation at the end of this task.

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<2> If the door does not unlatch, replace the Unlocked Limit Switch, S1162. These are the tasks:

- Door Latch Enable Switch Removal, AMM TASK 52-71-32-000-807
- Door Latch Enable Switch Installation, AMM TASK 52-71-32-400-810

<a> Do the Repair Confirmation at the end of this task.

b) If the Switch Roller is not in contact with the lock shaft, do the visual check of the Unlocked Limit Switch, S1162 and make sure that the lock nuts are tight and safety wire is installed.

<1> If the lock nuts are tight and safety wire is installed, replace the Unlocked Limit Switch, S1162. These are the tasks:

- Door Latch Enable Switch Removal, AMM TASK 52-71-32-000-807
- Door Latch Enable Switch Installation, AMM TASK 52-71-32-400-810

<a> Do the Repair Confirmation at the end of this task.

<2> If lock nuts are not tight and/or safety wire is not installed, adjust the lock nuts and install safety wire as necessary.

<a> Do the Repair Confirmation at the end of this task.

2) If the door locks did not clearly disengage from the Latch Cam, replace the Lock Actuator, M3280. These are the tasks:

- Lock Mechanism - Removal, AMM TASK 52-32-24-000-801
- Lock Mechanism - Installation, AMM TASK 52-32-24-400-801

a) Do the Repair Confirmation at the end of this task.

(b) If you find continuity between pins B9 and B10, continue.

(3) Do a check for 28V DC at pin B10 of the J48 Junction Box connector, D40956J (WDM 52-32-11).

(a) If you do not find 28V DC at pin B10, repair the wiring in the Junction Box.

1) Do the Repair Confirmation at the end of this task.

(b) If you find 28V DC at pin B10, install a jumper between pins B9 and B10 of the J48 Junction Box connector, D40956J, and do a check for 28V DC at pin A9 of the J48 Junction Box connector, D40956J.

1) If you do not find 28V DC at pin A9, repair the wiring between pins A9 and B9 in the Junction Box.

a) Do the Repair Confirmation at the end of this task.

2) If you find 28V DC at pin A9, remove the jumper, reconnect the J48 Junction Box connector D40956, and remove connector D43156P. Do a check for continuity between pins 15 and 28 of the Main Deck Cargo Door Control Panel connector, D43156J.

a) If you find no continuity, do a check for continuity between pins 2 and 3 of the ARM Switch, S1234 on the Main Deck Cargo Door Control Panel, P32.

<1> If you find no continuity, repair the wiring between pins 15 and 28.

<a> Do the Repair Confirmation at the end of this task.

<2> If you find continuity, replace the ARM Switch, S1234 on the Main Deck Cargo Door Control Panel, P32.

<a> Do the Repair Confirmation at the end of this task.

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b) If you find continuity, hold the ARM Switch to the Up ARM position and put the UP TO CANOPY/CLOSE Switch, S1235 to the “UP TO CANOPY” position. Do a check for continuity between pins 15 and 28 of the Main Deck Cargo Door Control Panel connector, D43156J.

<1> If you find continuity, replace the ARM Switch, S1234 on the Main Deck Cargo Door Control Panel, P32.

<a> Do the Repair Confirmation at the end of this task.

<2> If you find no continuity, continue.

(4) Hold the ARM Switch to the Up ARM position and put the UP TO CANOPY/CLOSE Switch, S1235 to the “UP TO CANOPY” position, and do a check for continuity between pins 15 and 29 of the Main Deck Cargo Door Control Panel connector, D43156J.

(a) If you find no continuity, do a check for continuity between pins 15 and 24 of the Main Deck Cargo Door Control Panel connector, D43156J.

1) If you find no continuity, repair the wiring between pin 10 of the ARM Switch, S1234 and pin 5 of the UP TO CANOPY/CLOSE Switch, S1235 on the Main Deck Cargo Door Control Panel, P32.

a) Do the Repair Confirmation at the end of this task.

2) If you find continuity, replace the UP TO CANOPY/CLOSE Switch, S1235 on the Main Deck Cargo Door Control Panel, P32.

a) Do the Repair Confirmation at the end of this task.

(b) If you find continuity, continue.

(5) Open this access panel:

Number	Name/Location
194DR	Aft Wing To Body Fairing Panel

(6) Put the Control Valve Lever to position 1, and do a check for continuity between pins 29 and 11 of the Main Deck Cargo Door Control Panel connector, D43156P.

(a) If you find continuity, reconnect all the connectors, hold the ARM Switch to the Up ARM position and put the UP TO CANOPY/CLOSE Switch, S1235 to the “UP TO CANOPY” position. Make sure that there is an audible “CLICK” sound from inside of the Main Deck Cargo Door Control Panel, P32.

1) If there is no audible “CLICK” sound, replace the Main Deck Cargo Door Latch Relay, R1231.

a) Do the Repair Confirmation at the end of this task.

2) If there is audible “CLICK” sound, do a check for 115V AC at pins 1, 2, and 3 of the Main Deck Cargo Door Latch connector, D16834.

a) If you find 115V AC, do a check for ground at pin 4 of the Main Deck Cargo Door Latch connector, D16834.

<1> If you find ground, replace the Main Deck Cargo Door Latch, M3279. These are the tasks:

- Latch Actuator Removal, AMM TASK 52-32-21-000-801
- Latch Actuator Installation, AMM TASK 52-32-21-400-801

<a> Do the Repair Confirmation at the end of this task.

<2> If you find no ground, replace the MDCD AIR/GND Relay, R1297.

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<a> Do the Repair Confirmation at the end of this task.

3) If you find no continuity between pins 29 and 11 of the MDCD Control Panel connector, D43156P, continue.

(7) Do a check for continuity between pins 31 and 11 of the Main Deck Cargo Door Control Panel connector, D43156P.

(a) If you find continuity, replace the CANOPY POSITION B Switch, S1163.

1) Do the Repair Confirmation at the end of this task.

(b) If you find no continuity, do a check for continuity between pins 29 and 20 of the Main Deck Cargo Door Control Panel connector, D43156P.

1) If you find continuity, replace the Unlatched Switch, S1227.

a) Do the Repair Confirmation at the end of this task.

2) If you find no continuity, continue.

(8) With the Motor Control Valve, V59 at position 1, remove the connector D1220 from the Motor Control Valve, V59, and do a check for continuity between pins A and C of the Motor Control Valve, V59.

(a) If you find no continuity, replace the Motor Control Valve, V59. These are the tasks:

- Control Valve Removal, AMM TASK 52-32-61-000-801
- Control Valve Installation, AMM TASK 52-32-61-400-801

1) Do the Repair Confirmation at the end of this task.

(b) If you find continuity, continue.

(9) Do a check for continuity between pins 29 and 11 of the Main Deck Cargo Door Control Panel connector, D43156P.

(a) If you find no continuity, repair the wiring between pins 29 and 11 of the Main Deck Cargo Door Control Panel connector, D43156P.

1) Do the Repair Confirmation at the end of this task.

(b) If you find continuity, replace the Motor Control Valve, V59. These are the tasks:

- Control Valve Removal, AMM TASK 52-32-61-000-801
- Control Valve Installation, AMM TASK 52-32-61-400-801

1) Do the Repair Confirmation at the end of this task.

F. Repair Confirmation

(1) On the Main Deck Cargo Door Control Panel, P32, hold the ARM Switch to the up ARM position and put the UP TO CANOPY/CLOSE Control Switch, S1235 to the “UP TO CANOPY” position to raise the door electrically.

(a) If the door raises correctly, then you corrected the problem.

1) Close this access panel:

Number	Name/Location
194DR	Aft Wing To Body Fairing Panel

(b) If the door does not raise correctly, continue the applicable Fault Isolation Procedure at the subsequent step.

END OF TASK

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816. Main Deck Cargo Door Does Not Lock - Fault Isolation

A. Description

(1) This task is for these Observed Faults:

(a) The Main Deck Cargo Door (MDCD) does not lock when you hold the LOCK/UNLOCK Switch to the “LOCK” position.

(b) The UNLOCKED Light is ON and the READY TO OPEN Light is ON.

(c) The UNLOCKED Light is ON and the READY TO OPEN Light is OFF.

B. Possible Causes

(1) Lock Switch, S1237

(2) Lock Actuator, M3280

(3) Lock Enable Switch, S1233

(4) Unlocked Light Relay, R1301

(5) Vent Door Proximity Sensor, S1229

(6) Wiring

C. Related Data

(1) WDM 52-32-12

(2) SSM 52-32-12

D. Initial Evaluation

(1) On the Main Deck Cargo Door Control Panel, P32, put the UNLOCK/LOCK Switch to the “LOCK” position and hold it to lock the door.

(a) Make sure that the LOCKED Light on the Main Deck Cargo Door Control Panel, P32, is ON.

NOTE: The Main Deck Cargo Door Control Panel, P32, is on the Rigid Cargo Barrier, adjacent to the Forward Entry Door.

1) If the door locks, the LOCKED Light is ON and the READY TO OPEN Light is OFF, then there was an intermittent fault.

2) If the UNLOCKED Light is ON and the READY TO OPEN Light is ON, do the “Fault Isolation - UNLOCKED Light is ON and READY TO OPEN Light is ON” below.

3) If the UNLOCKED Light is ON and the READY TO OPEN Light is OFF, do the “Fault Isolation - UNLOCKED Light is ON and READY TO OPEN Light is OFF” below.

E. Fault Isolation - UNLOCKED Light is ON and READY TO OPEN Light is ON

(1) Remove connector D16838 from the Lock Actuator, M3280 (WDM 52-32-12).

(a) Hold the UNLOCK/LOCK Switch to the “LOCK” position, and do a check for 115V AC at pin 4 of the connector D16838.

1) If you find 115V AC, do a check for Ground at pin 8 of connector D16838.

a) If you find ground, do a check for continuity between pins 9 to 10 of the connector D16838.

<1> If you find continuity, replace the Lock Actuator, M3280. These are the tasks:

- Lock Mechanism - Removal, AMM TASK 52-32-24-000-801
- Lock Mechanism - Installation, AMM TASK 52-32-24-400-801

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<a> Do the Repair Confirmation at the end of this task.

<2> If you do not find continuity, repair the wiring between pins 9 and 10 (WDM 52-32-12).

<a> Do the Repair Confirmation at the end of this task.

2) If you do not find 115V AC at pin 4 of the connector D16838, continue.

(2) Remove the connector D43158P of the MDCD Control Panel, P32, and do a check for continuity between pin 4 of connector D43158P and pin 4 of connector D16838 (WDM 52-32-12).

(a) If you do not find continuity, do a check for continuity between pins 4 and 6 of the Lock Enable Switch, S1233.

1) If you do not find continuity, replace the Lock Enable Switch, S1233 (WDM 52-32-12).

a) Do the Repair Confirmation at the end of this task.

2) If you find continuity between pins 4 and 6 of the Lock Enable Switch, S1233, repair the wiring between pin 4 of connector D43158P and pin 4 of connector D16838 (WDM 52-32-12).

a) Do the Repair Confirmation at the end of this task.

(b) If you find continuity between pin 4 of connector D43158P and pin 4 of connector D16838, replace the MDCD Lock Switch, S1237 (WDM 52-32-12).

F. Fault Isolation - UNLOCKED Light is ON and READY TO OPEN Light is OFF

(1) Remove the connector D16840 from the Vent Door Closed Proximity Sensor, S1229, and install a jumper between pins B and C (WDM 52-32-12).

(2) Disconnect connector D43158P from the MDCD Control Panel, P32 and do a check for ground at pin 7 of connector D43158P.

(a) If you find ground, replace the Unlocked Light Relay, R1301.

1) Remove the jumper and reconnect the connector D16840 to the Vent Door Proximity Sensor, S1229,

2) Do the Repair Confirmation at the end of this task.

(b) If you do not find ground, examine the wiring as follows (WDM 52-32-12):

	VENT DOOR CLOSED PROX
MDCD PNL	**SNSR**
D43158P	**D16840**
pin 7	pin C

VENT DOOR	
CLOSED PROX	
SNSR	
D16840	
pin B	GND

1) Repair the problems that you find.

a) Do the Repair Confirmation at the end of this task.

(3) Replace the Vent Door Closed Proximity Sensor, S1229 (WDM 52-32-12).

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(a) Do the Repair Confirmation at the end of this task.

G. Repair Confirmation

(1) On the Main Deck Cargo Door Control Panel, P32, put the UNLOCK/LOCK Switch to the “LOCK” position and hold it to lock the door.

(a) If the door locks correctly, then you corrected the problem.

(b) If the door does not lock, continue the applicable Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

817. Main Deck Cargo Door Does Not Latch - Fault Isolation

A. Description

(1) This task is for these Observed Faults:

(a) The Main Deck Cargo Door (MDCD) does not latch when you hold the ARM Switch, S1234 to the Down ARM position and put the UP TO CANOPY/CLOSE Control Switch, S1235 to the “CLOSE” position.

(b) The CLOSED & LATCHED Light is OFF.

B. Possible Causes

(1) Latch Actuator

(2) Latch Mechanism

(3) Main Deck Cargo Door Latch, M3279

(4) Main Deck Cargo Door Latch Relay, R1231

(5) Latch Delay Relay, R1232

(6) Latched Switch, S1226

(7) Wiring

C. Related Data

(1) WDM 52-32-11

(2) SSM 52-32-11

D. Initial Evaluation

(1) On the Main Deck Cargo Door Control Panel, P32, hold the ARM Switch to the Down ARM position and put the UP TO CANOPY/CLOSE Switch to the “CLOSE” position to close and latch the door electrically.

(a) Make sure that the Green CLOSED & LATCHED Light on the Main Deck Cargo Door Control Panel, P32, is ON.

NOTE: The Main Deck Cargo Door Control Panel, P32, is on the Rigid Cargo Barrier, adjacent to the Forward Entry Door.

1) If the CLOSED & LATCHED Light comes ON and the door closes and latches with the IN TRANSIT Light OFF, then there was an intermittent fault.

2) If the CLOSED & LATCHED Light stays OFF and the door does not latch, do the Fault Isolation Procedure below.

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E. Fault Isolation Procedure

(1) Open this access panel:

Number	Name/Location
194DR	Aft Wing To Body Fairing Panel

(2) Remove the connector D1220 from the Motor Control Valve, V59 and connector D43156P from the Main Deck Cargo Door Control Panel, P32 (WDM 52-32-11).

(a) Do a check for continuity between pin D of the connector D1220 and pin 13 of the connector D43156P.

1) If you do not find continuity, repair the wiring between pin D of the connector D1220 and pin 13 of the connector D43156P.

a) Do the Repair Confirmation at the end of this task.

2) If you find continuity, do a check for continuity between pin D of the connector D1220 to pin 14 of the connector D43156P.

a) If you find continuity, replace the Latched Switch, S1226.

<1> Do the Repair Confirmation at the end of this task.

b) If you do not find continuity, continue.

(3) Reinstall the connector D43156P to the Main Deck Cargo Door Control Panel, P32, and remove the connector D16834 from the Main Deck Cargo Door Latch, M3279 (WDM 52-32-11).

(a) Hold the ARM Switch, S1234 to the Down ARM position and put the UP TO CANOPY/CLOSE switch, S1235 to the “CLOSE” position, do a check for 115V AC at pins 1, 2, and 3 of the Main Deck Cargo Door Latch connector, D16834.

1) If you find 115V AC, do a check for Ground at pin 4 of the Main Deck Cargo Door Latch connector, D16834.

a) If you find Ground, replace the Main Deck Cargo Door Latch, M3279. These are the tasks:

- Latch Actuator Removal, AMM TASK 52-32-21-000-801
- Latch Actuator Installation, AMM TASK 52-32-21-400-801

<1> Do the Repair Confirmation at the end of this task.

b) If you do not find Ground, repair the wiring between the Main Deck Cargo Door Latch connector, D16834, pin 4 and Ground.

<1> Do the Repair Confirmation at the end of this task.

2) If you do not find 115V AC, continue.

(4) Remove the connector D43156P from the Main Deck Cargo Door Control Panel, P32 (WDM 52-32-11).

(a) Do a check for continuity as follows:

MDCD CTL PANEL, P32 D43156P		MDCD LATCH, M3279 D16834
pin 6	. . . . .	pin 1
pin 8	. . . . .	pin 3
pin 10	. . . . .	pin 2

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1) If you do not find continuity, repair the wiring as necessary.
   a) Do the Repair Confirmation at the end of this task.
2) If you find continuity, replace the Main Cargo Door Latch Relay, R1231, and do the Repair Confirmation at the end of this task.
   a) If the problem continue, replace the Latch Delay Relay, R1232.
      <1> Do the Repair Confirmation at the end of this task.

F. Repair Confirmation

(1) On the Main Deck Cargo Door Control Panel, P32, hold the ARM Switch, S1234 to the Down ARM position and put the UP TO CANOPY/CLOSE Control Switch, S1235 to the “CLOSE” position to close and latch the door electrically.

(a) If the door closes and latches correctly and the CLOSED & LATCHED Light is ON, then you corrected the problem.

1) Close this access panel:

Number	Name/Location
194DR	Aft Wing To Body Fairing Panel

(b) If the door does not close and latch correctly, continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

801. PSEU Forward Access Door Monitored Problem - Fault Isolation

A. Description

(1) This task is for these maintenance messages:

(a) 52-72003 FWD ACC DR OPEN

(2) This task is also for EQUIP Door Warning Light (P5) Indication during taxi or takeoff.

(3) For a detail description of the Door Warning System, refer to SDS SUBJECT 52-71-00.

B. Possible Causes

(1) Forward Access Door Indication Switch, S196

(2) Wiring

(3) Proximity Switch Electronic Unit (PSEU), M2061

C. Related Data

(1) WDM 52-71-12

(2) SSM 52-71-12

D. Initial Evaluation

(1) Close, latch and lock the Forward Access Door.

(2) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801

(a) If the maintenance message does not show in EXISTING FAULTS, then there was an intermittent fault.

(b) If the maintenance message shows in EXISTING FAULTS, then do the Fault Isolation Procedure below.

E. Fault Isolation Procedure

(1) Do this check of the PSEU:

(a) Disconnect the connector D10988 from the PSEU.

(b) With the door closed, latched and locked, do a continuity check between pin 6 of connector D10988 and Structure Ground.

1) If you find continuity between pin 6 and Structure Ground, then replace the PSEU, M2061. These are the tasks:

- Proximity Switch Electronics Unit (PSEU) Removal, AMM TASK 32-09-10-000-801
- Proximity Switch Electronics Unit (PSEU) Installation, AMM TASK 32-09-10-400-801

a) Do the Repair Confirmation at the end of this task.

2) If you do not find continuity between pin 6 and Structure Ground, then continue.

(2) De-actuate the Forward Access Door Indication Switch.

(3) Do this check of the Forward Access Door Indication Switch Wiring (WDM 52-71-12):

FWD ACCESS DOOR IND SW WIRING

PSEU	**S196**
D10988	**Terminal**
pin 6	term C

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S196
Terminal
term NO GD1210-DC

(a) If you find a problem in the wiring, then do these steps:
1) Repair the wiring as necessary.
2) Connect the connector D10988 to the PSEU.
3) Do the Repair Confirmation at the end of this task.

(b) If you do not find a problem in the wiring, then replace the Forward Access Door Indication Switch, S196. These are the tasks:
- Forward Access Door Indication Switch Removal, AMM TASK 52-71-41-000-801
- Forward Access Door Indication Switch Installation, AMM TASK 52-71-41-400-801

1) Connect the connector D10988 to the PSEU.
2) Do the Repair Confirmation at the end of this task.

F. Repair Confirmation

(1) Close, latch and lock the Forward Access Door.

(2) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(a) If the maintenance message does not show in EXISTING FAULTS then you corrected the problem.

1) Close and latch the access panel:

Number	Name/Location
112A	Forward Access Door

(b) If the maintenance message still shows in EXISTING FAULTS, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

802. PSEU Electronic Equipment Access Door Monitored Problem - Fault Isolation

A. Description

(1) This task is for these maintenance messages:

NOTE: The PSEU does not record this message when the airplane is on the ground.

(a) 52-72004 EE ACC DR OPEN

(2) This task is also for EQUIP Door Warning Light (P5) Indication during taxi or takeoff.

(3) For a detail description of the Door Warning System, refer to SDS SUBJECT 52-71-00.

B. Possible Causes

(1) Electronic Equipment Access Door Indication Switch, S197

(2) Wiring

(3) PSEU, M2061

C. Related Data

(1) WDM 52-71-12

(2) SSM 52-71-12

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Initial Evaluation

(1) Close and latch the Electronic Equipment Access Door.

(2) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801

(a) If the maintenance message does not show in EXISTING FAULTS, then there was an intermittent fault.

(b) If the maintenance message shows in EXISTING FAULTS, then do the Fault Isolation Procedure below.

E. Fault Isolation Procedure

(1) Do this check of the PSEU:

(a) Disconnect the connector D10986 from the PSEU.

(b) With the door closed, latched and locked, do a continuity check between pin 6 of the connector D10986 and Structure Ground.

1) If you find continuity between pin 6 and Structure Ground, then replace the PSEU, M2061. These are the tasks:

- Proximity Switch Electronics Unit (PSEU) Removal, AMM TASK 32-09-10-000-801
- Proximity Switch Electronics Unit (PSEU) Installation, AMM TASK 32-09-10-400-801

a) Do the Repair Confirmation at the end of this task.

2) If you do not find continuity between pin 6 and Structure Ground, then continue.

(2) De-actuate the Electronic Equipment Access Door Switch.

(3) Do this check of the Electronic Equipment Access Door Indication Switch Wiring (WDM 52-71-12):

EE ACCESS DOOR IND SW WIRING

PSEU D10986		**S197 Terminal**
pin 6	. . . . .	term C

S197 Terminal		
term NO	. . . . .	GD1322-DC

(a) If you find a problem in the wiring, then do these steps:

1) Repair the wiring as necessary.

2) Connect the connector D10986 to the PSEU.

3) Do the Repair Confirmation at the end of this task.

(b) If you do not find a problem in the wiring, then replace the Electronic Equipment Access Door Indication Switch, S197. These are the tasks:

- Electronic Equipment Access Door Indication Switch Removal, AMM TASK 52-71-42-000-801
- Electronic Equipment Access Door Indication Switch Installation, AMM TASK 52-71-42-400-801

1) Connect the connector D10986 to the PSEU.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Do the Repair Confirmation at the end of this task.

F. Repair Confirmation

(1) Close, latch and lock the Electronic Equipment Access Door Switch.

(2) Do this task: Proximity Switch Electronics Unit (PSEU) BITE Procedure, 32-09 TASK 801.

(a) If the maintenance message does not show in EXISTING FAULTS then you corrected the problem.

(b) If the maintenance message still shows in EXISTING FAULTS, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799, 802

801. Control Cabin Door - CAB DOOR UNLOCKED Light Does Not Come On When the Door is Unlocked - Fault Isolation

A. Description

(1) (SDS SUBJECT 52-51-00)

B. Possible Causes

(1) Control cabin door lock switch, S2

(2) Master dim relay, R34

(3) Annunciation and dimming module, M469

(4) Wiring

C. Circuit Breakers

(1) These are the primary circuit breakers related to the fault:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	11	C00133	INDICATOR MASTER DIM DIM/TST CONT
E	1	C00137	DOOR LOCK

D. Related Data

(1) (SSM 33-18-11)

(2) (SSM 33-18-36)

(3) (SSM 33-18-51)

(4) (SSM 33-18-61)

(5) (SSM 33-18-62)

(6) (SSM 52-51-11)

(7) (WDM 33-18-11)

(8) (WDM 33-18-36)

(9) (WDM 33-18-51)

(10) (WDM 33-18-61)

(11) (WDM 33-18-62)

(12) (WDM 52-51-11)

E. Initial Evaluation

(1) Push the CAB DOOR UNLOCKED switch S2, on the stab/trim/cabin door module P8-47, to the unlock position to open the control door.

(2) If the CAB DOOR UNLOCKED light does not come on, then do the Fault Isolation Procedure below.

(3) If the CAB DOOR UNLOCKED light comes on, then there was an intermittent fault.

F. Fault Isolation Procedure

(1) Do this check of the CAB DOOR UNLOCKED light:

(a) On the captain's instrument panel P1-3, set the DIM/BRT/TEST switch to TEST.

(b) If the CAB DOOR UNLOCKED light comes on, then do these steps:

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OMR 701-799, 802 (Continued)

1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	11	C00133	INDICATOR MASTER DIM DIM/TST CONT
E	1	C00137	DOOR LOCK

2) Do a wiring check between these pins of connector D1344 at the annunciation and dimming module M469 and connector D40816P at the load control center P6-3:

D1344	D40816P
pin 45	pin 9

3) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	11	C00133	INDICATOR MASTER DIM DIM/TST CONT
E	1	C00137	DOOR LOCK

   c) Push the CAB DOOR UNLOCKED switch S2, on the stab/trim/cabin door module P8-47, to the unlock position to open the control door.
   d) If the CAB DOOR UNLOCKED light comes on, then you corrected the fault.

4) If you do not find a problem with the wiring, then do these steps:
   a) Replace the control cabin door lock switch S2.
   b) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	11	C00133	INDICATOR MASTER DIM DIM/TST CONT
E	1	C00137	DOOR LOCK

   c) Push the CAB DOOR UNLOCKED switch S2, on the stab/trim/cabin door module P8-47, to the unlock position to open the control door.
   d) If the CAB DOOR UNLOCKED light comes on, then you corrected the fault.

(c) If the CAB DOOR UNLOCKED light does not come on with either the door open or the DIM/BRT/TEST switch to TEST, then continue.

(2) Do this check of the wiring:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	11	C00133	INDICATOR MASTER DIM DIM/TST CONT
E	1	C00137	DOOR LOCK

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OMR 701-799, 802 (Continued)

(b) Do a wiring check between these pins on connector D1344 at the annunciation and dimming module, M469 and connector D40800P at the load control center P6 connector:

D1344	D40800P
pin 46	pin 6

1) If you find a problem with the wiring, then do these steps:
   a) Repair the wiring.
   b) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	11	C00133	INDICATOR MASTER DIM DIM/TST CONT
E	1	C00137	DOOR LOCK

   c) Push the CAB DOOR UNLOCKED switch S2, on the stab/trim/cabin door module P8-47, to the unlock position to open the control door.
   d) If the CAB DOOR UNLOCKED light comes on, then you corrected the fault.
2) If you do not find a problem with the wiring, then continue.

(3) Replace the control cabin door lock switch S2.
   (a) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	11	C00133	INDICATOR MASTER DIM DIM/TST CONT
E	1	C00137	DOOR LOCK

   (b) Push the CAB DOOR UNLOCKED switch S2, on the stab/trim/cabin door module P8-47, to the unlock position to open the control door.
   (c) If the CAB DOOR UNLOCKED light comes on, then you corrected the fault.
   (d) If the CAB DOOR UNLOCKED light is still off, then the fault is in load control center P6 master dim relay R34 circuit.

OMR ALL

——— END OF TASK ———

802. Control Cabin Door Difficult to Open/Close - Fault Isolation

A. Description

(1) This task is for these Control Cabin Door problems:
   (a) The Control Cabin Door does not open easily.
   (b) The Control Cabin Door does not close easily.
   (c) Adjustment of the Control Cabin Door is necessary.

B. Possible Causes

(1) Not a sufficient quantity of lubricant on the latch.
(2) Header

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(3) Trim angle

(4) Hinge

(5) Control Cabin Door lock

C. Related Data

(1) (SDS SUBJECT 52-51-00).

D. Initial Evaluation

(1) Open and close the Control Cabin Door.

(a) If it is easy to open and close the Control Cabin Door, then you have an intermittent fault.

(b) If it is difficult to open and close the Control Cabin Door, then do the Fault Isolation Procedure below.

E. Fault Isolation Procedure

(1) Do a check of the latch assembly to see if it is necessary to lubricate it.

(a) Spray a light layer of dry lubricant on the parts of the latch assembly that move.

(b) Do the Repair Confirmation at the end of this task.

(2) Look for unwanted material on the door lock and hinges.

(a) If unwanted material is found on the door lock and hinges, then remove the unwanted material from the door lock and hinges.

(b) Do the Repair Confirmation at the end of this task.

(3) Look and listen for points of interference between door and frame.

(a) Do these steps to adjust the door header and the trim angle:

OMR 701-799, 802

1) For the adjustment, refer to Crew Door Installation, AMM TASK 52-51-00-400-801.

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

2) For the adjustment, refer to Flight Compartment Door - Installation, AMM TASK 52-51-01-400-801.

OMR ALL

3) Do the Repair Confirmation at the end of this task.

F. Repair Confirmation

(1) Open and close the Control Cabin Door.

(a) If it is easy to open and close the Control Cabin Door, then you have corrected the fault.

(b) If it is difficult to open and close the Control Cabin Door, then continue the Fault Isolation Procedure at the subsequent step.

——— END OF TASK ———

OMR 701-799, 802

803. Control Cabin Door Does Not Release Electrically - Fault Isolation

A. Description

(1) (SDS SUBJECT 52-51-00)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

OMR 701-799, 802 (Continued)

B. Possible Causes

(1) Door lock, M117

(2) Control cabin door lock switch S2, P8-47

(3) Wiring

C. Circuit Breakers

(1) This is the circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	1	C00137	DOOR LOCK

D. Related Data

(1) (SSM 52-51-11)

(2) (WDM 52-51-11)

E. Initial Evaluation

(1) Manually open door.

(2) Operate control cabin door lock switch S2 to lock and then to unlock positions.

(3) Monitor that the control cabin door lock unlocks and locks.

(4) If the control cabin door lock does not operate correctly, then do the Fault Isolation Procedure below.

(5) If the door lock operates correctly, then there was an intermittent fault.

F. Fault Isolation Procedure

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	1	C00137	DOOR LOCK

(2) Find the splice which connects the door/lock solenoid red wire to the airplane wiring.

(3) With the control cabin door lock switch S2 in the Unlock position, do a wiring check between the door lock solenoid wire and circuit breaker panel P6-3 connector D40816P pin:

S2 Soleniod	D40816P
red wire	pin 9

(a) Wiring check should show an open circuit

1) If wiring check shows continuity, do this:

a) Replace the control cabin door lock switch S2.

2) If wiring check shows an open circuit, do this:

a) Replace the door lock, M117.

These are the tasks:Crew Door Lock Removal, AMM TASK 52-51-21-000-801,Crew Door Lock Installation, AMM TASK 52-51-21-400-801.

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OMR 701-799, 802 (Continued)

3) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	1	C00137	DOOR LOCK

(4) If the door releases electrically, then you corrected the fault.

——— END OF TASK ———

OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999

804. Door Does Not Lock in the AUTO Position - Fault Isolation

A. Description

(1) This task is for this observed fault:

(a) Flight deck door (enhanced security door) does not lock with switch set to AUTO.

B. Possible Causes

(1) Electric Strike, M2538

(2) Flight Compartment Door Lock Switch, S2

(3) Door Access Controller/Chime Module, M2537

(4) Wiring

C. Circuit Breakers

(1) This is the circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	1	C00137	DOOR LOCK

D. Related Data

(1) SSM 52-51-11

(2) WDM 52-51-11

E. Initial Evaluation

(1) Open the flight compartment door.

(2) Put the power switch on the Door Access Controller/Chime Module, M2537 to the NORM position.

(3) Operate Flight Compartment Door Lock Switch, S2 to the UNLOCK and then to the AUTO positions.

(4) Make sure that the Electric Strike, M2538 releases and engages.

(5) If the Electric Strike, M2538 did not go to the engage position, then do the Fault Isolation Procedure-Electrical Problem.

(6) If the Electric Strike, M2538 did go to the engage position, then do these steps:

(a) Make sure that the Flight Compartment Door Lock Switch, S2 is in the AUTO position.

(b) Close the flight compartment door.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION

DO NOT USE FORCE TO OPEN THE DOOR. IF YOU DO, YOU CAN CAUSE DAMAGE TO THE EQUIPMENT.

(c) Carefully try to open the flight compartment door without turning the handle.

(d) If the flight compartment door did not open, then there was an intermittent fault.

(e) If the flight compartment door did open, then do the Fault Isolation Procedure-Out of Tolerance.

F. Fault Isolation Procedure-Out of Tolerance

(1) Close the flight compartment door.

(2) Make sure that the Flight Compartment Door Lock Switch, S2 is in the AUTO position.

(3) If the Electric Strike, M2538 is in the engage position, do these steps:

(a) Make sure that the Electric Strike, M2538 and door latch engagement is 0.330 in. (8.382 mm) - 0.360 in. (9.144 mm).

(b) If the electric strike/door latch engagement is correct, then replace the Electric Strike, M2538. These are the tasks:

- Electric Strike Removal, AMM TASK 52-51-03-000-801
- Electric Strike Installation, AMM TASK 52-51-03-400-801

1) If the flight compartment door locks with the Flight Compartment Door Lock Switch, S2 in the AUTO position you have corrected the problem.

G. Fault Isolation Procedure-Electrical Problem

(1) Look at the LOCK FAIL light on the Switch/Light Module, P8-47.

(2) If the LOCK FAIL light is on, then replace the Electric Strike, M2538. These are the tasks:

- Electric Strike Removal, AMM TASK 52-51-03-000-801
- Electric Strike Installation, AMM TASK 52-51-03-400-801

(a) If the Electric Strike, M2538 engages with the Flight Compartment Door Lock Switch, S2 in the AUTO position you have corrected the problem.

(b) If the Electric Strike, M2538 did not engage with the Flight Compartment Door Lock Switch, S2 in the AUTO position, install a new Door Access Controller/Chime Module, M2537. These are the tasks:

- Chime Module Removal, AMM TASK 52-51-06-000-801
- Chime Module Installation, AMM TASK 52-51-06-400-801

1) If the Electric Strike, M2538 engages with the flight compartment door switch in the AUTO position you have corrected the problem.

2) If the Electric Strike, M2538 did not engage with the flight compartment door switch in the AUTO position, repair the wiring between the Electric Strike, M2538 and the chime module, M2537 (WDM 52-51-11).

a) Install the Door Access Controller/Chime Module, M2537. This is the task: Chime Module Installation, AMM TASK 52-51-06-400-801.

b) Install the Electric Strike, M2538. This is the task: Electric Strike Removal, AMM TASK 52-51-03-000-801.

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(3) If the LOCK FAIL light is not on, then do these steps:

(a) Remove the Door Access Controller/Chime Module, M2537. This is the task: Chime Module Removal, AMM TASK 52-51-06-000-801.

(b) Do a check for 28V DC between Pin 1 and structure ground of connector D13730.

(c) If there is not 28V DC between Pin 1 and structure ground, then do this step:

1) Repair the wiring.

2) If the flight compartment door locks with the Flight Compartment Door Lock Switch, S2 in the AUTO position you have corrected the problem.

(d) If there is 28V DC between Pin 1 and structure ground of connector D13730, then continue.

(e) Install a new Door Access Controller/Chime Module, M2537. This is the task: Chime Module Installation, AMM TASK 52-51-06-400-801.

1) If the flight compartment door locks with the Flight Compartment Door Lock Switch, S2 in the AUTO position you have corrected the problem.

2) If the flight compartment door does not lock with the Flight Compartment Door Lock Switch, S2 in the AUTO position, then continue.

(f) Replace the Flight Compartment Door Lock Switch, S2. These are the tasks:

- Cockpit Control Panel Switch/Light (P8-47) Removal, AMM TASK 52-51-07-020-801
- Cockpit Control Panel Switch/Light (P8-47) Installation, AMM TASK 52-51-07-420-801

1) If the flight compartment door locks with the Flight Compartment Door Lock Switch, S2 in the AUTO position you have corrected the problem.

2) If the flight compartment door does not lock with the Flight Compartment Door Lock Switch, S2 in the AUTO position, then continue.

(g) Repair the wiring between connector D2 on the Door Access Controller/Chime Module, M2537 and the Electric Strike, M2538 (WDM 52-51-11).

1) If the flight compartment door locks with the Flight Compartment Door Lock Switch, S2 in the AUTO position, you have corrected the problem.

2) If the flight compartment door does not lock with the Flight Compartment Door Lock Switch, S2 in the AUTO position, then continue.

(h) Do a check for continuity in the wiring that goes between the Stab Trim/Light Module, P8-47 and the Door Access Controller/Chime Module, M2537.

1) If you find a problem with wiring then repair the wiring (WDM 52-51-11).

——— END OF TASK ———

805. Door Does Not Unlock in the UNLOCK Position - Fault Isolation

A. Description

(1) (SDS SUBJECT 52-51-00)

B. Possible Causes

(1) Control Panel Switch/Light Module, P8-47

(2) Door Access Controller/Chime Module, M2537

(3) Wiring

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C. Circuit Breakers

(1) This is the circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	1	C00137	DOOR LOCK

D. Related Data

(1) (SSM 52-51-11)

(2) (WDM 52-51-11)

E. Initial Evaluation

(1) Open the flight compartment door.

(2) Put the power switch on the chime module, M2537 to the NORM position.

(3) Put the flight compartment door lock switch, S2, to the UNLOCK position.

(4) Make sure that the electric strike M2538 goes to the release position.

(5) If the electric strike did not go to the release position, then do the Fault Isolation Procedure below.

(6) If the electric strike did go to the release position, then there was an intermittent fault.

F. Fault Isolation Procedure

(1) Install a new switch/light module, P8-47.

These are the tasks:

Cockpit Control Panel Switch/Light (P8-47) Removal, AMM TASK 52-51-07-020-801,

Cockpit Control Panel Switch/Light (P8-47) Installation, AMM TASK 52-51-07-420-801.

(2) Put the flight compartment door lock switch, S2, to the UNLOCK position.

(a) If the electric strike, M2538 went to the release position you have corrected the problem.

(b) If the electric strike, M2538, did not go to the to the release position then continue.

(3) Do these steps to do a check for continuity between Pin 10 on connector D3428 and Pin 3 on connector D13730.

(a) Remove connector D3428 from the switch/light module, P8-47.

(b) Remove connector D13730 from the chime module M2537. To remove connector D13730 from the chime module, do this task: Chime Module Installation, AMM TASK 52-51-06-400-801

(c) Do a check for continuity between Pin 10 on connector D3428 and Pin 3 on connector D13730. (WDM 52-51-11)

(d) If there is not continuity between Pin 10 on connector D3428 and Pin 3 on connector D13730, do these steps:

1) Repair the wire, (WDM 52-51-11).

2) Install the chime module, M2537. To install the chime module, do this task: Chime Module Installation, AMM TASK 52-51-06-400-801

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3) Install the switch/light module, P8-47. To install the P8-47 panel, do this task: Cockpit Control Panel Switch/Light (P8-47) Installation, AMM TASK 52-51-07-420-801

(e) If there is continuity between Pin 10 on connector D3428 and Pin 3 on connector D13730, then do these steps:

1) Install a new chime module, M2537. To install a new chime module, do this task: Chime Module Installation, AMM TASK 52-51-06-400-801.

2) Install the switch/light module P8-47. To install the P8-47 module, do this task: Cockpit Control Panel Switch/Light (P8-47) Installation, AMM TASK 52-51-07-420-801

——— END OF TASK ———

806. Door Does Not Unlock in the AUTO Position After The Correct Code Has Been Entered - Fault Isolation

A. Description

(1) (SDS SUBJECT 52-51-00)

B. Possible Causes

(1) Keypad, M2536

(2) Door Access Controller/Chime Module, M2537

(3) Wiring

C. Circuit Breakers

(1) This is the circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	1	C00137	DOOR LOCK

D. Related Data

(1) (SSM 52-51-11)

(2) (WDM 52-51-11)

E. Initial Evaluation

(1) Open the flight compartment door.

(2) Put the power switch on the chime module, M2537 to the NORM position.

(3) Make sure that the flight compartment door lock switch, S2 is in the AUTO position.

(4) Make sure the electric strike M2538 is in the engage position.

(5) Enter the correct access code on the keypad, M2536 and push the ENT button.

(6) If the electric strike M2538 did not go to the release position, then do the Fault Isolation Procedure below.

(7) If the electric strike M2538 did go to the release position, then there was an intermittent fault.

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F. Fault Isolation Procedure

(1) Make sure that the flight compartment door lock switch, S2 is in the AUTO position.

(2) Put the flight compartment door lock switch, S2 to the UNLOCK position.

(3) If the electric strike, M2538 did not go to the release position, do this task: Door Does Not Unlock in the UNLOCK Position - Fault Isolation, 52-50 TASK 805.

(4) If the electric strike, M2538 did go to the release position, do these steps:

(a) Program a new access code in the keypad. To program a new access code in the keypad, do this task: Program the Access Code, AMM TASK 52-51-00-900-801

(b) Enter the new code and press the ENT button.

(c) If the electric strike went to the release position you have corrected the problem.

1) Make sure that the keypad is set to the correct access code.

(d) If the electric strike did not go to the release position, then continue.

(e) If the amber light on the keypad came on after the correct code was entered, install a new chime module M2537.

These are the tasks:

Chime Module Removal, AMM TASK 52-51-06-000-801,

Chime Module Installation, AMM TASK 52-51-06-400-801.

1) If the electric strike went to the release position you have corrected the problem.

(f) If the amber light on the keypad did not come on after the correct code was entered, install a new keypad M2536.

These are the tasks:

Keypad Removal, AMM TASK 52-51-05-000-801,

Keypad Installation, AMM TASK 52-51-05-400-801.

1) If the electric strike went to the release position you have corrected the problem.

2) If the electric strike did not go to the release position, do this step:

a) Repair the wiring between the chime module M2537 and the keypad M2536. (WDM 52-51-11)

——— END OF TASK ———

807. Door Unlocks or Chime Sounds in the DENY Position - Fault Isolation

A. Description

(1) (SDS SUBJECT 52-51-00)

B. Possible Causes

(1) Control Panel Switch/Light Module P8-47

(2) Door Access Controller/Chime Module, M2537

(3) Wiring

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OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999 (Continued)

C. Circuit Breakers

(1) This is the circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	1	C00137	DOOR LOCK

D. Related Data

(1) (SSM 52-51-11)

(2) (WDM 52-51-11)

E. Initial Evaluation

(1) Open the flight compartment door.

(2) Put the power switch on the chime module, M2537 to the NORM position.

(3) Make sure that the flight compartment door lock switch, S2 is in the AUTO position.

(4) Check that electric strike M2538 is in the engage position.

(5) Put the flight compartment door lock switch, S2 to the DENY position.

(6) If the electric strike M2538 went to the release position or the chime sounds, then do the Fault Isolation Procedure below.

(7) If the electric strike M2538 did not go to the release position and the chime module, M2537 did not sound, then there was an intermittent fault.

F. Fault Isolation Procedure

(1) Install a new control panel switch/light module, P8-47.

These are the tasks:

Cockpit Control Panel Switch/Light (P8-47) Removal, AMM TASK 52-51-07-020-801,

Cockpit Control Panel Switch/Light (P8-47) Installation, AMM TASK 52-51-07-420-801.

(2) If the electric strike, M2538 did not go to the release position and the chime module, M2537 did not sound you have corrected the problem.

(3) If the electric strike M2538 went to the release position or the chime sounds, then do these steps:

(a) Remove connector D3428 from the switch/light module, P8-47.

(b) Remove connector D13730 from the chime module, M2537.

(c) Check the continuity between pin 13 of connector D3428 and pin 4 of connector D13730.

(d) If there is not continuity between pin 13 of connector D3428 and pin 4 of connector D13730, repair the wiring. (WDM 52-51-11)

1) Install the chime module, M2537. To install the chime module, do this task: Chime Module Installation, AMM TASK 52-51-06-400-801

(e) If there is continuity between pin 13 of connector D3428 and pin 4 of connector D13730, install a new chime module, M2537.

1) To install a new chime module, do this task: Chime Module Installation, AMM TASK 52-51-06-400-801

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(f) Install the control panel switch/light, P8-47. To install the P8-47 panel, do this task: Cockpit Control Panel Switch/Light (P8-47) Installation, AMM TASK 52-51-07-420-801

END OF TASK

808. Chime Does Not Sound When the Correct Code is Entered on the Keypad in the AUTO Mode - Fault Isolation

A. Description

(1) (SDS SUBJECT 52-51-00)

B. Possible Causes

(1) Keypad, M2536

(2) Door Access Controller/Chime Module, M2537

(3) Wiring

C. Circuit Breakers

(1) This is the circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	1	C00137	DOOR LOCK

D. Related Data

(1) (SSM 52-51-11)

(2) (WDM 52-51-11)

E. Initial Evaluation

(1) Open the flight compartment door.

(2) Put the power switch on the chime module, M2537 to the NORM position.

(3) Make sure that the flight compartment door lock switch, S2 is in the AUTO position.

(4) Check that electric strike M2538 is in the engage position.

(5) Enter the correct access code on the keypad, M2536 and push the ENT button.

(6) If the chime module did not sound, then do the Fault Isolation Procedure below.

NOTE: The chime and the keypad can be disabled for up to 30 minutes after the door lock switch is turned to the DENY position. The DENY mode can be ended by turning the door lock switch to the UNLOCK position.

(7) If the chime module did sound, then there was an intermittent fault.

F. Fault Isolation Procedure

(1) Make sure that the flight compartment door lock switch, S2 is in the AUTO position.

(2) Enter the correct access code on the keypad, M2537.

(3) If the electric strike, M2538 did not go to the release position, do this task: Door Does Not Unlock in the AUTO Position After The Correct Code Has Been Entered - Fault Isolation, 52-50 TASK 806

(4) If the electric strike, M2538 did go to the release position, do this step:

(a) Install a new chime module, M2537. To install a new chime module,

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OMR 805, 807, 809, 810, 812, 813, 815, 816, 818-826, 901-999 (Continued)

These are the tasks:
AMM TASK 52-51-06-000-801,
Chime Module Installation, AMM TASK 52-51-06-400-801,
.

——— END OF TASK ———

809. Flight Compartment Door LOCK FAIL light is On - Fault Isolation

A. Description

(1) (SDS 52-50-00)

B. Possible Causes

(1) Electric Strike, M2538
(2) Door Access Controller/Chime Module, M2537
(3) Control Panel Switch/Module, P8-47
(4) Wiring

C. Circuit Breakers

(1) This is the circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	1	C00137	DOOR LOCK

D. Related Data

(1) (SSM 52-51-11)
(2) (WDM 52-51-11) (WDM 52-51-11)

E. Initial Evaluation

(1) Open the flight compartment door.
(2) Put the power switch on the chime module, M2537 to the NORM position.
(3) Make sure that the flight compartment door lock switch, S2 is in the AUTO position.
(4) Make sure that the LOCK FAIL light on the P8-47 panel is on.
(5) If the LOCK FAIL light is on, then do the Fault Isolation Procedure below.
(6) If the LOCK FAIL light is not on, then there was an intermittent fault.

F. Fault Isolation Procedure

(1) Make sure that the flight compartment door lock switch, S2 is in the AUTO position.
(2) If the electric strike, M2538 is in the release position, do this step:
    (a) Do this task: Door Does Not Lock in the AUTO Position - Fault Isolation, 52-50 TASK 804.
(3) If the electric strike, M2538 is in the engage position, do these steps:
    (a) Install a new control panel switch/light module, P8-47. These are the tasks:
    Cockpit Control Panel Switch/Light (P8-47) Removal, AMM TASK 52-51-07-020-801,
    Cockpit Control Panel Switch/Light (P8-47) Installation, AMM TASK 52-51-07-420-801.

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(b) If the LOCK FAIL light went out with the door lock control switch in the AUTO position, you have corrected the problem.

(c) If the LOCK FAIL light did not go out with the door lock control switch in the AUTO position, continue:

(d) Install a new chime module, M2537. To install a new chime module. These are the tasks:

Chime Module Removal, AMM TASK 52-51-06-000-801,

Chime Module Installation, AMM TASK 52-51-06-400-801.

(e) If the LOCK FAIL light did not go out with the door lock control switch in the AUTO position, do these steps:

1) Repair the wiring between the chime module, M2537 and the control panel switch/light, P8-47 (WDM 52-51-11).

2) Install the chime module, M2537. To install the chime module, do this task: Chime Module Installation, AMM TASK 52-51-06-400-801.

3) Install the control panel switch/light module, P8-47. To install the P8-47 panel, do this task: Cockpit Control Panel Switch/Light (P8-47) Installation, AMM TASK 52-51-07-420-801.

——— END OF TASK ———

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801. Airline Method Procedure - Fault Isolation

A. Initial Evaluation

NOTE: Use the standard method of your airline to correct this fault.

END OF TASK

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